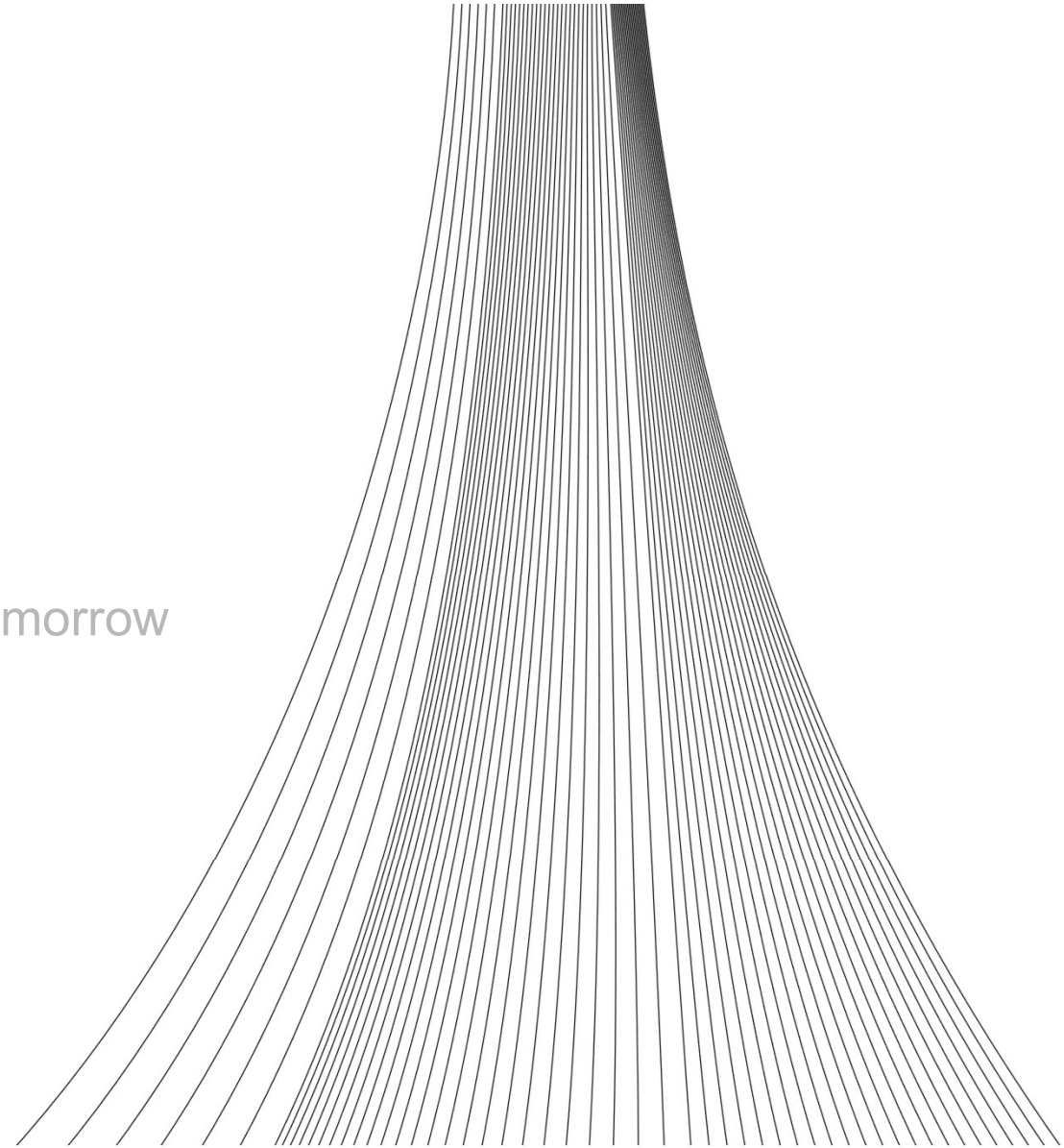
FORTNA

OptiSweep – Training

Redefining consolidation for tomorrow



Contents

1. Overview
2. Geek+ RMS
3. Aveva

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Overview

What is Optisweep?

1

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What is OptiSweep

Traditional vs Fortna's Solution

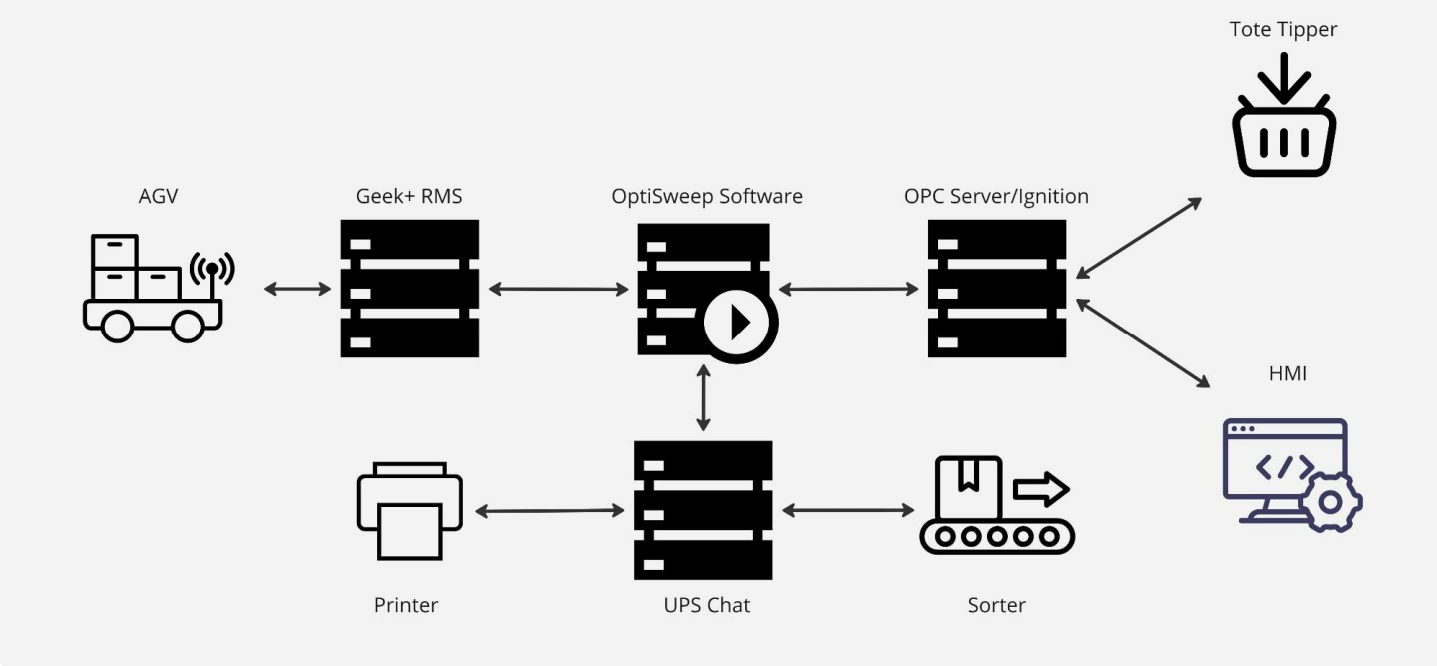
- FORTNA developed an automated Consolidated Bagging (ACB) solution utilizing AGVs and tippers to create a flexible and scalable bagging system.
- In a traditional small sort system. Operators are assigned sections of destinations (20-25) on a sorter. Parcels divert down the destination and fall into a bag. When the bag becomes full an operator closes the destination, prints a label, applies the label to the bag, zips up the bag, places the bag on a takeaway belt, and then reopens the chute.
- In a Belted ACB solution when a destination is full the bin is dropped on a takeaway conveyor that conveys the packages as a slug that go down a chute into a bag station where the user applies a label, zips up the bag and places it on a takeaway belt.
- Fortna wanted a more modular and adaptive design that would need little modification to existing sorters. By utilizing AGVs and tippers we can scale solutions and tailor maps to existing building sorters.

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Overview

Core Elements

- UPS Chat
- Tote Tip Stations
- Geek+ AGVs
- Hospital Controller
- OptiSweep Software
- Aveva HMI

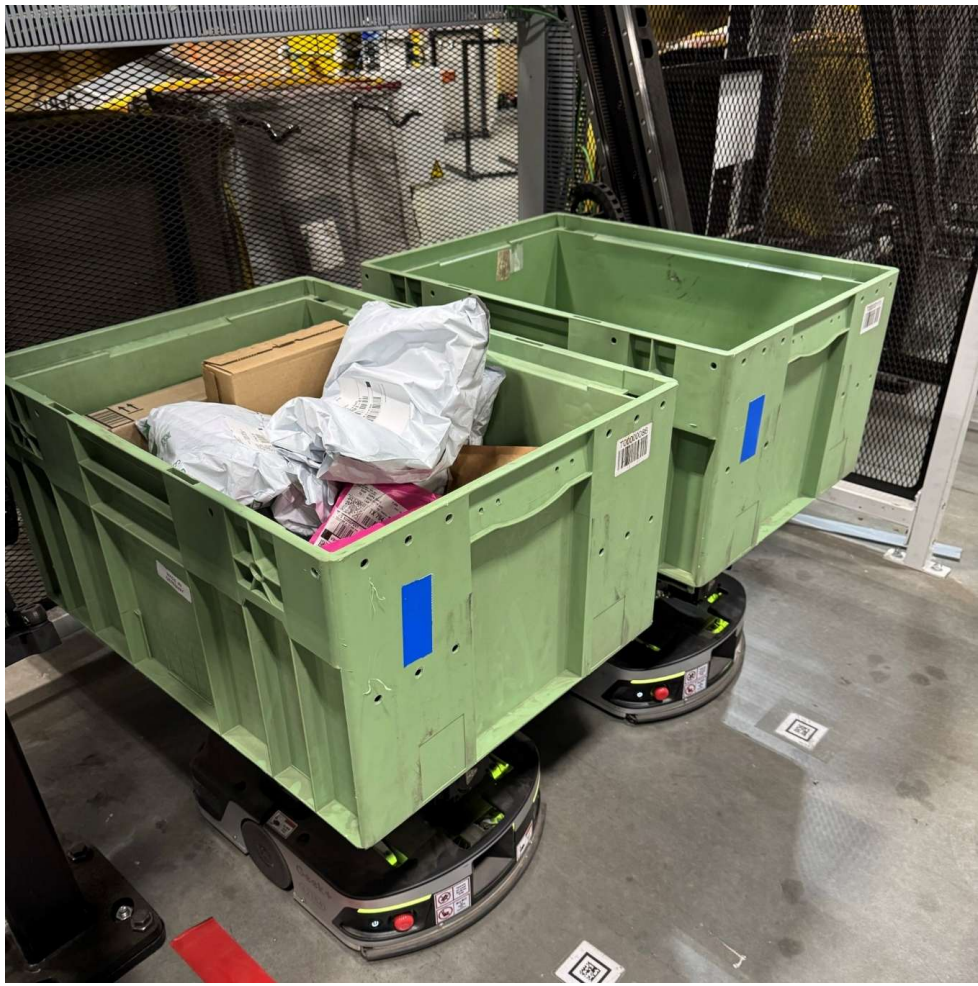


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AGVs

Automated Guided Vehicles

- Geek+ AGVs
- Operate over QR codes
- Directly controlled by Geek+ RMS
- Geek+ handles all movement
- Geek+ determines the best route to nagivate from point A to point B
- These AGVs have forward collision sensors
- This version of AGVs are intended to operate in a fenced system.



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Tote Racking

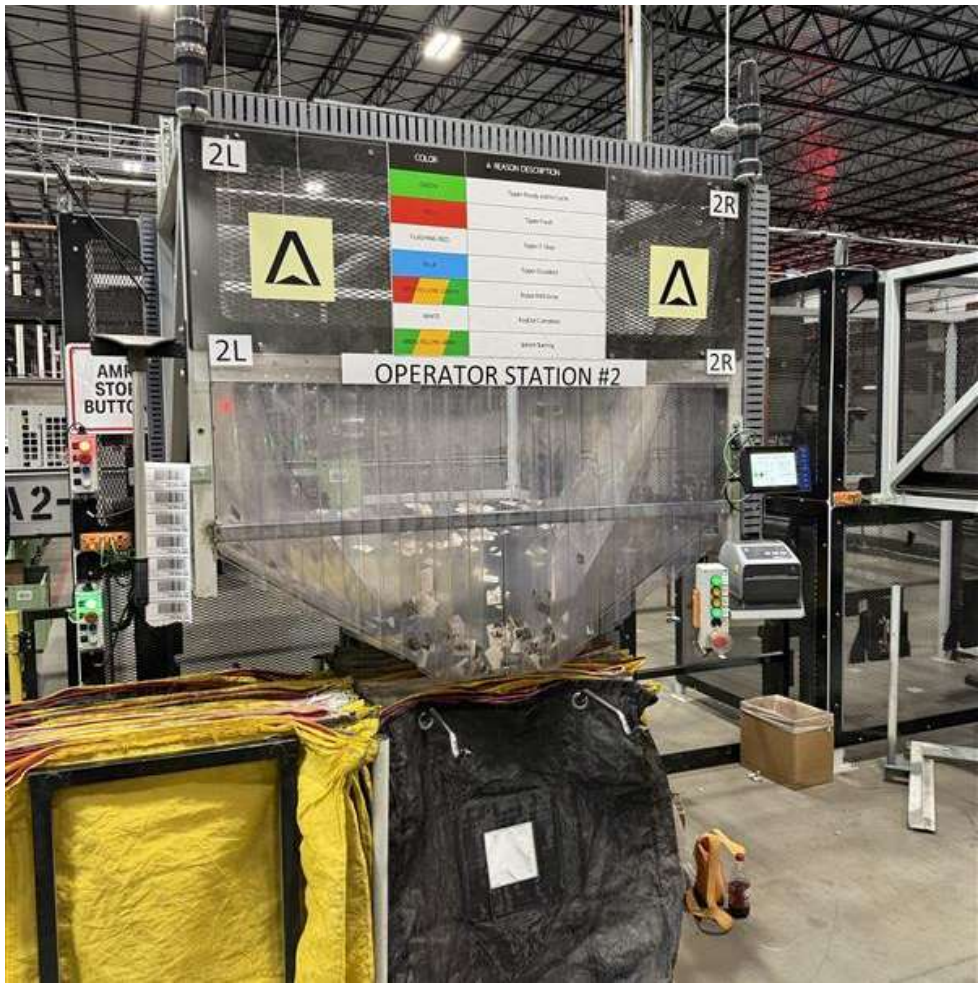
- Fortna designed and installed
- Beveled design to securely position tote for parcel impact



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Tipper Station

- Fortna R&D Design
- Each station has two tippers
- Tipper will clamp tote from AGV
- Actuates tipping arm to dump contents onto chute
- Places tote back onto AGV top
- Self-contained HMI for operation and fault recovery
- Operators equipped with Estop and ready to tip buttons.



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Software

- OptiSweep software
- Aveva - HMI
- Ignition - OPC Server
- Chat xLE – UPS communications chat
- S-Tunnel – Encryption from OTNet to UPS Corp

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Lifecycle of a Sort

Start-up

- Takes AGVs from a power cycling state to being staged for the sort

Running

- Bulk of the sort where items are sorted and AGVs take full totes to tippers for bagging

Bagout

- Cycles all locations that have any items in a tote to finish out the sort

Shutdown

- Stages AGVs in a position where they can cycle to chargers to prepare for the next sort

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Lifecycle of a parcel

Before parcels reach our system, they are unloaded from trailers in bulk onto a series of belts that lead to sorter induction.

Those parcels are singulated and travel

- This triggers the first message from Chat with a tracking Id, dimensions, and package type
- If this message fails and we see something unknown inducted, it is assigned default dimensions by OptiSweep

Parcels then travel through auto-inducts onto the sorter

- This triggers the second message from Chat with a tracking Id, barcode data, dimensional data, and potential destination data.

At this point, a parcel is on the sorter, we have dimensional data, and we have an indication where it could be going.

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Lifecycle of a parcel

A parcel will sort down to a chute location and into a tote (hopefully)

- This triggers the next message from Chat that an item has sorted
- This message is critical. If not received or processed, we don't have a record of it being in a tote.

Software adds that parcel to the tote located in that rack

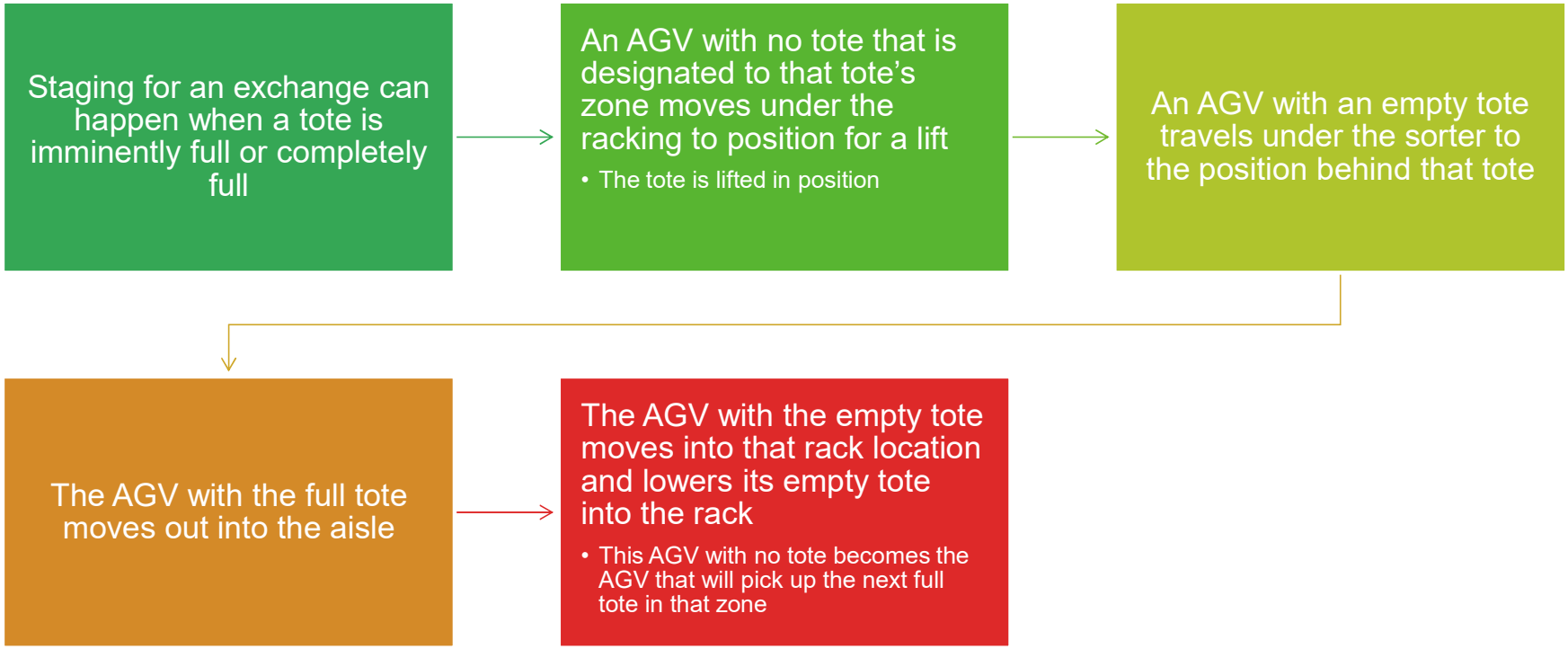
- Weight and dimensions are accumulated as items sort
- Dimensions are scaled up or down by software based on package type

Software determines how full a tote is

- Two thresholds exist
  - Pretty full – Not full, but getting close enough to start considering an exchange
  - Full – No more product can be diverted to this location until an exchange is done
    - When a tote is full, we tell Chat that the chute is unavailable so that no packages come down while doing an exchange

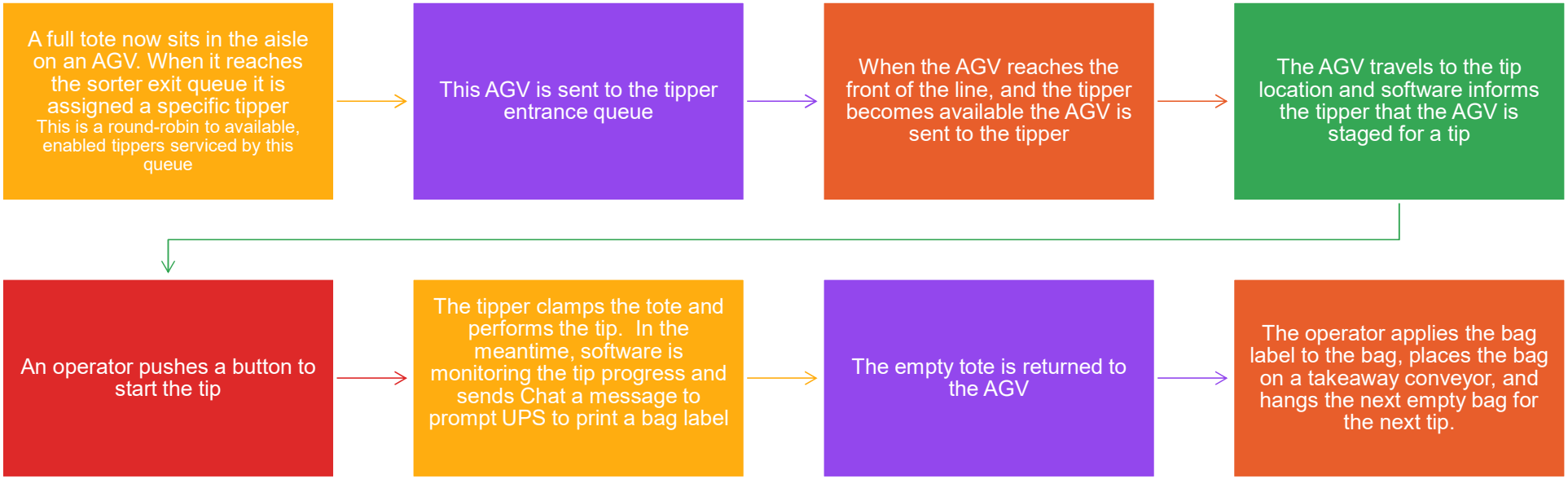
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The Excahnge



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Tipping



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The Return

When the full cycle is complete, the tipper informs software that the AGV is clear to move out

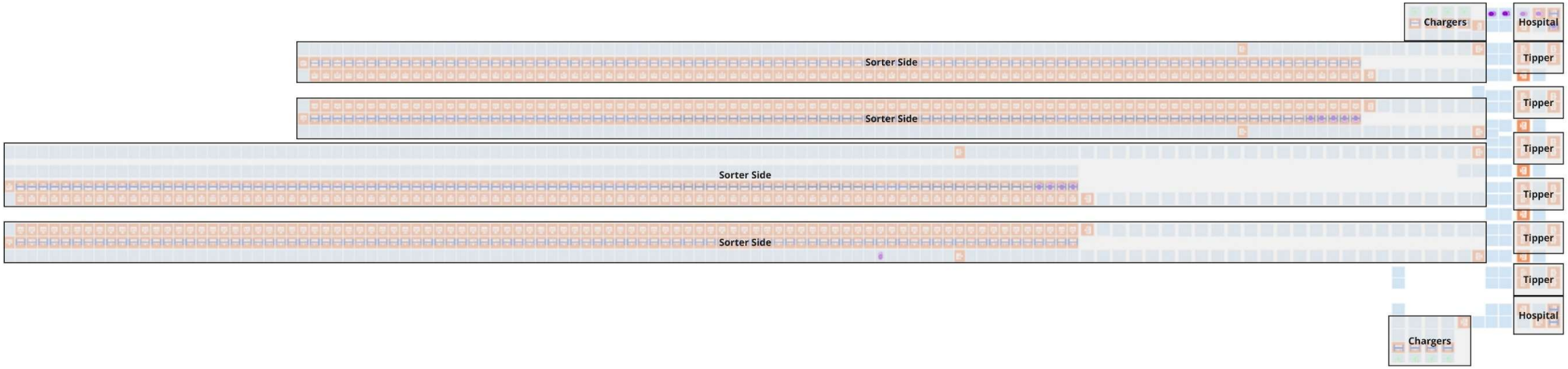
The AGV can return to the sorter, go to the hospital, or go to charge

- Failed tip leads to Hospital
- Low battery level leads to the charger
- If all is good, the AGV returns to the sorter entrance queue with an empty tote that will be used for its next exchange

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Layout Design

- The layout is designed for modularity and scalability, with distinct areas connected by a common entrance and exit. This ensures that to move from one area to another, an AGV must go to the exit before entering the next.
- Areas: Hospital, SorterSide, Charger, Tipper



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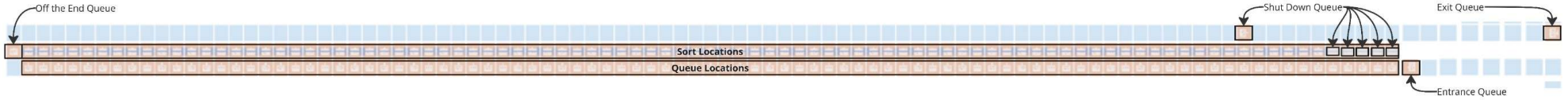
Slide 16

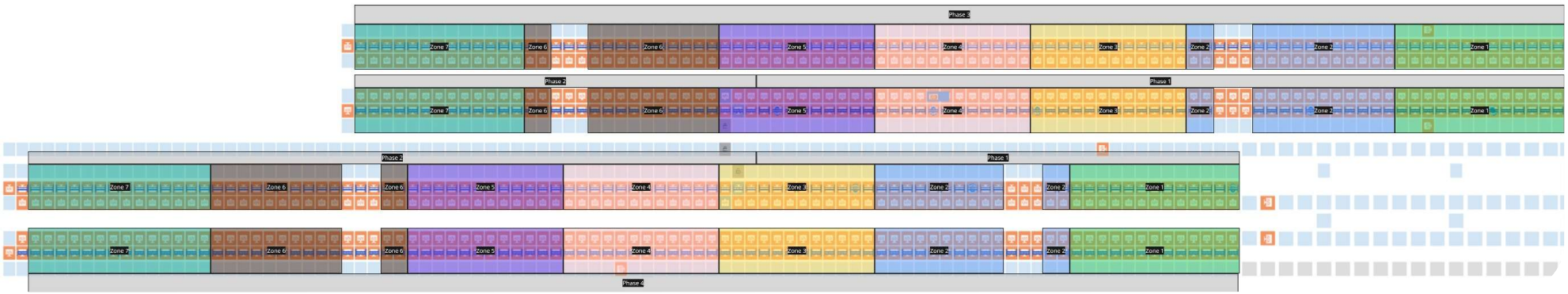
MF1 Maybe Add GLD

Mitchel Flynn, 2026-03-11T17:51:17.457

Layout Design

- Sorter



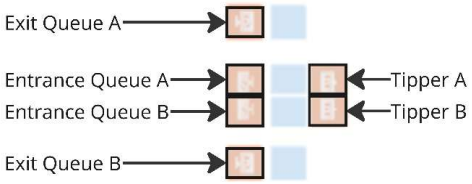


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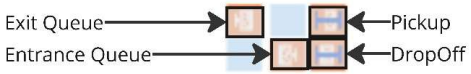
Layout Design

Modularization

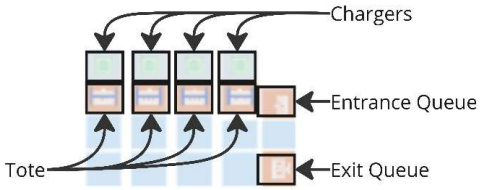
- Tipper



- Hospital



- Charger



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Geek+ RMS Walkthrough

2

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Home



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Home Breakdown

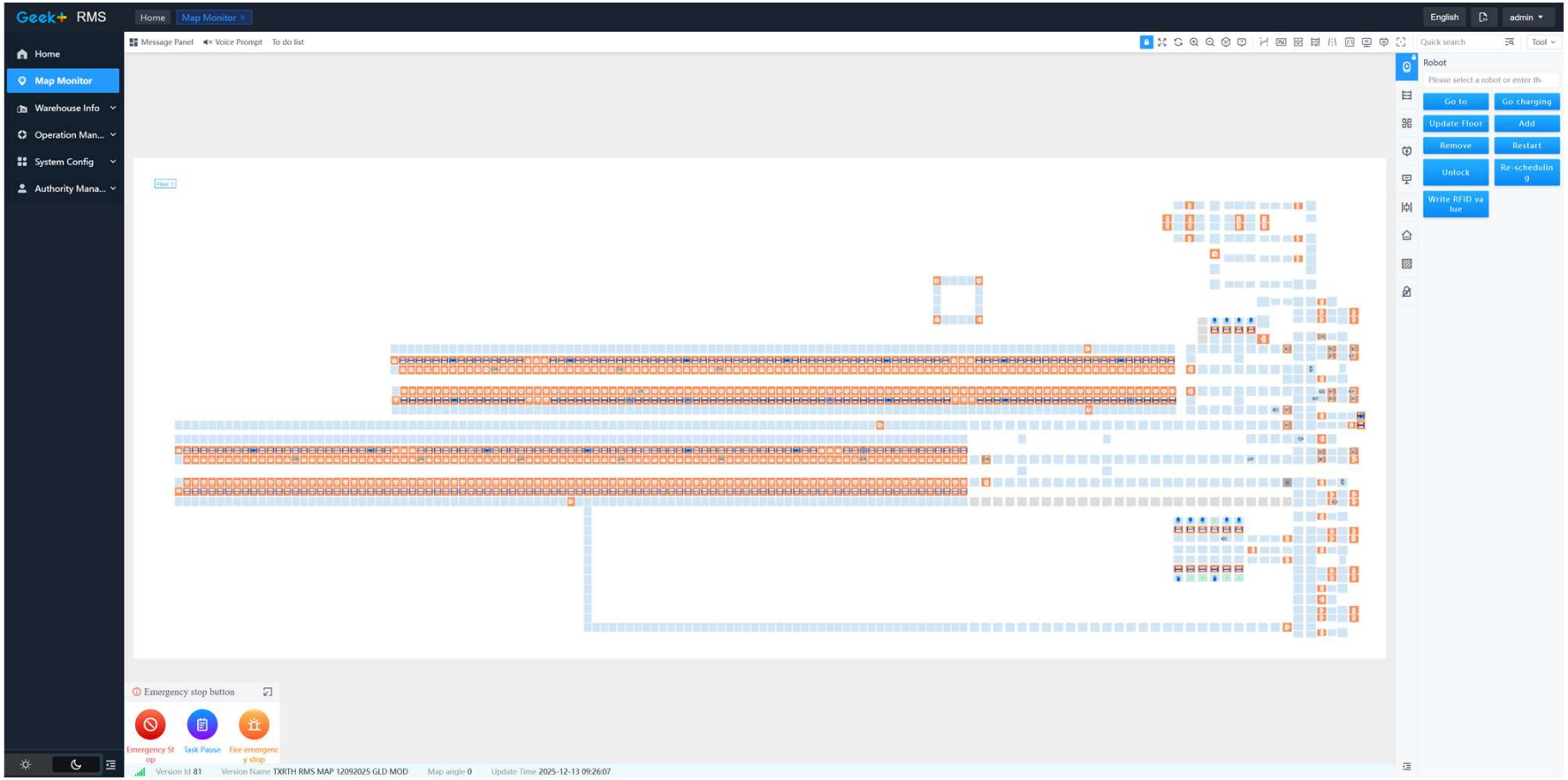
1. In the right-hand corner is your language selection
2. On the right side you can expand menu options as well as select map monitor.
   - Map Monitor allows you to view the map and AGV movements
3. You can expand the menu you or collapse it.



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Map Monitor

Map Overview

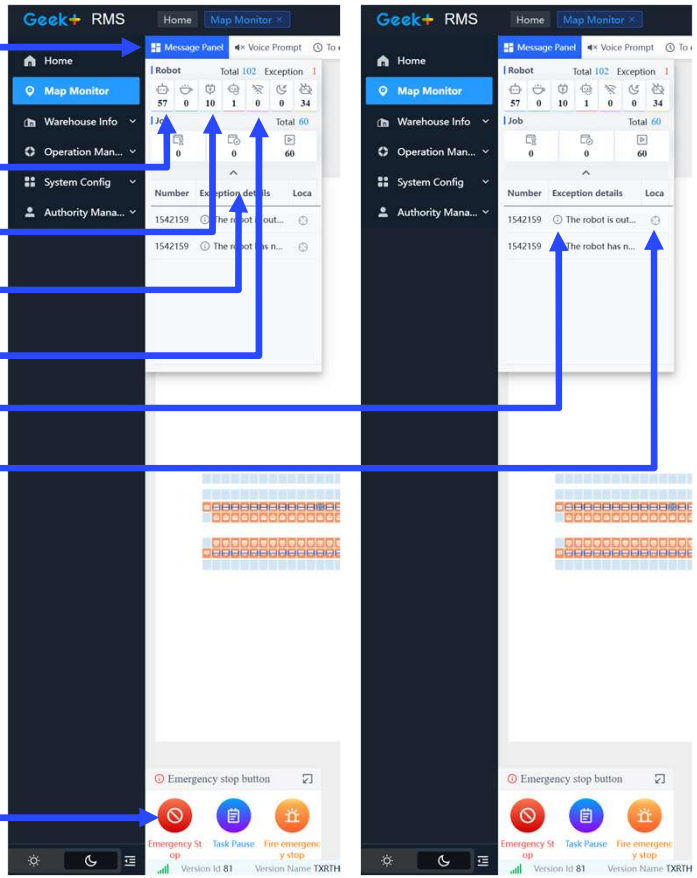


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Map Monitor

Message Panel

- Open message panel
- Count of active Robots
- Count of Robots charging
- Drop down to show AGV faults
- Count of AGVs disconnected
- Inspection to view details of fault
- Locate Faulted AGV
- E stop the system

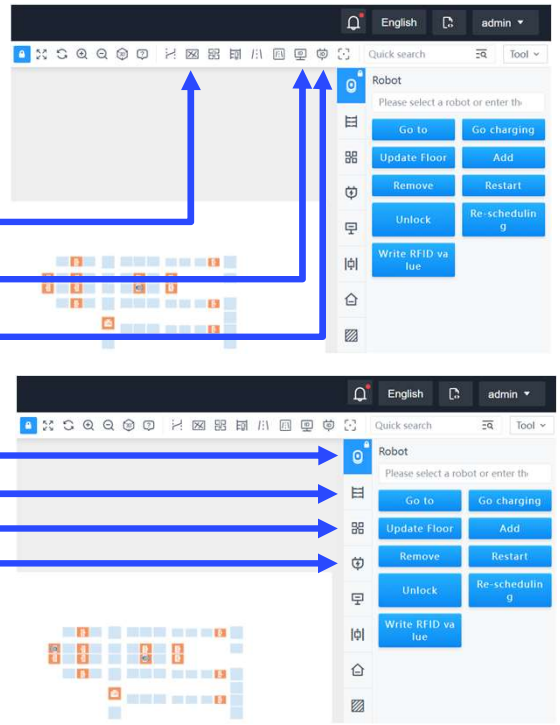


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Map Monitor

Quick Select

- View all AGV active paths
- View Station Ids
- View Charger Ids
- Robot Selection
- Container Selection
- Cell Selection
- Charger Selection



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Robot Selection

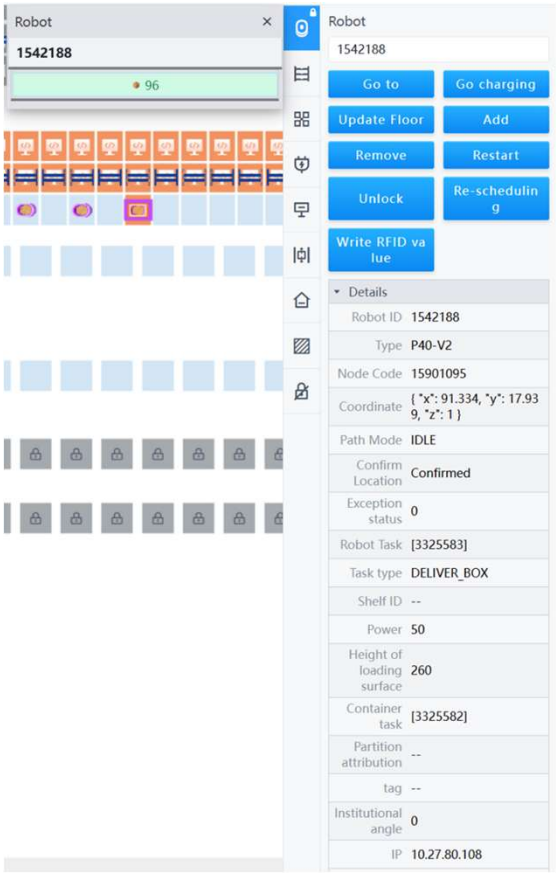
Overview

When Robot option is selected, select a robot to view robot details.

The robot selected will have a box around it showing its selected.

If the robot has a tote it will show under the robot Id. If the tote has a “?” before it, that means the robot has been removed but the tote has not. If a robot doesn’t have a tote it will say layer_1

- Robot colors
  - Purple: robot disconnected
  - Blue/Flashing Blue: robot in task/searching for label
  - Green: robot not in a task
  - Red: fault/low battery



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Robot Selection

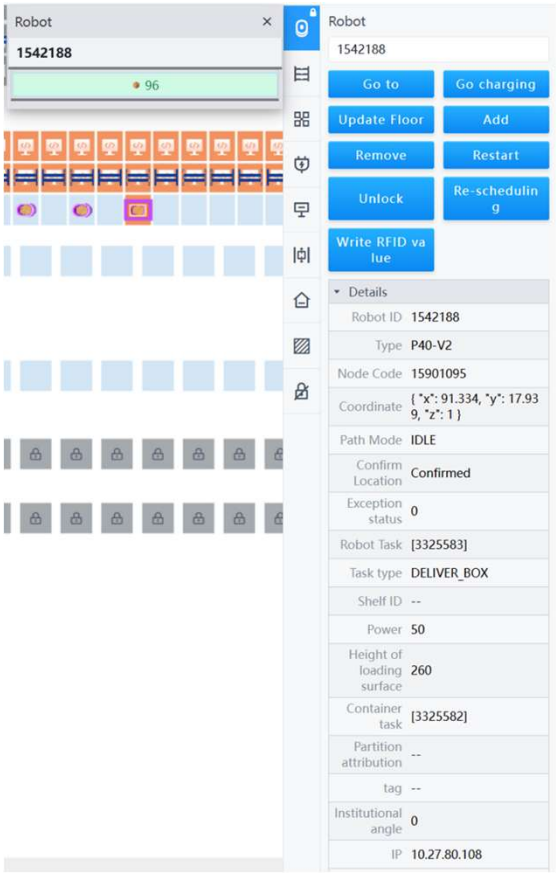
Add and Remove

- Add Robot

1. Enter the corresponding robot ID
2. Click “Add”
3. The system will show that the adding was successful.

- Remove Robot

1. Enter the corresponding robot Id or select the robot.
2. Click “Remove”
3. The system will show that the removal was successful.



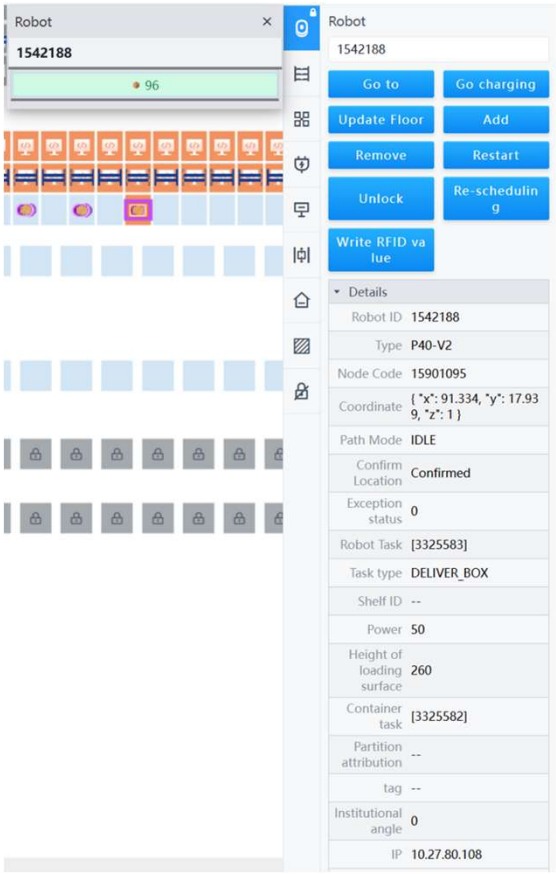
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Robot Selection

Restart

- Restart Robot

1. Enter the corresponding robot Id or select the robot.
2. Click “Restart”
3. The robot will restart (it is at the restart point or on the QR code cell).
4. After starting, the robot that has been positioned will be displayed on the map.



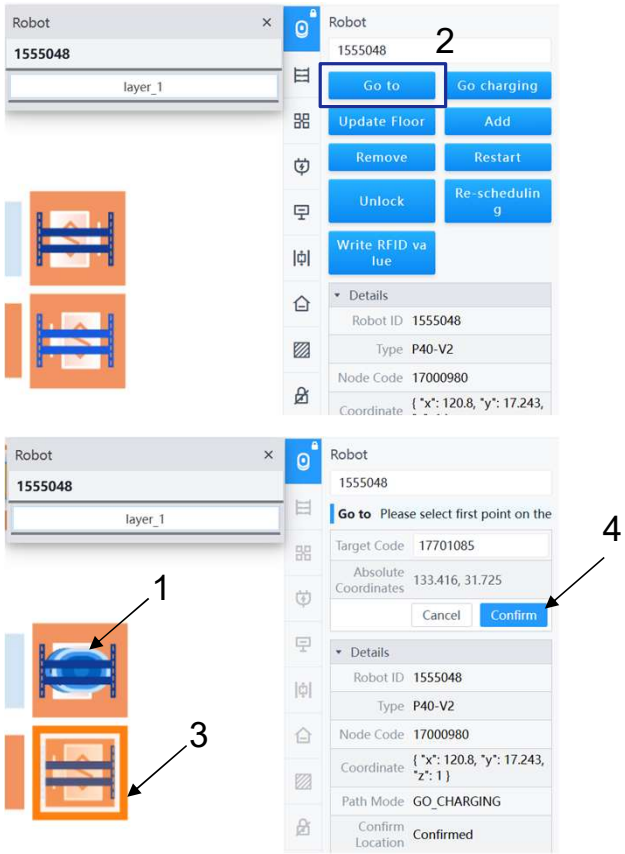
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Robot Selection

Go-to function

An AGV not in a task can be sent to any square. This is helpful when trying to recover quickly. Example if a Tipper goes down and an AGV will not leave. You can remove and add the agv and use the go to function to drop it off at the hospital. The AGV will pickup a new task once it's tote has been removed

1. Select an AGV
2. Select Go to
3. Select which square you want the AGV to go to
4. Select Confirm

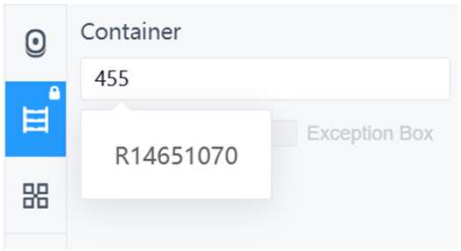


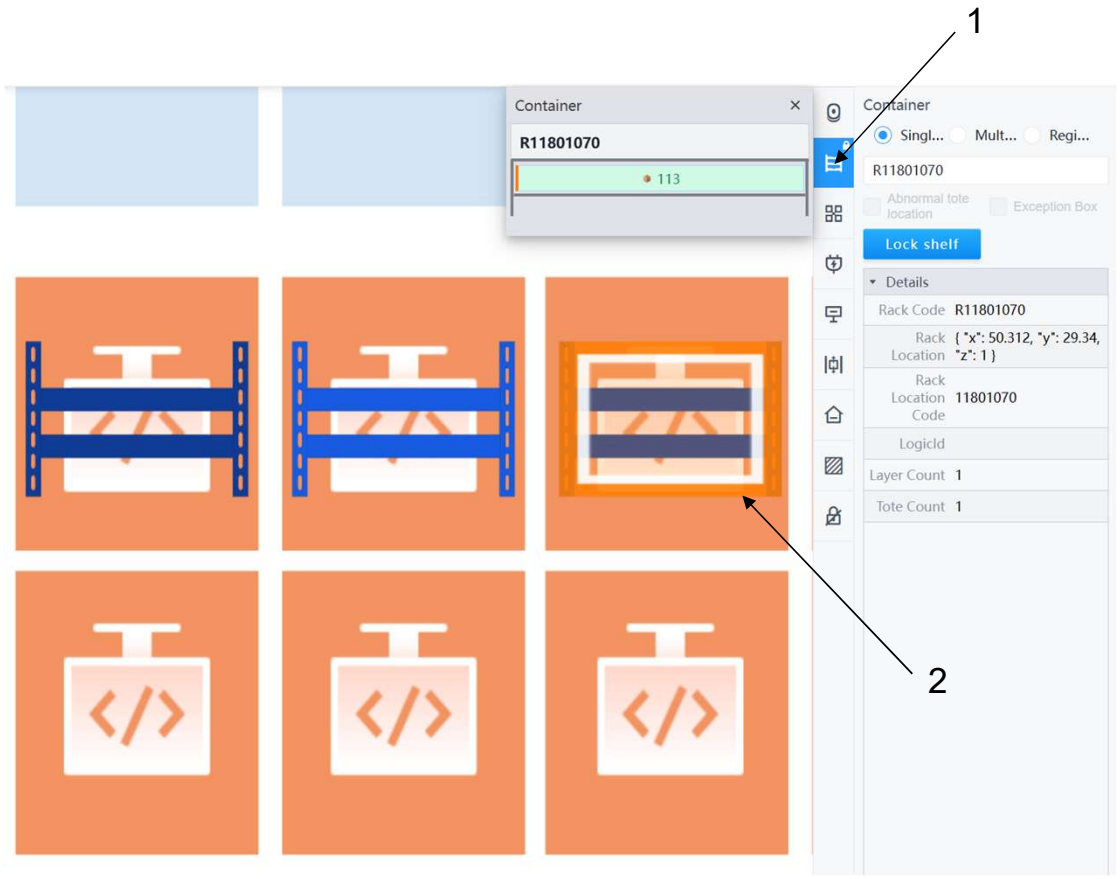
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Container Selection

Totes

- Dark blue indicates the presence of a tote. While light blue represents empty rack. If you select it, a box icon will show as well as the tote number
- You can also find a tote by typing the number in and selecting the rack or AGV it's located on if its in the system.



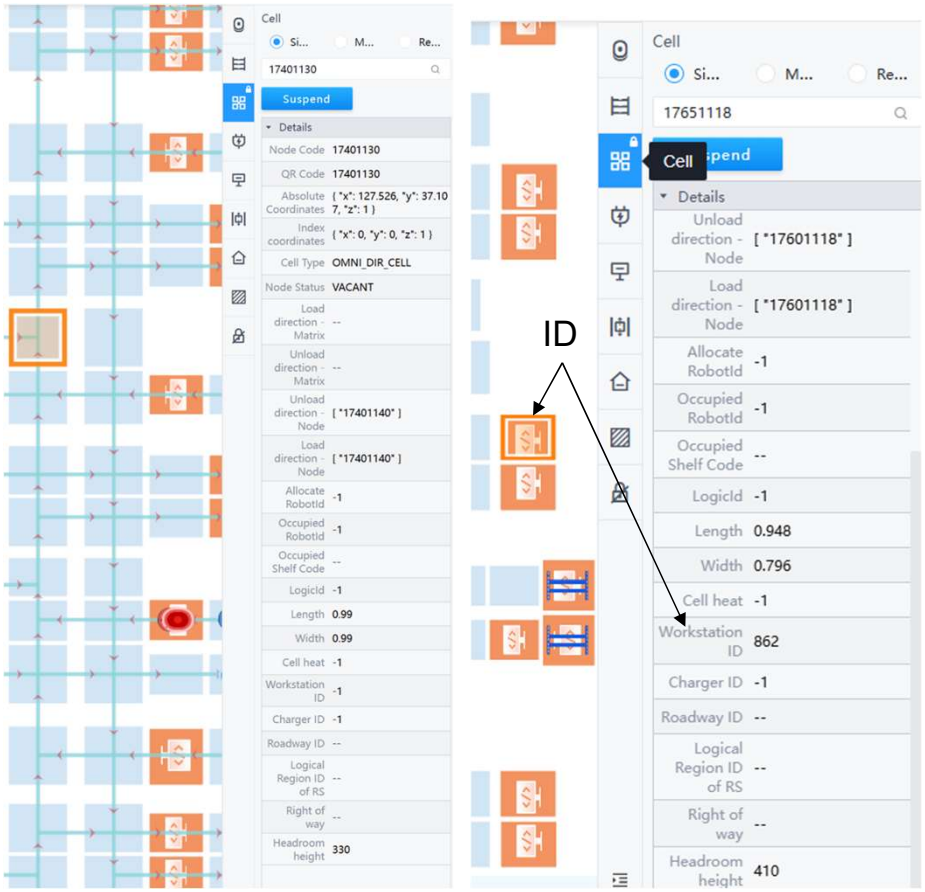


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Cell Selection

QR codes

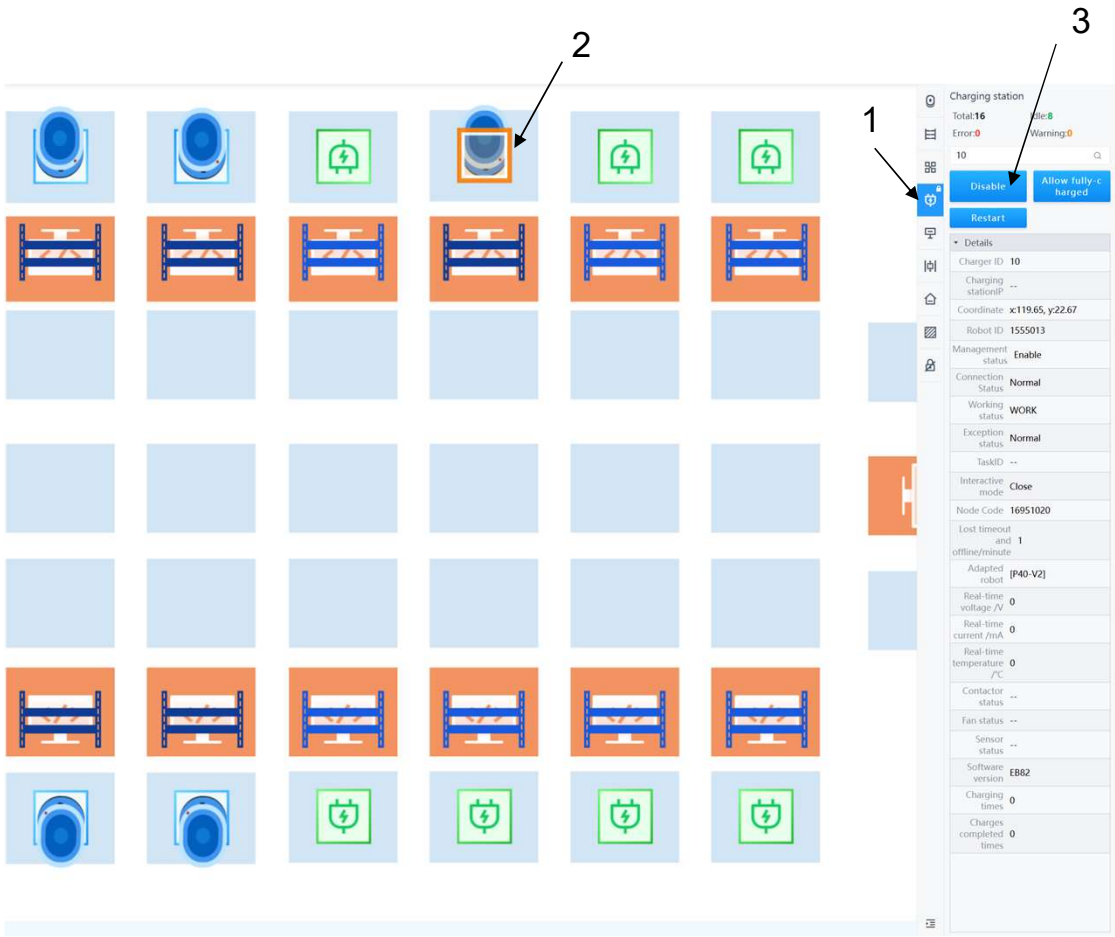
- When cell selection is selected select a cell to view cell information including QR code.
- Orange cells represents stations. If you select an orange cell it will show you the station ID along with the QR code.



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Charger Selection

- When charger selection is selected select a charger to view charger information.
- If you need to use a charger to manually charge a robot you can disable a charger so that no other robot is sent to that charger.
- Once the charger is disable, desired AGV can be pushed in to the disabled charging station.



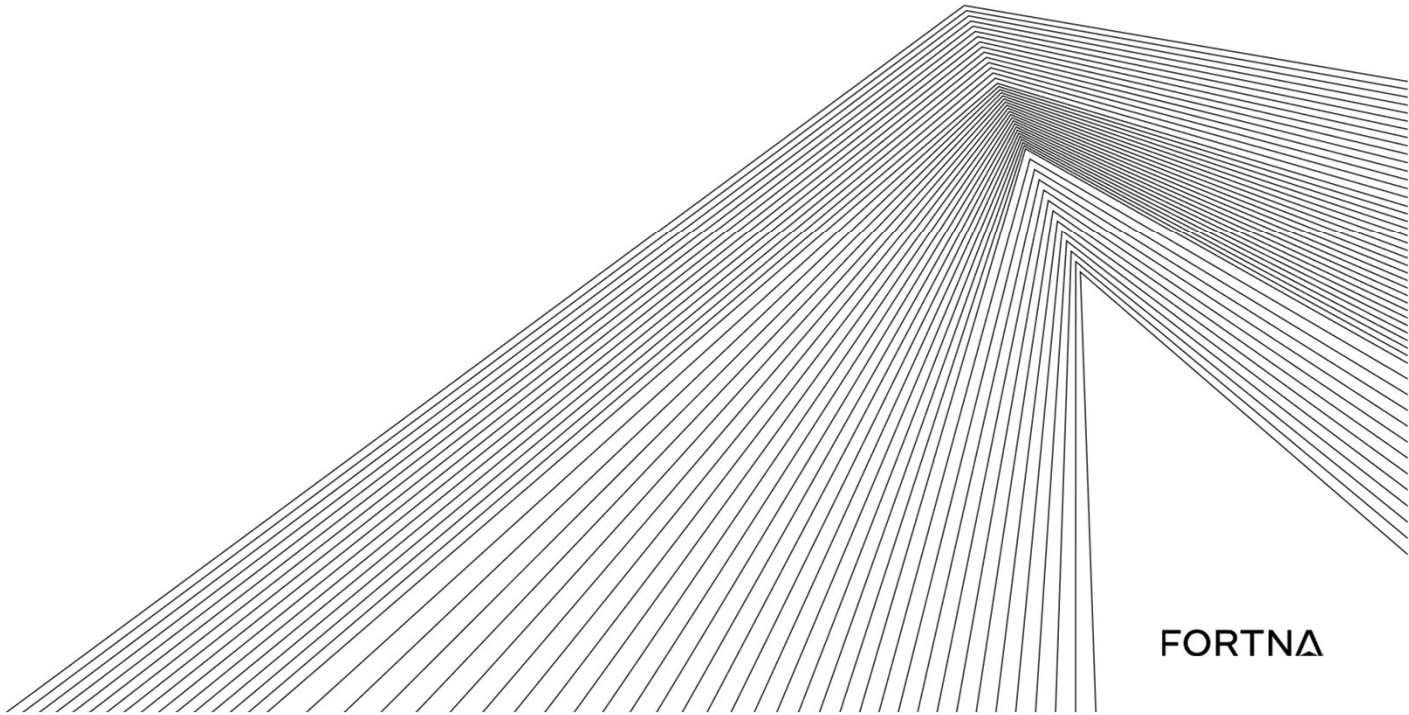
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Aveva Walkthrough

3

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HMI



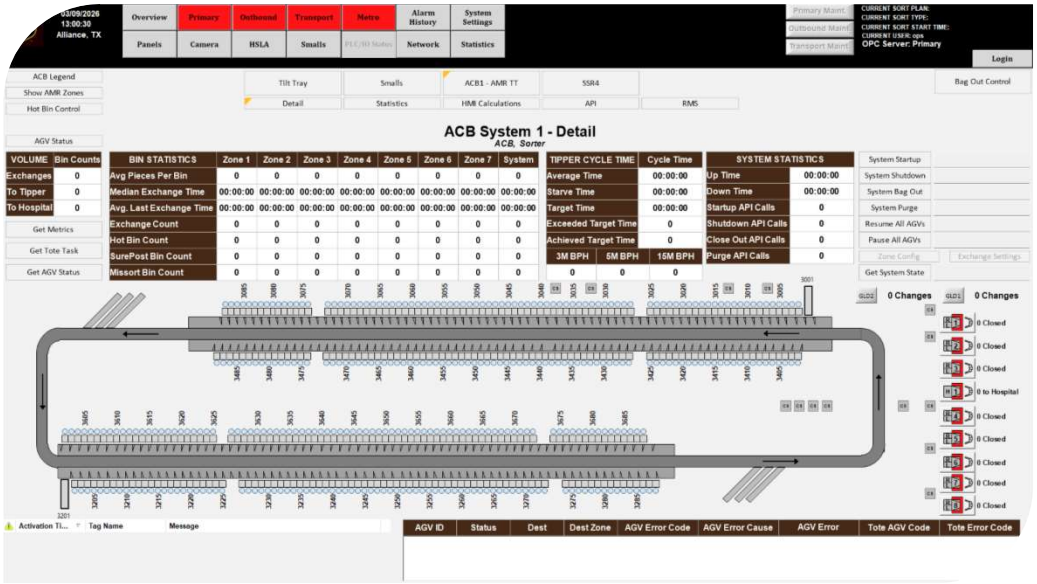
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Overall System

System Detail

- **Main Menu:** Navigate subscreens – Detail, Statistics, HMI Calculations, API, RMS
  - *Tip:* Click **“ACB1 – AMR TT”** if a subscreen does not update
- **System Layout:** Shows Sorter, Tippers, Hospital, GLDS, CS Estops and AGV Errors
  - Click icons for more info on Tippers/Hospitals/GLDS
- **Controls & Info:** Start/Stop system, Bag Out, Purge, view Metrics (Destinations), AGV status, Tote
- **Purpose:** Home screen for system overview and navigation

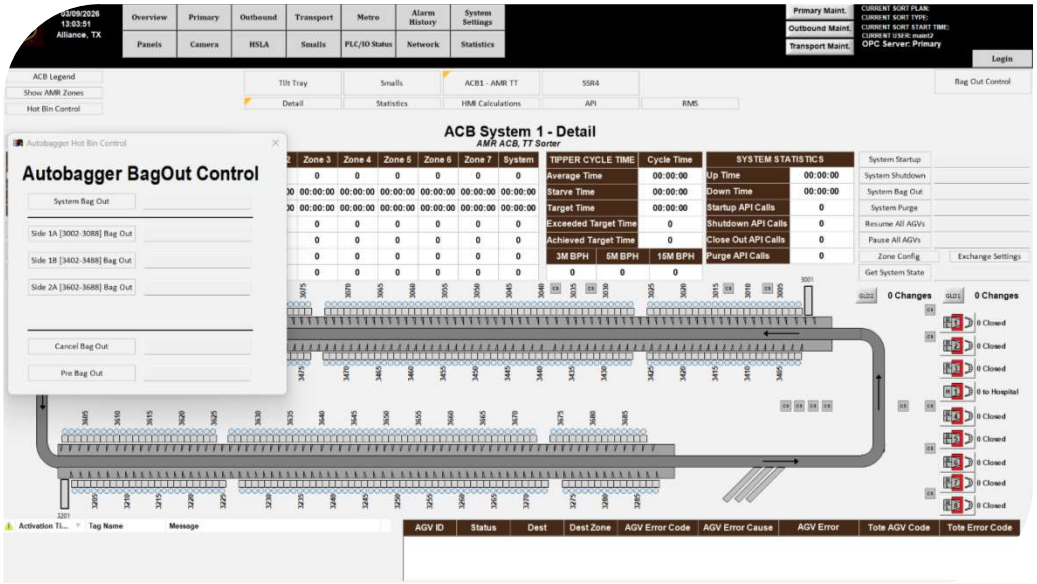


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Overall System

Bag Out Control

- System Bag out will close all destinations and exchange all totes with parcels in them.
- A user could also choose to bag out by leg of the sorter.
- In the event of a manual bag out Fortna offers a cancel bag out that will cancel any exchange task that is not in process.
- Pre bag out exchanges the lowest 25% of destinations per leg of the sorter.

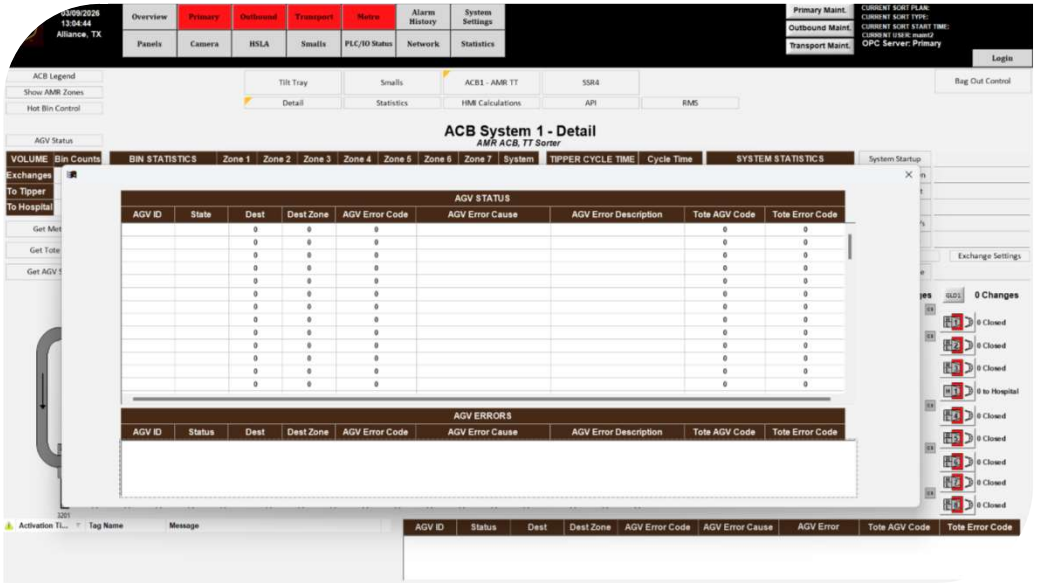


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Overall System

AGV Status

- **Top Section:** Detailed info for all AGVs – ID, Status, Destination, and Errors
- **Bottom Section:** Only AGVs with errors, with additional details will appear

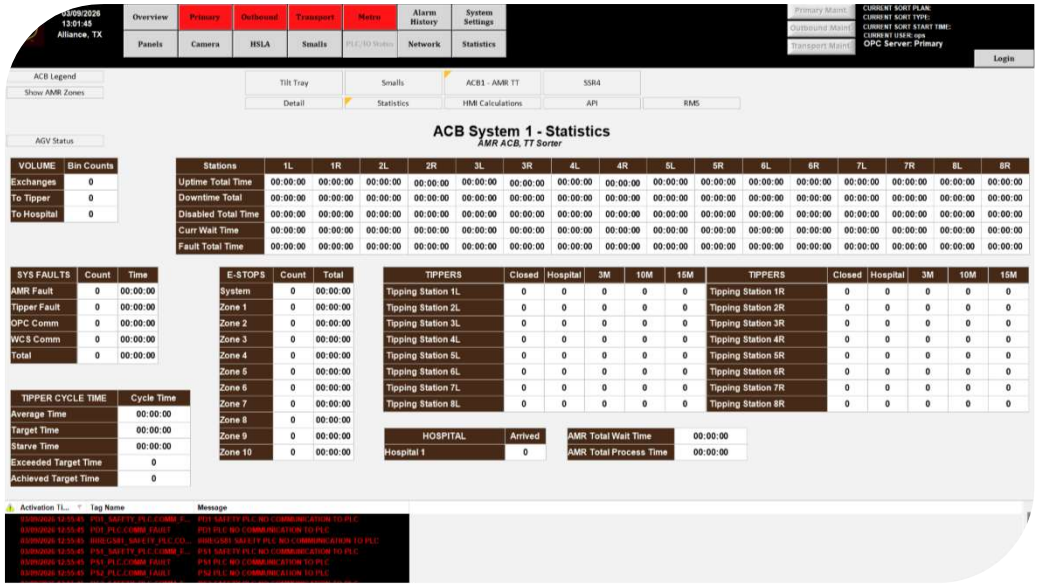


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Overall System

Statistics

- **Main Statistics:** Core system statistics
- **Types of Statistics:**
  - Tippers
  - Estop
  - Volume (Total Exchanges, To Tipper, To Hospital)

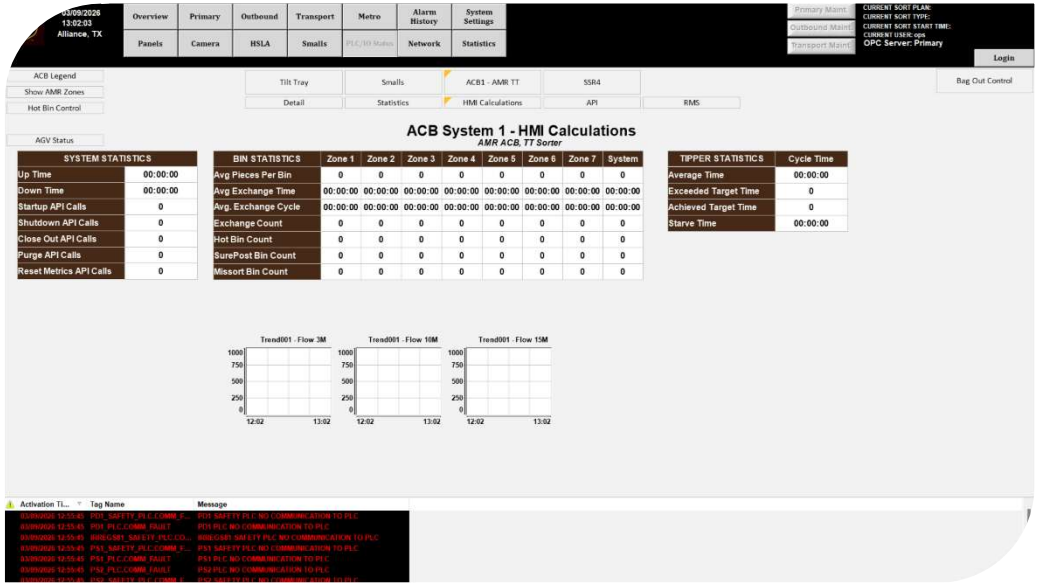


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Overall System

HMI Calculations

- **Customer-Defined Calculations:** Metrics calculated by the customer
- **Types of Calculations:**
  - System statistics
  - Bin statistics
  - Tipper statistics

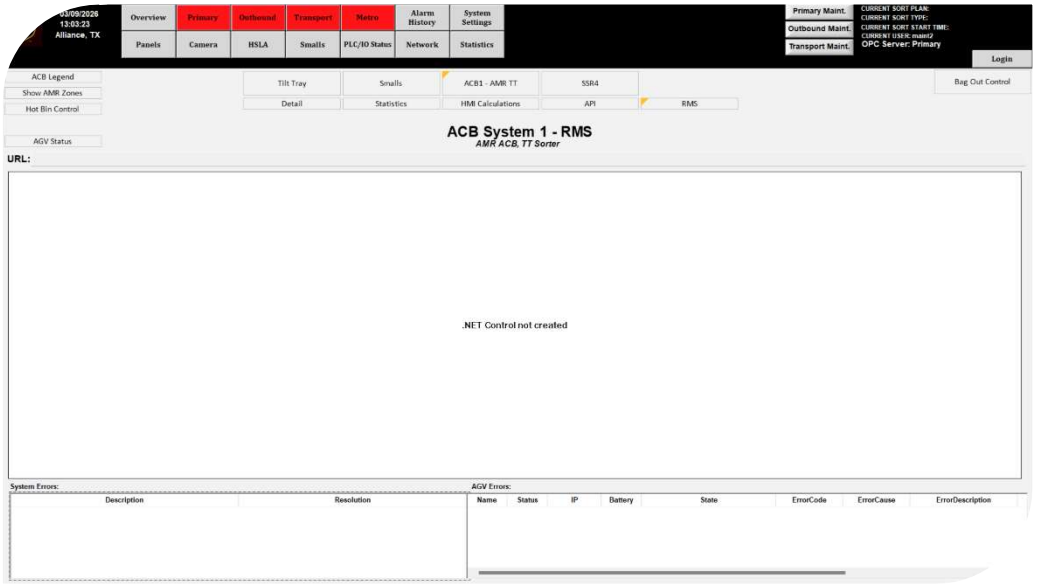


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Overall System

RMS

- This page will show the Geek+ RMS web page when appropriate firewall rules are open.
- I will also show system faults on the left and any active AGV faults on the right.

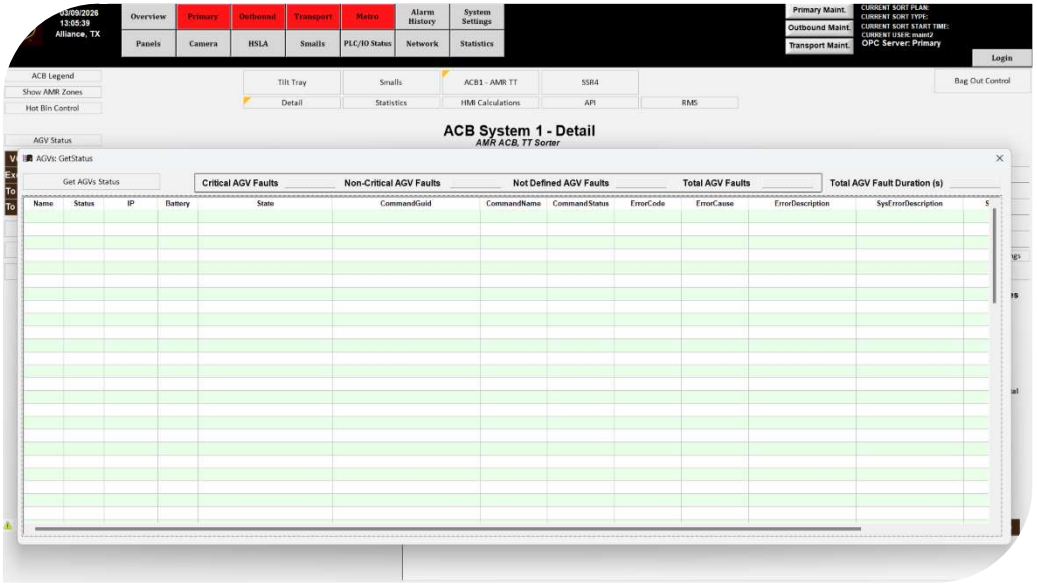


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Overall System

AGV Status

- This page will show the current status of the AGVs

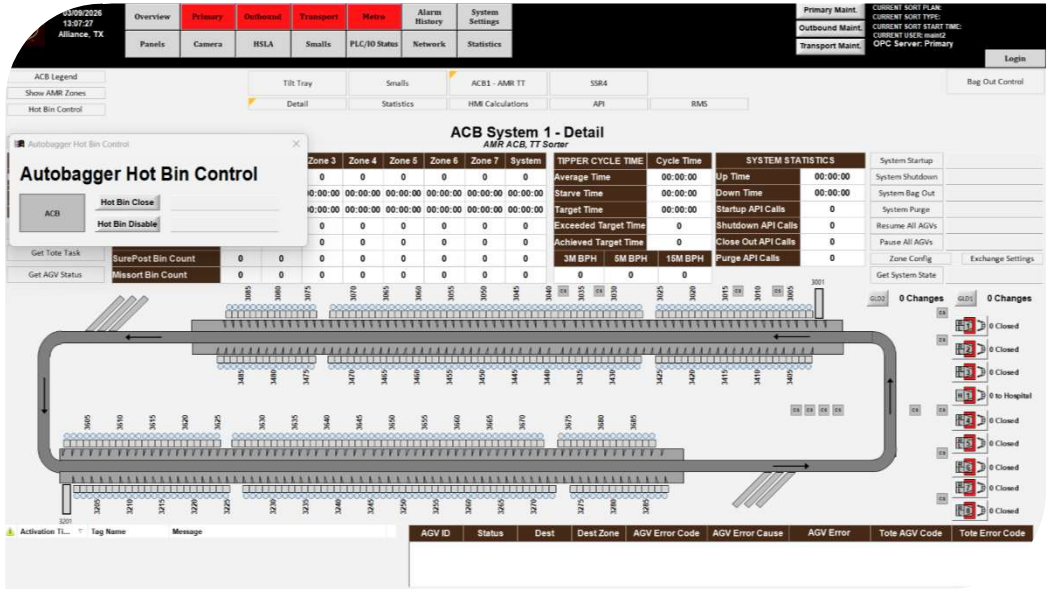


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Overall System

Hot bin

- Hot bins will trigger all totes that are marked as hot bin to close chute and exchange.
- Hot bin disable will not reopen chutes after the exchange is complete.

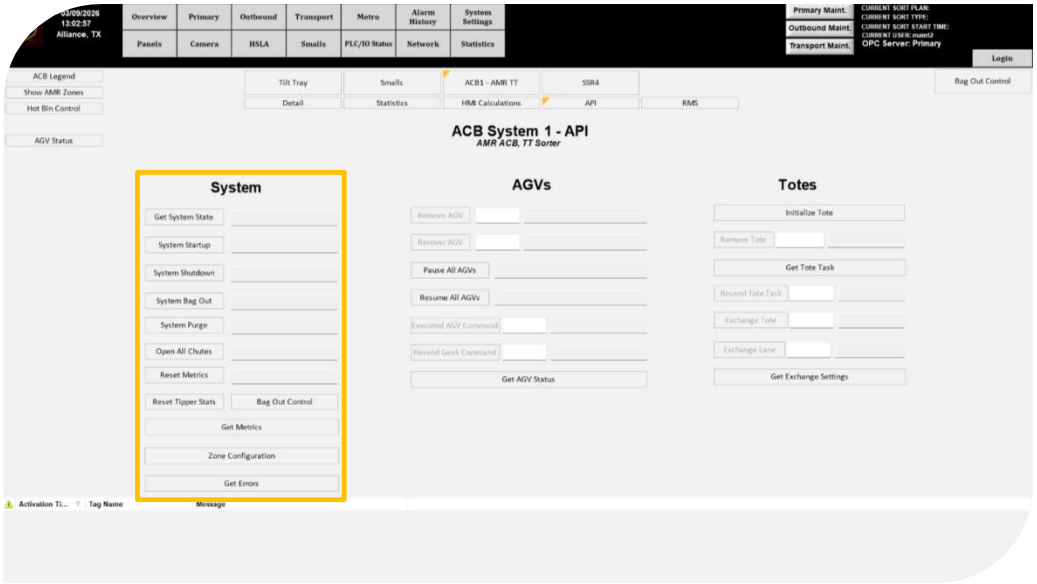


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Overall System

Aveva API page

- System State
  - Shows the current state the system is in
- System Startup
  - AGVs are kept in a shut down state between sorts to allow them to charge. System startup triggers OptiSweep to stage the AGVs for exchanges.
- System Bag Out
  - At the end of a sort the close out is used to close all chutes and exchange all totes with items.
- System Shutdown
  - When the sort is done the system should be put in a shutdown state for AGVs to rotate on the charger.
- System Purge
  - Exchanges all totes and does not print a label at the tipper.

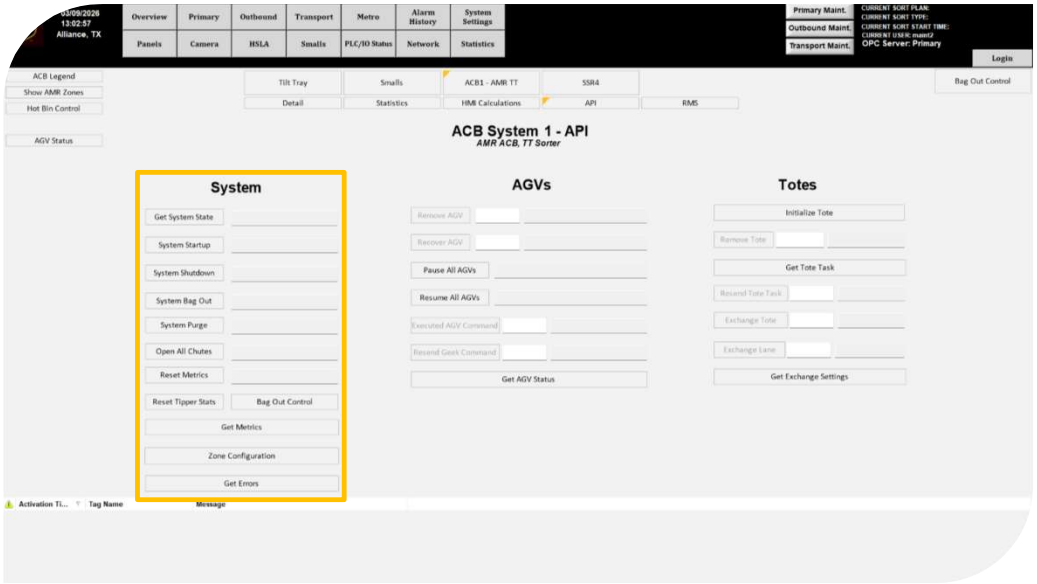


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Overall System

Aveva API page

- Reset Metrics
  - Metrics are reset when a new sort plan is loaded but can also be manually reset.
- Reset Tipper Stats
  - Metrics are reset when a new sort plan is loaded but can also be manually reset.
- Bag Out control
  - Pops up additional bag out controls
- Get metrics
  - Pops up the metrics page
- Zone configuration
  - Pops up the zone configuration page
- Get errors
  - Pops up the current errors

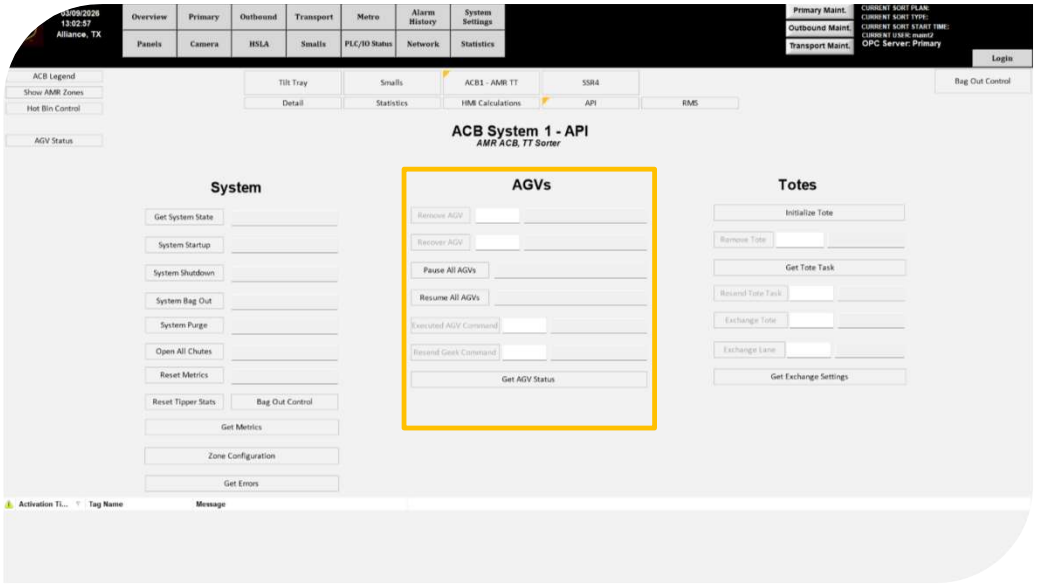


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AGVs

Aveva API page

- For ease of use and continuity, we incorporated select features from Geek+ RMS into our APIs.
  - Remove AGV
  - Recover AGV
  - Pause all AGVs
  - Resume all AGVs
- In the event of an RMS crash, we have also implemented recovery mechanisms.
  - Execute AGV command
    - Enables users to specify an AGV ID that has completed its task when RMS fails to send a callback.
  - Resend Geek Command
    - Allows users to specify a command GUID that was sent to Geek but encountered an error.

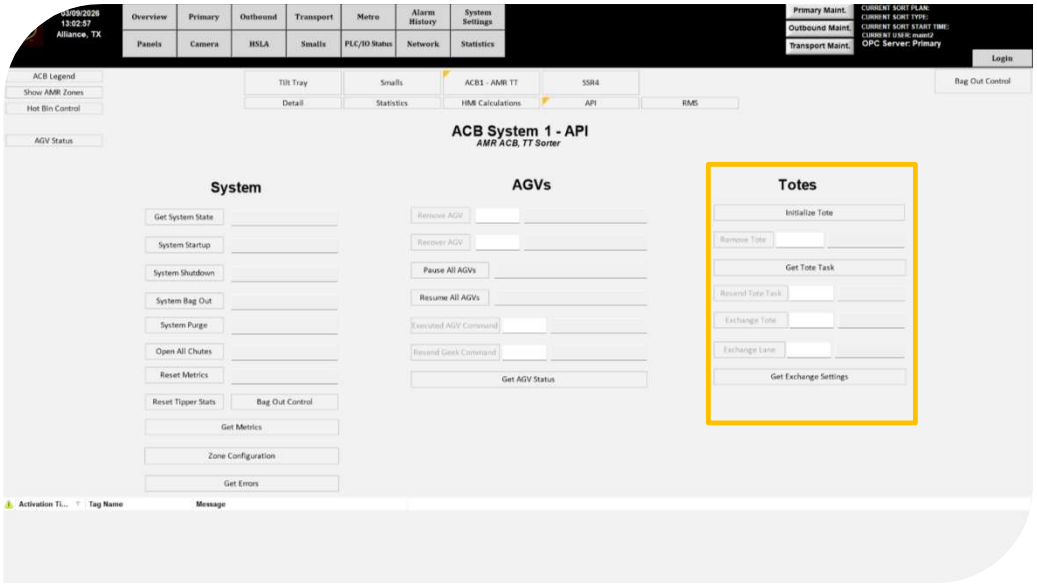


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Totes

Aveva API page

- API Features for Tote Management
  - Initialize Tote
    - Adds a specified tote to the system.
  - Remove Tote
    - Removes a specified tote from the system.
  - Get Tote Task
    - Retrieves the current tote task.
  - Resend Tote Task
    - Resends a command to a tote that encountered an error or failed to execute.
  - Exchange Tote
    - Manually trigger a tote exchange by tote Id.
  - Exchange Lane
    - Manually trigger a tote exchange by lane number.
  - Get Exchange Settings
    - Retrieves settings and configurations for tote exchanges.

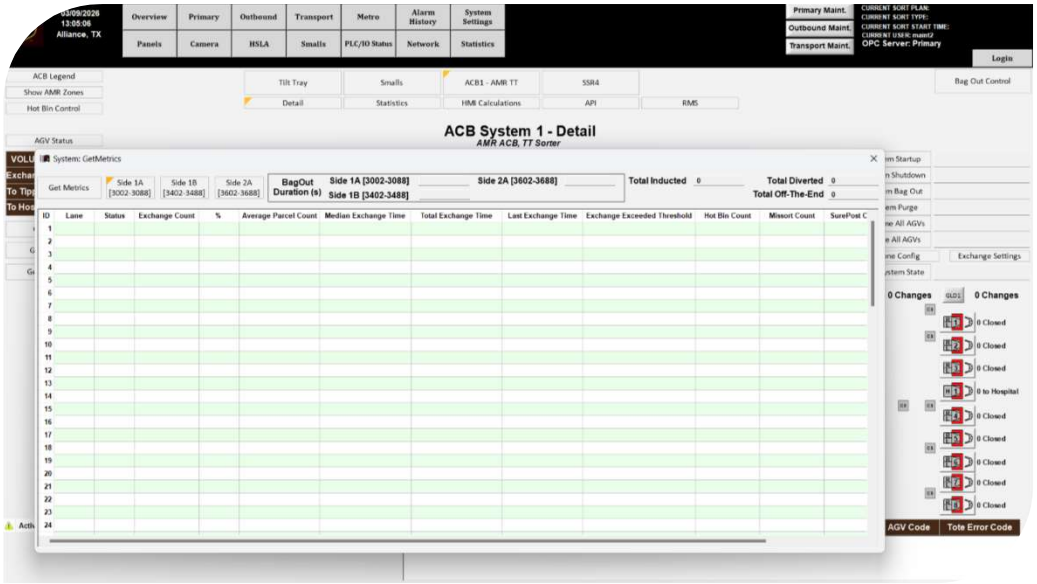


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System Metrics

Aveva API page

- Real-time tracking of KPIs for item exchanges in an automated sorting system.
  - Average Item Count Per Exchange
  - Median System Generated Exchange(s)
  - Lane Name
  - Last System Generated Exchange
  - Number of Exchanges
  - Number of HotBins
  - Number of MissSorts
  - Number of SurePost
  - Percent of Exchanges
  - Status
  - Zone

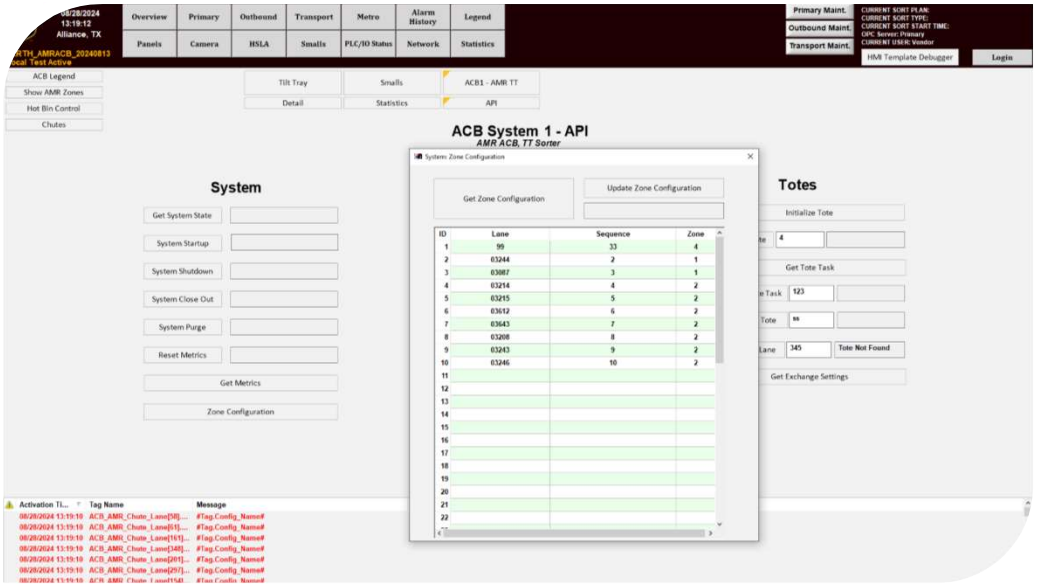


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Zone Config

Aveva API page

- This interface provides an overview of lane assignments, their respective zones, and the processing sequence numbers.
  - Get Zone Configuration
    - Fetches the current configuration of all lanes, offering visibility into how each lane is set up within the system.
  - Update Zone Configuration
    - Allows users to modify zone settings, ensuring flexibility to adapt operations based on system needs or changing requirements.

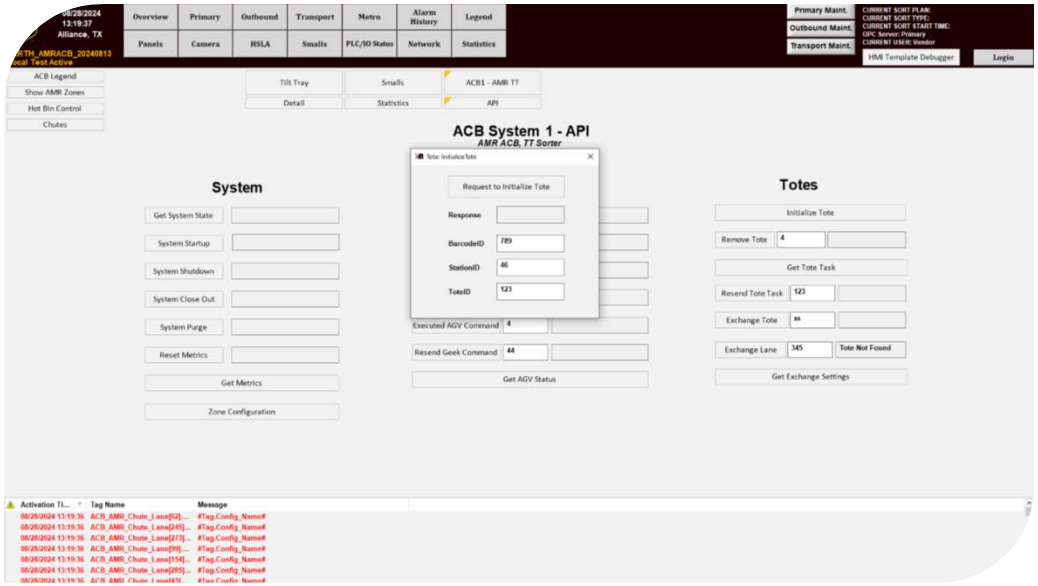


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Initialize Tote

Aveva API page

- This interface allows users to initialize a tote for processing within the system by specifying key information (BarcodeID, StationID, ToteID). The form fields and button interact as follows:
  - Request to Initialize Tote
    - This button sends a command to the system to initialize the tote with the provided details. Upon clicking, the system processes the request and displays the response in the "Response" field.
  - Response
    - Displays the result of the initialization request. If successful, this field will show a confirmation message. If the initialization fails, it will display an error message.

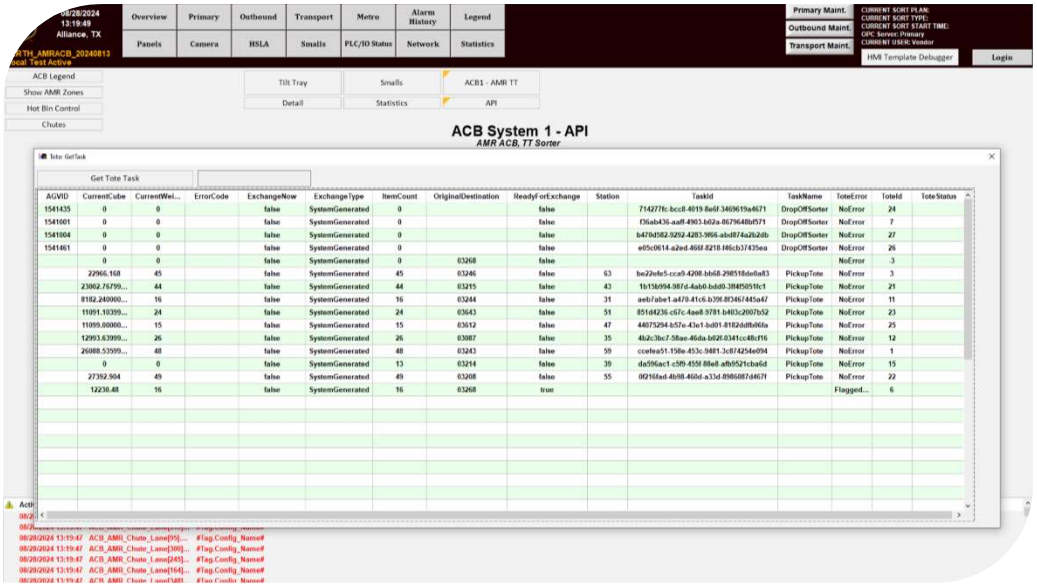


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Tote Tasks

Aveva API page

- The Tote Task interface offers insight into tote movements within the warehouse. It provides information such as:
  - Tote Id
  - AGV Id
  - Current Cube
  - Current Weight
  - Item Count
  - Original Destination
  - Station
  - Task id
  - Task Name
  - Tote Error

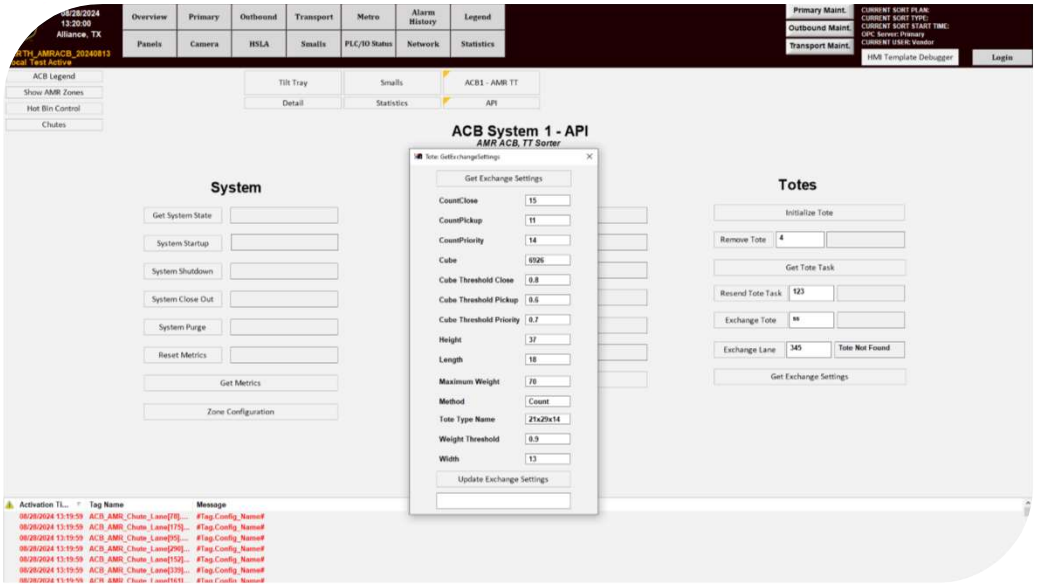


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Tote Exchange Settings

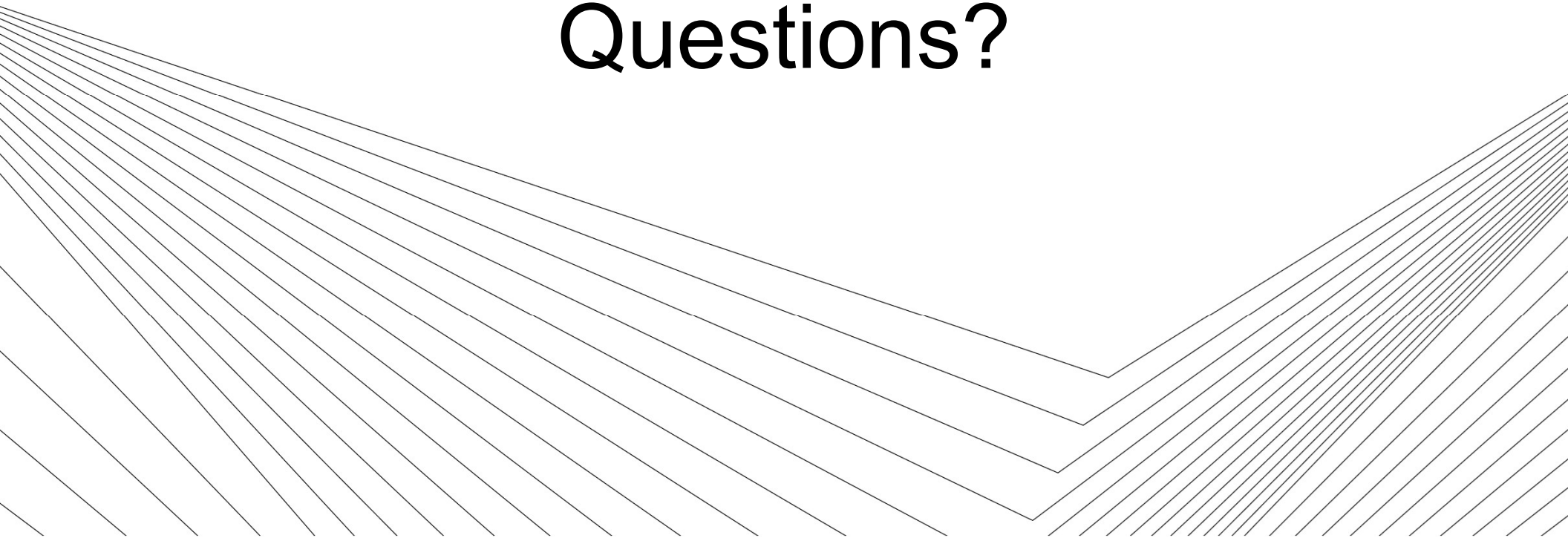
Aveva API page

- The Tote Exchange Settings interface enables users to view and manage parameters related to the tote exchange process. Some key components include:
  - Cube Threshold Pickup
  - Cube Threshold Priority
  - Cube Threshold Close
  - Count Threshold Pickup
  - Count Threshold Priority
  - Count Threshold Close
  - Maximum Weight
  - Weight Threshold
  - Method
    - Count, Cube, Cube and Count



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Questions?

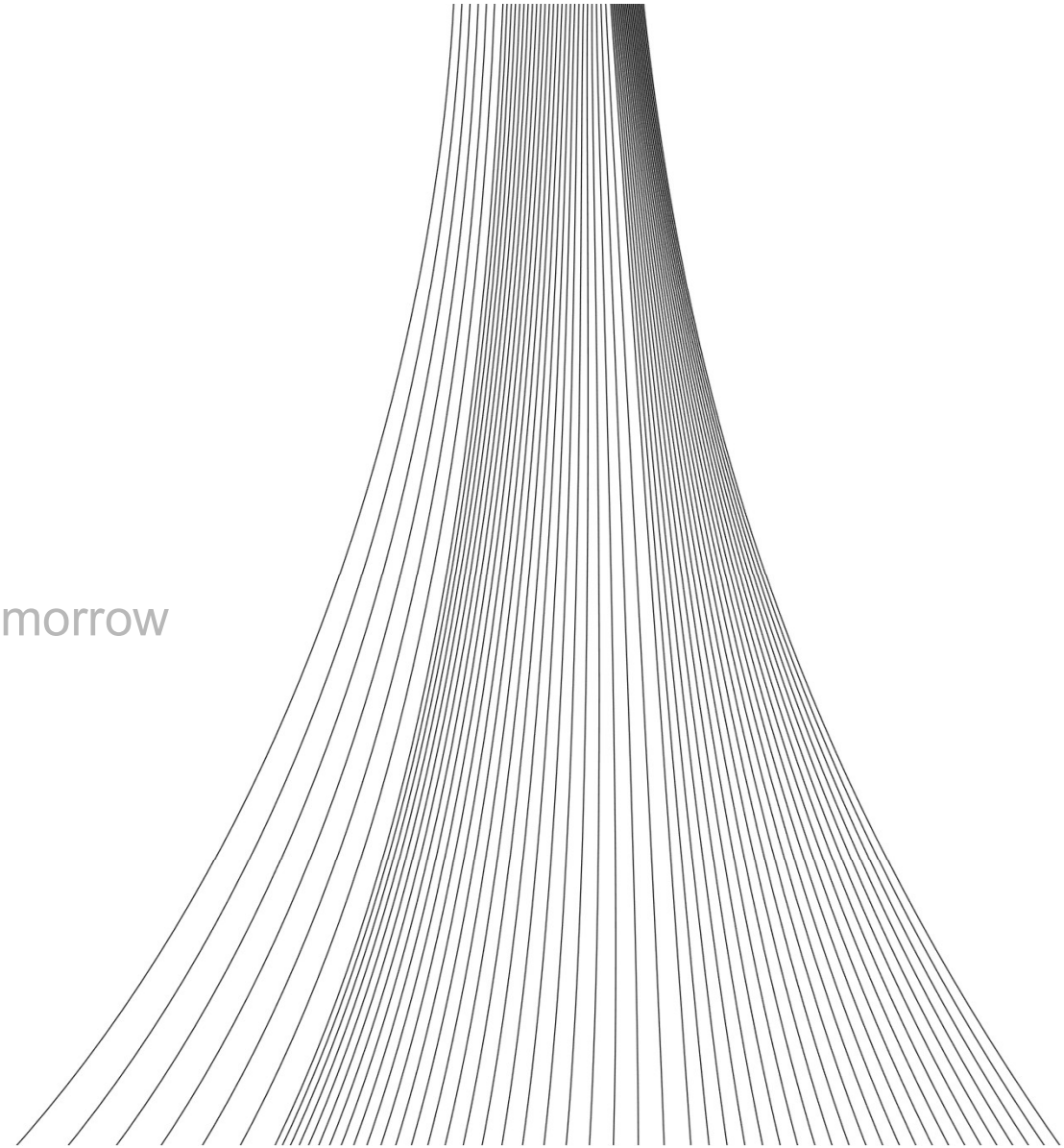




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OptiSweep – Training

Redefining consolidation for tomorrow



Contents

4. Message Flow
5. Database
6. API
7. Ignition
8. Service

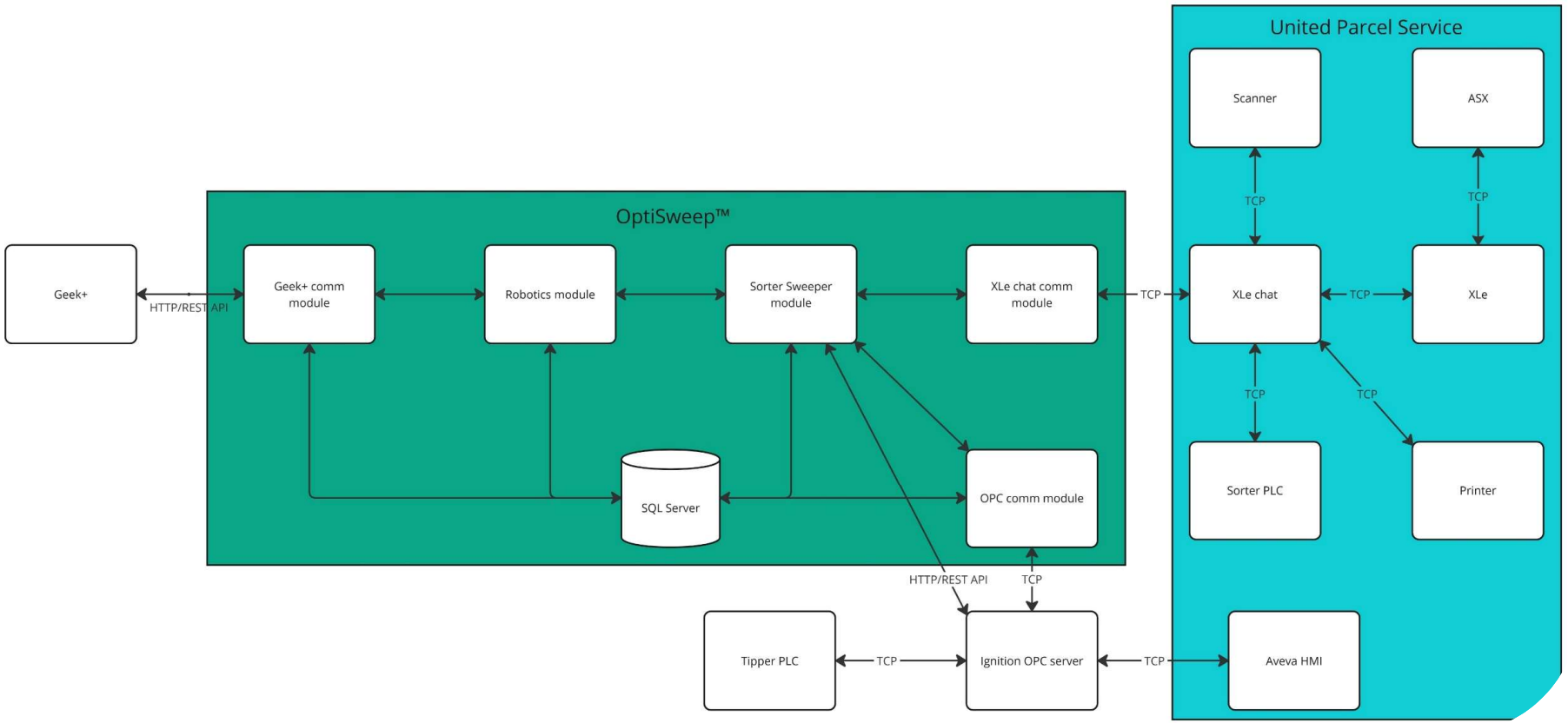
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Message Flow Walkthrough

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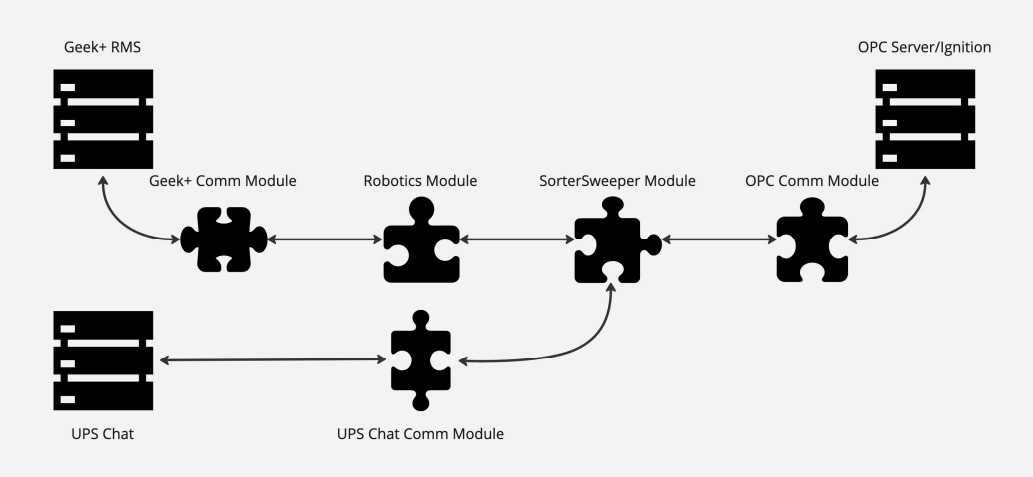
Architecture Design



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Module Breakdown

- **SorterSweeper Module**
  - Orchestrates movements of Totes, tote tippers, and sorter
- **Robotics Module**
  - Manages AGV tasks based on SorterSweeper directives
- **OPC Comm Module**
  - Manages communication between the SorterSweeper module and the OPC server.
- **Geek+ Comm Module**
  - Manages communication between the Robotics module and the Geek+ RMS server.
- **XLe Comm Module**
  - Manages communication between the SorterSweeper module and the UPS chat server.

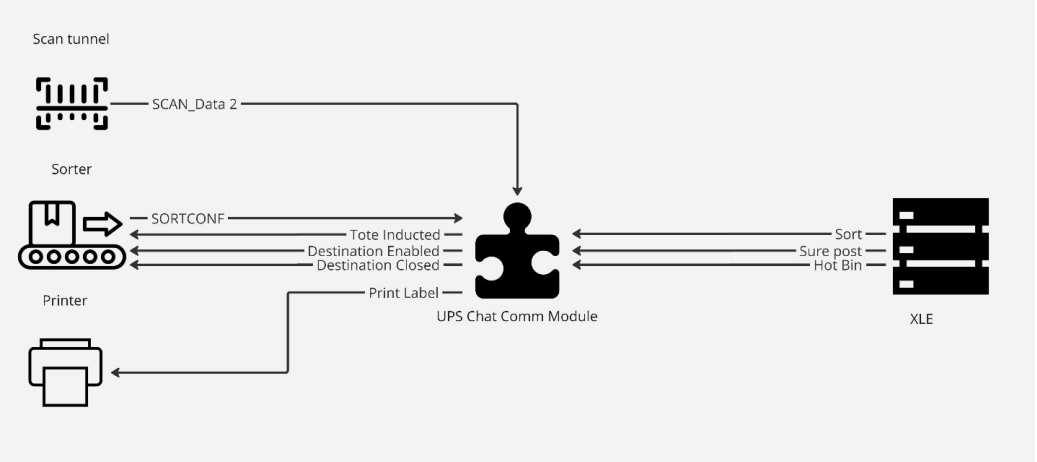


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UPS Chat Module

Detailed Design

- Tote arrives to the sorter
  - We send a tote inducted and a destination enabled message to the chat
- UPS message we listen to
  - We listen to message on the chat for parcels
    - Scan data 2 is sent from the camera tunnel onto the chat
      - We collect the dimensions of the parcel and the load category
    - Sort is sent from the xle (UPS version of WCS) onto the chat
      - We collect the possible destinations for the parcel
    - Sort confirmation is sent from the sorter onto the chat
      - We collect the final destination of the parcel
  - UPS can trigger a hot bin close out by sending us a hot bin message for all totes that are a hot bin and we do an exchange immediately.
  - UPS notifies us if it the load for a destination is a sure post. We notify the tipper before the tip so they can switch out the bag to a poly bag
- Tote ready to leave sorter
  - We send destination closed message onto the chat before doing an exchange
- Tote tipping at tipper
  - We send a print label message to the printer so it can print a label provide by the xle

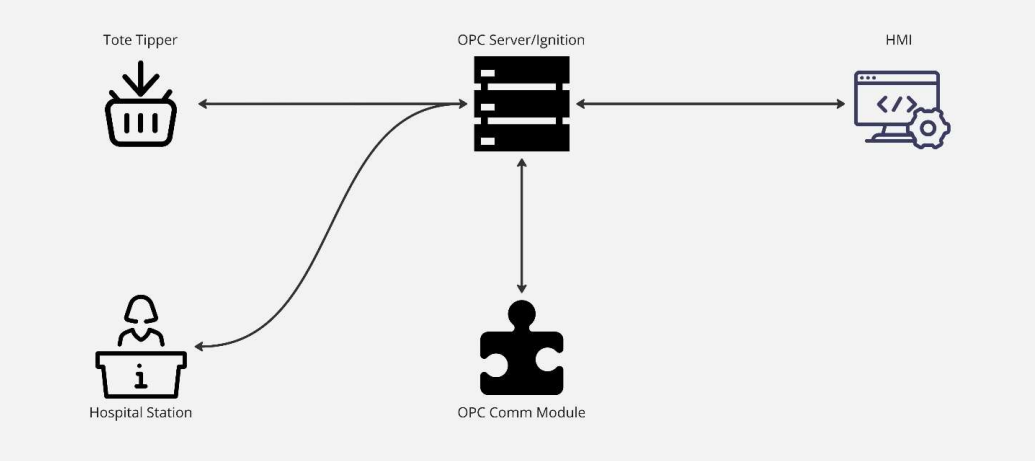


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OPC Comm Module

Detailed Design

- Ignition's built-in tag provider is used to connect to the hospital and tippers, relaying the relevant tags.
- The OPC communication module connects to the OPC server and subscribes to tags.
- When a tag is updated, it sends the update to the SorterSweeper module for processing.
- SorterSweeper also sends messages to the OPC communication module, which writes the data to the OPC server.

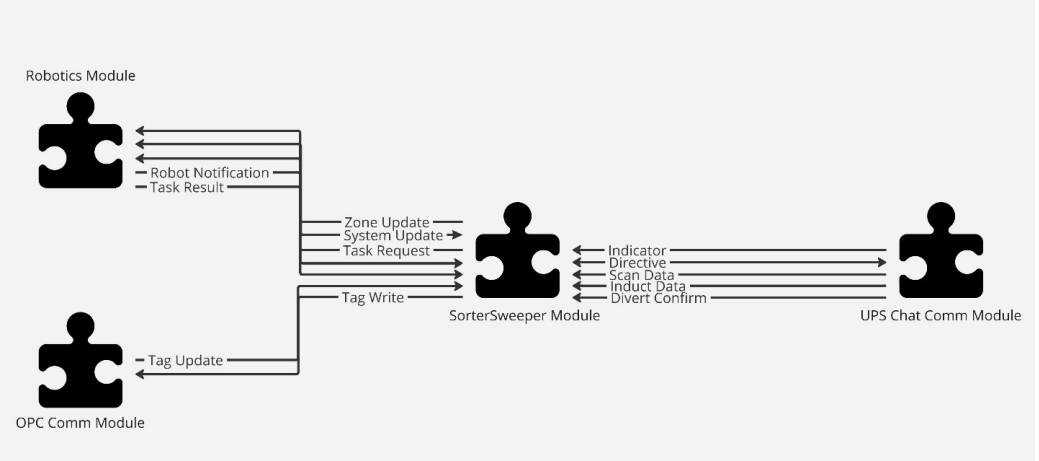


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Sorter Sweeper Module

Detailed Design

- The Sorter Sweeper module functions as the state machine for totes. It receives data from the sorter, tippers, hospital station, and HMI, creating tasks for totes with the robotics modules to execute. It also notifies tippers when a tote arrives or when there is a robot fault, and processes data from the hospital.
- Tote Task:
  - Induct Tote
  - Remove Tote
  - Pickup Tote
  - Move Out Sorter
  - Drop Off Sorter
  - Drop Off Hospital

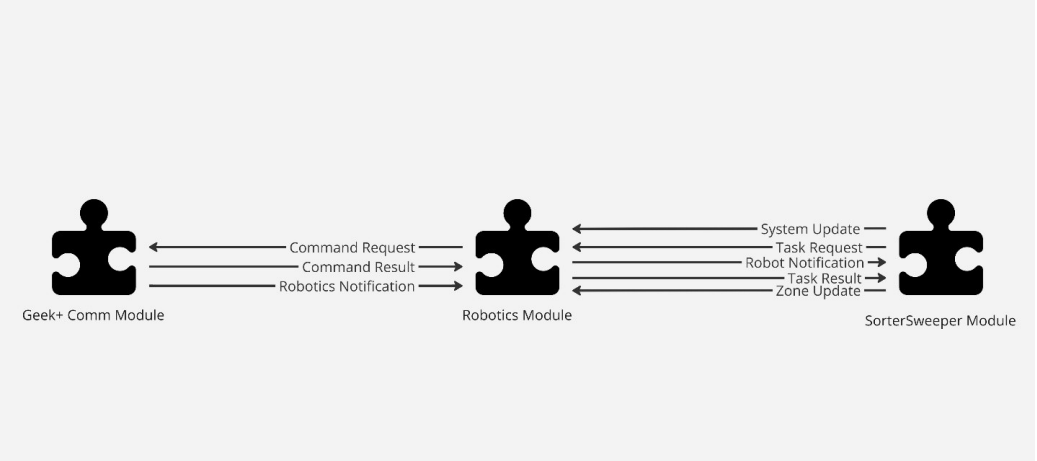


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Robotics Module

Detailed Design

- The Robotics module acts as the state machine for AGVs. It receives tote tasks from the Sorter Sweeper module and generates commands for Geek+ to execute those tasks.
- AGV commands:
  - Induct tote
  - Remove tote
  - Initialize AGV
  - Empty go to
  - Pickup tote
  - Move tote
  - Drop off tote
  - Cancel Request
  - Go Charge



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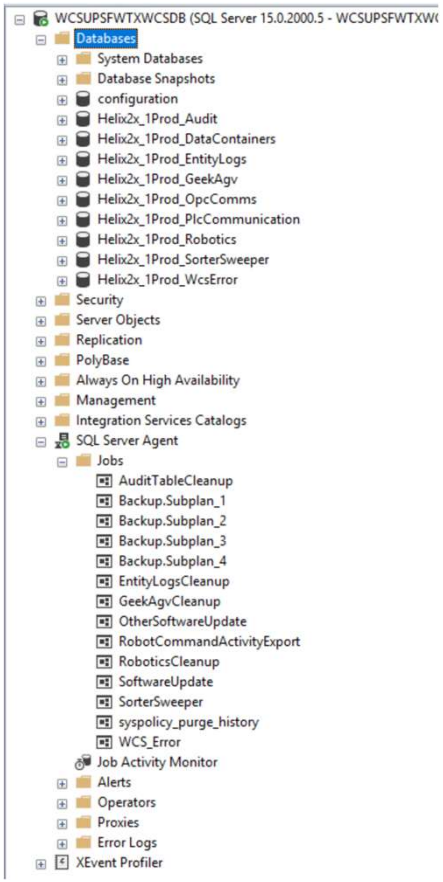
Database Walkthrough

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Overview

- WCS Databases
  - DataContainers
  - EntityLogs
  - PLCCommunication
- OptiSweep Databases
  - Audit (Geek Comm Module)
  - GeekAgv (Geek Comm Module)
  - OPCComms
  - Robotics
  - SorterSweeper
- SQL Jobs



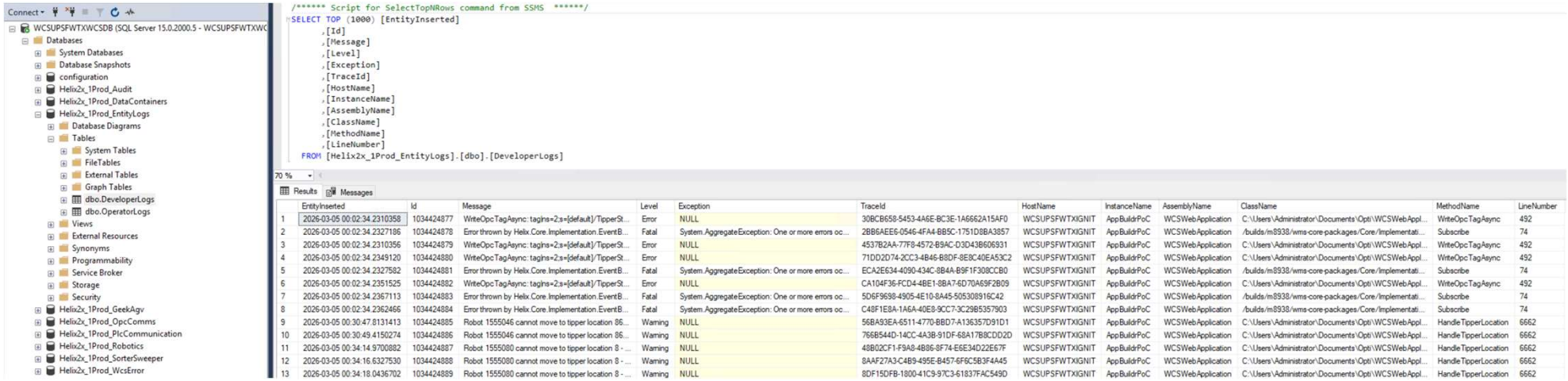
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OptiSweep Databases

EntityLogs

At install Fortna's WCS creates several databases as default. OPTISweep only utilizes entity logs for logging.

- DeveloperLogs



	EntityInserted	Id	Message	Level	Exception	TraceId	HostName	InstanceName	AssemblyName	ClassName	MethodName	LineNumber
1	2026-03-05 00:02:34.2310358	1034424877	WriteOpcTagAsync: tagIns=2;s=[default]/TipperSt...	Error	NULL	30BCB658-5453-4A6E-BC3E-1A6662A15AF0	WCSUPSFWTXIGNIT	AppBuildrPoC	WCSWebApplication	C:\Users\Administrator\Documents\Opti\WCSWebAppl...	WriteOpcTagAsync	492
2	2026-03-05 00:02:34.2327186	1034424878	Error thrown by Helix.Core.Implementation.EventB...	Fatal	System.AggregateException: One or more errors oc...	2BB6AEE6-0546-4FA4-BB5C-1751D8BA3857	WCSUPSFWTXIGNIT	AppBuildrPoC	WCSWebApplication	/builds/m8938/wms-core-packages/Core/Implementati...	Subscribe	74
3	2026-03-05 00:02:34.2310356	1034424879	WriteOpcTagAsync: tagIns=2;s=[default]/TipperSt...	Error	NULL	4537B2AA-77F8-4572-B9AC-D3D43B606931	WCSUPSFWTXIGNIT	AppBuildrPoC	WCSWebApplication	C:\Users\Administrator\Documents\Opti\WCSWebAppl...	WriteOpcTagAsync	492
4	2026-03-05 00:02:34.2349120	1034424880	WriteOpcTagAsync: tagIns=2;s=[default]/TipperSt...	Error	NULL	71DD2D74-2CC3-4B46-B8DF-8E8C40EA53C2	WCSUPSFWTXIGNIT	AppBuildrPoC	WCSWebApplication	C:\Users\Administrator\Documents\Opti\WCSWebAppl...	WriteOpcTagAsync	492
5	2026-03-05 00:02:34.2327582	1034424881	Error thrown by Helix.Core.Implementation.EventB...	Fatal	System.AggregateException: One or more errors oc...	ECA2E634-4090-434C-8B4A-B9F1F308CCB0	WCSUPSFWTXIGNIT	AppBuildrPoC	WCSWebApplication	/builds/m8938/wms-core-packages/Core/Implementati...	Subscribe	74
6	2026-03-05 00:02:34.2351525	1034424882	WriteOpcTagAsync: tagIns=2;s=[default]/TipperSt...	Error	NULL	CA104F36-FCD4-4BE1-8BA7-6D70A69F2B09	WCSUPSFWTXIGNIT	AppBuildrPoC	WCSWebApplication	C:\Users\Administrator\Documents\Opti\WCSWebAppl...	WriteOpcTagAsync	492
7	2026-03-05 00:02:34.2367113	1034424883	Error thrown by Helix.Core.Implementation.EventB...	Fatal	System.AggregateException: One or more errors oc...	5D6F9698-4905-4E10-8A45-505308916C42	WCSUPSFWTXIGNIT	AppBuildrPoC	WCSWebApplication	/builds/m8938/wms-core-packages/Core/Implementati...	Subscribe	74
8	2026-03-05 00:02:34.2362466	1034424884	Error thrown by Helix.Core.Implementation.EventB...	Fatal	System.AggregateException: One or more errors oc...	C48F1E8A-1A6A-40E8-9CC7-3C29B5357903	WCSUPSFWTXIGNIT	AppBuildrPoC	WCSWebApplication	/builds/m8938/wms-core-packages/Core/Implementati...	Subscribe	74
9	2026-03-05 00:30:47.8131413	1034424885	Robot 1555046 cannot move to tipper location 86...	Warning	NULL	56BA93EA-6511-4770-BBD7-A136357D91D1	WCSUPSFWTXIGNIT	AppBuildrPoC	WCSWebApplication	C:\Users\Administrator\Documents\Opti\WCSWebAppl...	HandleTipperLocation	6662
10	2026-03-05 00:30:49.4150274	1034424886	Robot 1555046 cannot move to tipper location 86...	Warning	NULL	766B544D-14CC-4A3B-91DF-68A17B8CDD2D	WCSUPSFWTXIGNIT	AppBuildrPoC	WCSWebApplication	C:\Users\Administrator\Documents\Opti\WCSWebAppl...	HandleTipperLocation	6662
11	2026-03-05 00:34:14.9700882	1034424887	Robot 1555080 cannot move to tipper location 8 - ...	Warning	NULL	48B02CF1-F9A8-4B86-8F74-E6E34D22E67F	WCSUPSFWTXIGNIT	AppBuildrPoC	WCSWebApplication	C:\Users\Administrator\Documents\Opti\WCSWebAppl...	HandleTipperLocation	6662
12	2026-03-05 00:34:16.6327530	1034424888	Robot 1555080 cannot move to tipper location 8 - ...	Warning	NULL	8AAF27A3-C4B9-495E-B457-6F6C5B3F4A45	WCSUPSFWTXIGNIT	AppBuildrPoC	WCSWebApplication	C:\Users\Administrator\Documents\Opti\WCSWebAppl...	HandleTipperLocation	6662
13	2026-03-05 00:34:18.0436702	1034424889	Robot 1555080 cannot move to tipper location 8 - ...	Warning	NULL	8DF15DFB-1800-41C9-97C3-61837FAC549D	WCSUPSFWTXIGNIT	AppBuildrPoC	WCSWebApplication	C:\Users\Administrator\Documents\Opti\WCSWebAppl...	HandleTipperLocation	6662

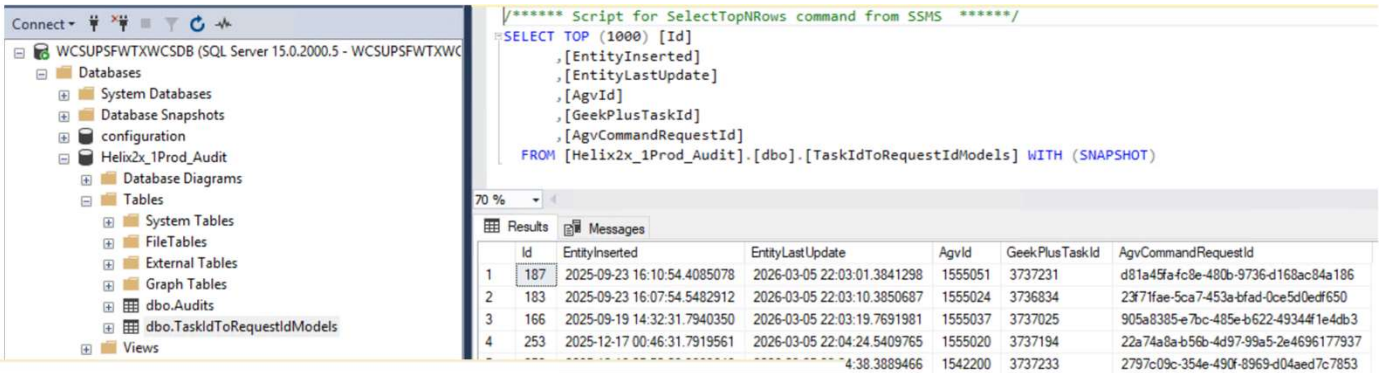
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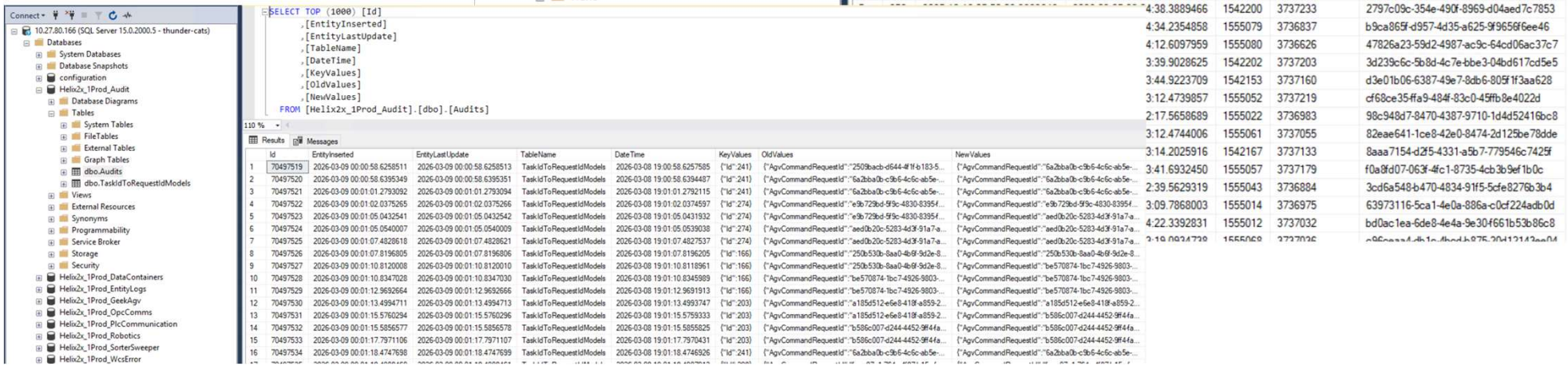
Audit Database

Task Id To Request Id

Tracks the AgvId, the current Geek Task ID, and the command guid sent by robotics engine.

Changes to this table are tracked in the Audits table.



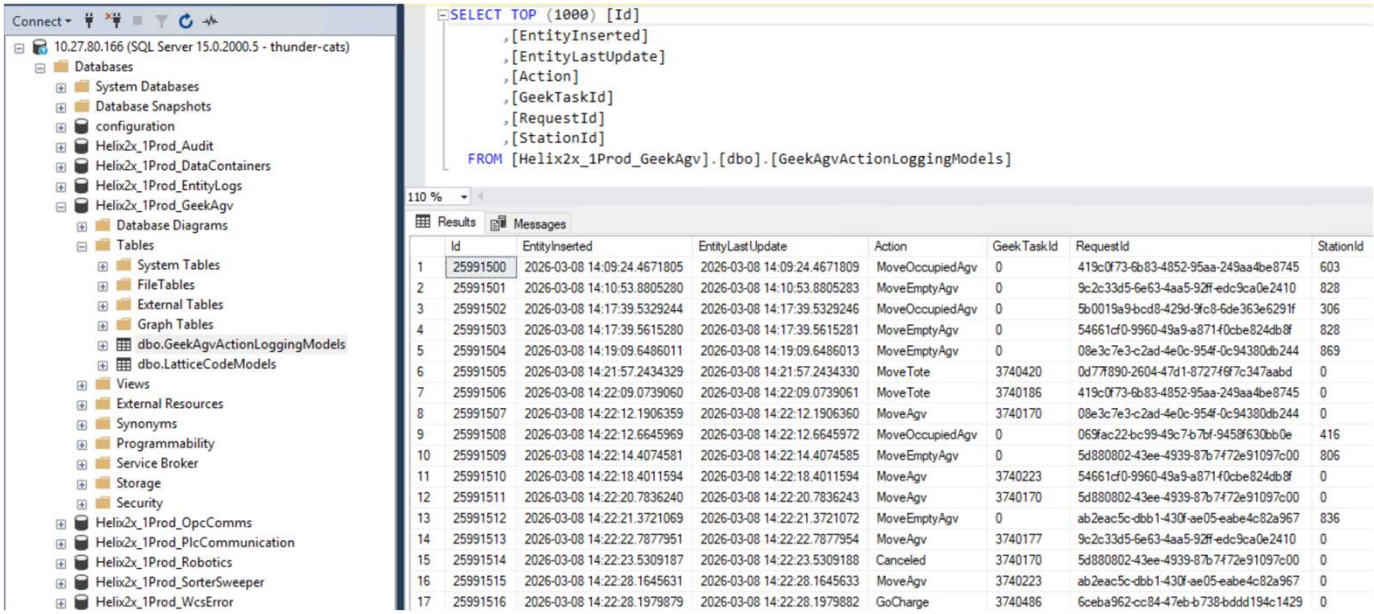


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Geek AGV Database

Geek AGV Action Logging Models

The GeekAgvActionLoggingModel logs all task to Geek+.



	Id	EntityInserted	EntityLastUpdate	Action	GeekTaskId	RequestId	StationId
1	25991500	2026-03-08 14:09:24.4671805	2026-03-08 14:09:24.4671809	MoveOccupiedAgv	0	419c0f73-6b83-4852-95aa-249aa4be8745	603
2	25991501	2026-03-08 14:10:53.8805280	2026-03-08 14:10:53.8805283	MoveEmptyAgv	0	9c2c33d5-6e63-4aa5-92ff-edc9ca0e2410	828
3	25991502	2026-03-08 14:17:39.5329244	2026-03-08 14:17:39.5329246	MoveOccupiedAgv	0	5b0019a9-bcd8-429d-9fc8-6de363e6291f	306
4	25991503	2026-03-08 14:17:39.5615280	2026-03-08 14:17:39.5615281	MoveEmptyAgv	0	54661cf0-9960-49a9-a871-f0cbe824db8f	828
5	25991504	2026-03-08 14:19:09.6486011	2026-03-08 14:19:09.6486013	MoveEmptyAgv	0	08e3c7e3-c2ad-4e0c-954f-0c94380db244	869
6	25991505	2026-03-08 14:21:57.2434329	2026-03-08 14:21:57.2434330	MoveTote	3740420	0d77f890-2604-47d1-8727-f6f7c347aabd	0
7	25991506	2026-03-08 14:22:09.0739060	2026-03-08 14:22:09.0739061	MoveTote	3740186	419c0f73-6b83-4852-95aa-249aa4be8745	0
8	25991507	2026-03-08 14:22:12.1906359	2026-03-08 14:22:12.1906360	MoveAgv	3740170	08e3c7e3-c2ad-4e0c-954f-0c94380db244	0
9	25991508	2026-03-08 14:22:12.6645969	2026-03-08 14:22:12.6645972	MoveOccupiedAgv	0	069fac22-bc99-49c7-b7bf-9458f630bb0e	416
10	25991509	2026-03-08 14:22:14.4074581	2026-03-08 14:22:14.4074585	MoveEmptyAgv	0	5d880802-43ee-4939-87b7-f72e91097c00	806
11	25991510	2026-03-08 14:22:18.4011594	2026-03-08 14:22:18.4011594	MoveAgv	3740223	54661cf0-9960-49a9-a871-f0cbe824db8f	0
12	25991511	2026-03-08 14:22:20.7836240	2026-03-08 14:22:20.7836243	MoveAgv	3740170	5d880802-43ee-4939-87b7-f72e91097c00	0
13	25991512	2026-03-08 14:22:21.3721069	2026-03-08 14:22:21.3721072	MoveEmptyAgv	0	ab2eac5c-dbb1-430f-ae05-eabe4c82a967	836
14	25991513	2026-03-08 14:22:22.7877951	2026-03-08 14:22:22.7877954	MoveAgv	3740177	9c2c33d5-6e63-4aa5-92ff-edc9ca0e2410	0
15	25991514	2026-03-08 14:22:23.5309187	2026-03-08 14:22:23.5309188	Canceled	3740170	5d880802-43ee-4939-87b7-f72e91097c00	0
16	25991515	2026-03-08 14:22:28.1645631	2026-03-08 14:22:28.1645633	MoveAgv	3740223	ab2eac5c-dbb1-430f-ae05-eabe4c82a967	0
17	25991516	2026-03-08 14:22:28.1979879	2026-03-08 14:22:28.1979882	GoCharge	3740486	6ceba962-cc84-47eb-b738-bddd194c1429	0

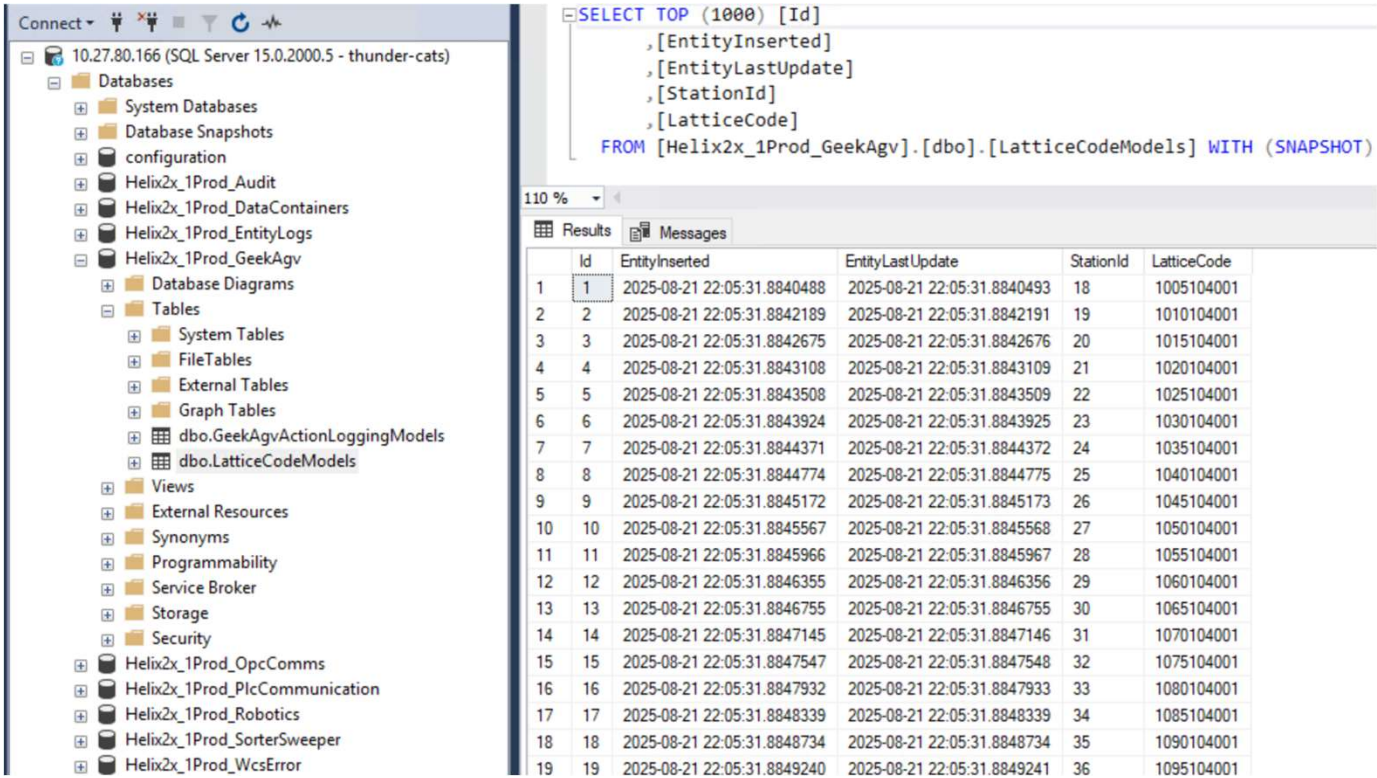
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Geek AGV Database

Lattice Code Models

The LatticeCodeModel stores the lattice codes and their respective StationId.

When a tote is inducted or dropped off. Geek requires a lattice code of the location. Geek Comm module uses this table to covert the StationId from robotics module request to the lattice code Geek requires.



	Id	EntityInserted	EntityLastUpdate	StationId	LatticeCode
1	1	2025-08-21 22:05:31.8840488	2025-08-21 22:05:31.8840493	18	1005104001
2	2	2025-08-21 22:05:31.8842189	2025-08-21 22:05:31.8842191	19	1010104001
3	3	2025-08-21 22:05:31.8842675	2025-08-21 22:05:31.8842676	20	1015104001
4	4	2025-08-21 22:05:31.8843108	2025-08-21 22:05:31.8843109	21	1020104001
5	5	2025-08-21 22:05:31.8843508	2025-08-21 22:05:31.8843509	22	1025104001
6	6	2025-08-21 22:05:31.8843924	2025-08-21 22:05:31.8843925	23	1030104001
7	7	2025-08-21 22:05:31.8844371	2025-08-21 22:05:31.8844372	24	1035104001
8	8	2025-08-21 22:05:31.8844774	2025-08-21 22:05:31.8844775	25	1040104001
9	9	2025-08-21 22:05:31.8845172	2025-08-21 22:05:31.8845173	26	1045104001
10	10	2025-08-21 22:05:31.8845567	2025-08-21 22:05:31.8845568	27	1050104001
11	11	2025-08-21 22:05:31.8845966	2025-08-21 22:05:31.8845967	28	1055104001
12	12	2025-08-21 22:05:31.8846355	2025-08-21 22:05:31.8846356	29	1060104001
13	13	2025-08-21 22:05:31.8846755	2025-08-21 22:05:31.8846755	30	1065104001
14	14	2025-08-21 22:05:31.8847145	2025-08-21 22:05:31.8847146	31	1070104001
15	15	2025-08-21 22:05:31.8847547	2025-08-21 22:05:31.8847548	32	1075104001
16	16	2025-08-21 22:05:31.8847932	2025-08-21 22:05:31.8847933	33	1080104001
17	17	2025-08-21 22:05:31.8848339	2025-08-21 22:05:31.8848339	34	1085104001
18	18	2025-08-21 22:05:31.8848734	2025-08-21 22:05:31.8848734	35	1090104001
19	19	2025-08-21 22:05:31.8849240	2025-08-21 22:05:31.8849241	36	1095104001

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OPC Comms Database

The OpcConfiguration table stores the connection parameters required for the software to bridge the gap to the OPC.

OpcTags stores the tags that on startup will be subscribed to.

```
SELECT TOP (1000) [Id]
      ,[EntityInserted]
      ,[EntityLastUpdate]
      ,[Default]
      ,[Enabled]
      ,[ConfigName]
      ,[ConfigDescription]
      ,[OpcServerUrl]
      ,[OpcXmlConfiguration]
  FROM [Helix2x_1Prod_OpcComms].[dbo].[OpcConfiguration]
```

110 %

Results | Messages

	Id	EntityInserted	EntityLastUpdate	Default	Enabled	ConfigName	ConfigDescription	OpcServerUrl	OpcXmlConfiguration
1	1	2025-08-21 22:04:45.9862224	2025-08-21 22:04:45.9862829	1	1	DefaultConfig	Default Config for Helix OPC Client Engine	opc.tcp://localhost:62541/discovery	<?xml version="1.0" encoding="utf-8"?> <Applic...

```
SELECT TOP (1000) [Id]
      ,[EntityInserted]
      ,[EntityLastUpdate]
      ,[Enabled]
      ,[TagPath]
      ,[OpcConfigurationId]
  FROM [Helix2x_1Prod_OpcComms].[dbo].[OpcTags]
```

110 %

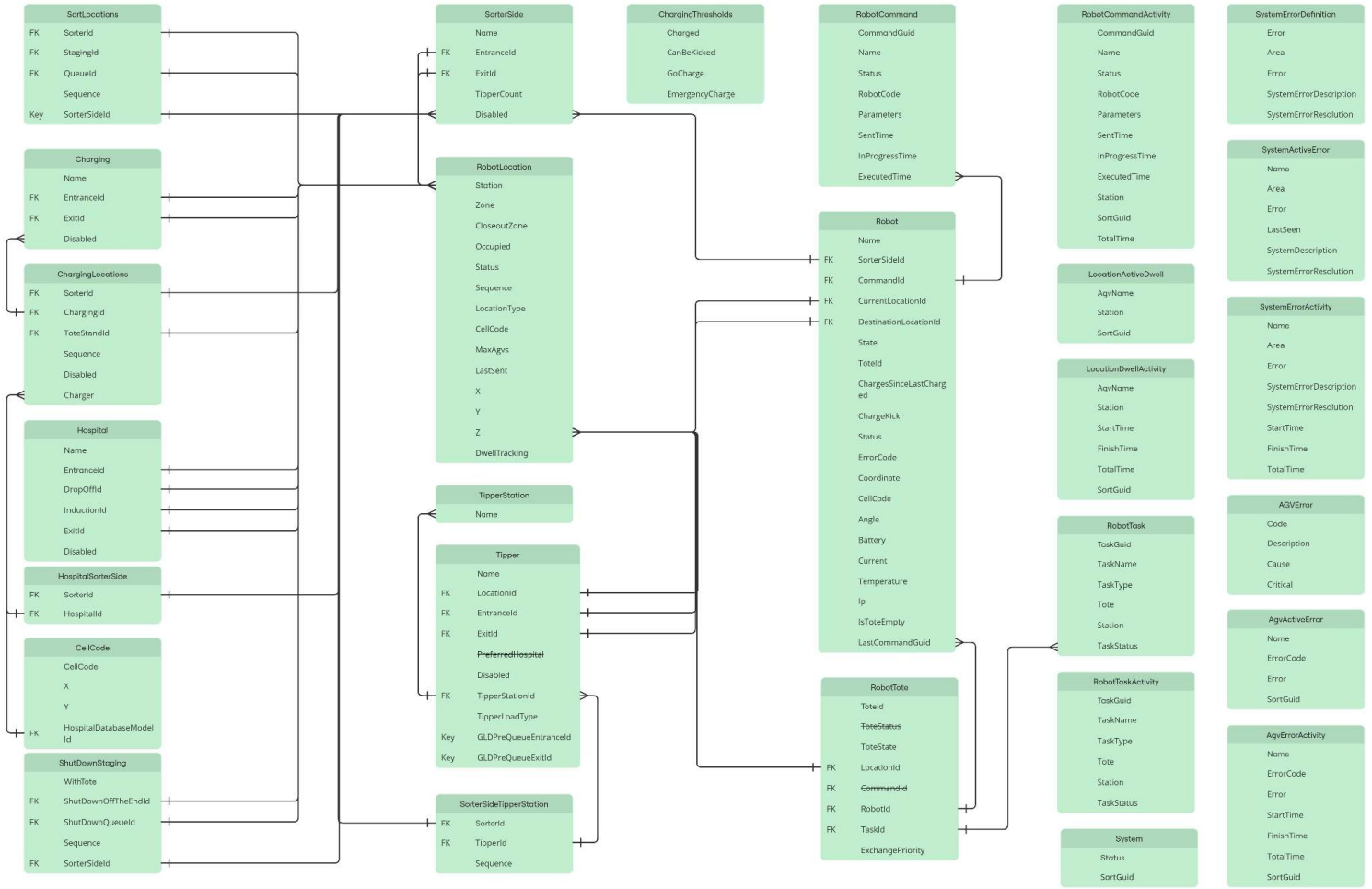
Results | Messages

	Id	EntityInserted	EntityLastUpdate	Enabled	TagPath	OpcConfigurationId
1	1	2025-08-21 22:04:46.1611165	2025-08-21 22:04:46.1611172	1	ns=2;s=[default]/TipperStation8b/In/wCommand	1
2	2	2025-08-21 22:04:46.1625841	2025-08-21 22:04:46.1625844	1	ns=2;s=[default]/TipperStation8b/In/wCommand2	1
3	3	2025-08-21 22:04:46.1626831	2025-08-21 22:04:46.1626832	1	ns=2;s=[default]/TipperStation8b/In/iRMS_Status	1
4	4	2025-08-21 22:04:46.1627094	2025-08-21 22:04:46.1627095	1	ns=2;s=[default]/TipperStation8b/In/abRFID	1
5	5	2025-08-21 22:04:46.1627381	2025-08-21 22:04:46.1627382	1	ns=2;s=[default]/TipperStation8b/In/diRFID	1
6	6	2025-08-21 22:04:46.1627581	2025-08-21 22:04:46.1627582	1	ns=2;s=[default]/TipperStation8b/In/wSortStatus	1
7	7	2025-08-21 22:04:46.1627774	2025-08-21 22:04:46.1627774	1	ns=2;s=[default]/TipperStation8b/In/iHeartBeat	1
8	8	2025-08-21 22:04:46.1627998	2025-08-21 22:04:46.1627999	1	ns=2;s=[default]/TipperStation8b/Out/iHeartBeat	1
9	9	2025-08-21 22:04:46.1628196	2025-08-21 22:04:46.1628197	1	ns=2;s=[default]/TipperStation8b/Out/wFaultCode	1
10	10	2025-08-21 22:04:46.1628387	2025-08-21 22:04:46.1628387	1	ns=2;s=[default]/TipperStation8b/Out/abRFID	1
11	11	2025-08-21 22:04:46.1628574	2025-08-21 22:04:46.1628575	1	ns=2;s=[default]/TipperStation8b/Out/wState	1
12	12	2025-08-21 22:04:46.1628793	2025-08-21 22:04:46.1628794	1	ns=2;s=[default]/TipperStation8b/Out/wCommand	1

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Robotics Database

Overview



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Robotics Database

Robot Location Database Model

The RobotLocationDatabaseModel table contains all the information about the locations including the station number, whether it is occupied or not, zone, its close out zone, the location type, the max number of agvs allowed at that location, and whether the location should be tracked for dwell time.

Station	Zone	CloseOutZone	Occupied	Status	Sequence	LocationType	CellCode	MaxAgvs	LastSent	X	Y	Z	DwellTracking
424	7	7	0	Enabled	86	Sorter	11701110	1	NULL	48.737	34.074	NULL	0
591	1	1	0	Enabled	12	SorterQueue	15801115	1	NULL	107.006	34.916	NULL	0
554	4	4	0	Enabled	49	SorterQueue	13551115	1	NULL	77.871	34.916	NULL	0
858	0	0	0	Enabled	NULL	SorterExitQueue	17351095	10	NULL	126.516	33.154	NULL	1
343	5	5	0	Enabled	63	Sorter	11601070	1	NULL	47.162	29.34	NULL	0
853	0	0	0	Enabled	NULL	SorterExitQueue	17351085	27	NULL	126.516	31.725	NULL	1
586	2	2	0	Enabled	17	SorterQueue	15401115	1	NULL	103.069	34.916	NULL	0
658	4	4	0	Enabled	43	SorterQueue	13851130	1	NULL	82.595	37.107	NULL	0
223	7	7	0	Disabled	83	SorterQueue	10601060	1	NULL	31.414	28.515	NULL	0
867	0	0	0	Enabled	0	TipperQueue	17551145	2	NULL	130.705	38.921	NULL	0
844	0	0	0	Enabled	0	Tipper	17651030	1	NULL	132.774	24.514	NULL	0
827	0	0	0	Enabled	NULL	Undefined	17350995	4	NULL	126.516	19.014	NULL	0

Location types

- Sorter
- SorterEntranceQueue
- SorterQueue
- SorterExitQueue
- Tipper
- TipperQueue
- TipperExitQueue
- HospitalDropOff
- HospitalEntranceQueue
- HospitalPickup
- HospitalExitQueue
- ShutDownOffTheEnd
- ShutDownQueue
- GLDPreQueueEntrance
- GLDPreQueueExit

Location Status

- Enabled
- Stopped
- Disabled

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Robotics Database

Sorter Side Database Model

The SorterSideDatabaseModel defines the legs of the sorter. The table includes the enterance and exit queue locations of each soterside, as well as the tipper count and whether the sorterside is disabled or not.

Name	EntranceId	ExitId	TipperCount	Disabled
SorterSide1a	46	44	4	0
SorterSide1b	246	244	4	0
SorterSide2a	446	444	4	0
SorterSide2b	648	646	4	1

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Robotics Database

Sort Locations Database Model

The SortLocationsDatabaseModel table contains all the sorter locations and their respective queue locations. The queue locations are the locations behind the sorter location where the agv with an empty tote will move to fill when an exchange happens. This table also includes which sorterside those locations are associated with as well as their sequence from the sorter entrance queue.

SorterId	StagingId	QueueId	Sequence	SorterSideDatabaseModelId
59	NULL	58	93	1
61	NULL	60	92	1
63	NULL	62	91	1
65	NULL	64	90	1
67	NULL	66	89	1
69	NULL	68	88	1
71	NULL	70	87	1
73	NULL	72	86	1
75	NULL	74	85	1
77	NULL	76	84	1
79	NULL	78	83	1
81	NULL	80	82	1
83	NULL	82	81	1
85	NULL	84	80	1
87	NULL	86	79	1
89	NULL	88	78	1
91	NULL	90	77	1
93	NULL	92	76	1
95	NULL	94	75	1
97	NULL	96	74	1
99	NULL	98	73	1
101	NULL	100	72	1
103	NULL	102	71	1
105	NULL	104	70	1
107	NULL	106	69	1
109	NULL	108	68	1
111	NULL	110	67	1
113	NULL	112	66	1

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Robotics Database

Shut Down Staging Database Model

The ShutDownStaging table maps where AGVs should go during a shutdown state. AGVs with totes have an off the end queue and a queue in the center to stage at. AGVs without totes are mapped to sorter locations.

WithTote	ShutDownOffTheEndId	ShutDownQueueId	Sequence	SorterSideDatabaseModelId
1	47	45	1	1
0	NULL	48	10	1
0	NULL	49	9	1
0	NULL	50	8	1
0	NULL	51	7	1
0	NULL	52	6	1
0	NULL	53	5	1
0	NULL	54	4	1
0	NULL	55	3	1
0	NULL	56	2	1
0	NULL	57	1	1
0	NULL	248	10	2
0	NULL	249	9	2
0	NULL	250	8	2
0	NULL	251	7	2
0	NULL	252	6	2
0	NULL	253	5	2
0	NULL	254	4	2
0	NULL	255	3	2
0	NULL	256	2	2
0	NULL	257	1	2
0	NULL	448	10	3
0	NULL	449	9	3
0	NULL	450	8	3
0	NULL	451	7	3

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Robotics Database

Charging

- The ChargingDatabaseModel table stores the separate charging areas as well as their entrance and exit locations. Each area has several chargers

Name	EntranceId	ExitId	Disabled
ChargerArea1	852	44	0
ChargerArea2	859	860	0
ChargerArea3	859	867	0

- The ChargingLocationDatabaseModel table stores the individual chargers, the stations they are located at, and the tote stands associated with the chargers.

SorterId	Charger	ToteStandId	Sequence	Disabled	ChargingDatabaseModelId
4	12	858	12	0	2
1	15	850	15	0	1
3	11	857	11	0	2
3	2	862	2	0	3
1	14	849	14	0	1
2	4	864	4	0	3
2	7	853	7	0	2
2	1	861	1	0	3
1	13	848	13	0	1
1	16	851	16	0	1
4	6	866	6	0	3

- The ChargingThresholdsDatabaseModel table stores values we use in the code to decide whether an agv needs charged or can be kicked off the charger or need an emergency charge.

Charged	CanBeKicked	GoCharge	EmergencyCharge
80	50	40	30

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Robotics Database

Hospital

- The HospitalDatabaseModel table stores the hospitals and their entrance and exit queue location as well as the drop off and induction locations, this is where it drops off a tote and picks one up.
- The HosptialSorterSideDatabaseModel table stores the mapping of sortersides and its associated hospital.
- The CellCodeDatabaseModel stores all cell codes in the map. When a new AGV is added to the system if we are in a running state we will first check if the AGVs current cell code is in a missing zone under the sorter. If not we use the CellCode table to lookup what hospital its connected to and we send a Initialize command to Geek.

Name	EntranceId	DropOffId	InductionId	ExitId	Disabled
Hospital1	868	869	870	12	0

SorterId	HospitalDatabaseModelId
1	1
2	1
3	1
4	1

CellCode	X	Y	HospitalDatabaseModelId
13850945	82.595	12.674	1
13851035	82.595	24.514	1
13851075	82.595	30.401	1
13851085	82.595	31.725	1
13851095	82.595	33.154	1
13851145	82.595	38.921	1
13900945	83.383	12.674	1
13901035	83.383	24.514	1
13901075	83.383	30.401	1
13901085	83.383	31.725	1
13901095	83.383	33.154	1

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Robotics Database

Tipper

- The TipperStationDatabaseModel table stores the name of the tipper station.
- The TipperDatabaseModel table contains all the location id of the tippers as well as the entrance and exit id's. There is also a TipperLoadType to determine what kind of tipper. GLD (gaylord) or Standard normal tipper. The GLDPreQueueEnteranceId and GLDPreQueueExitId are the enterance and exit id's for the GLD. They can be ignored for the standard tippers.

Name
GLDStation1
GLDStation2
TipperStation1
TipperStation2
TipperStation3
TipperStation4
TipperStation5
TipperStation6
TipperStation7
TipperStation8

Name	LocationId	EntranceId	ExitId	PreferredHospitalId	Disabled	TipperStationDatabaseModelId	TipperLoadType	GLDPreQueueEntranceId	GLDPreQueueExitId
TipperStation7a	35	34	28	NULL	0	7	Standard	NULL	884
TipperStation7b	37	36	33	NULL	0	7	Standard	NULL	884
TipperStation8a	41	40	38	NULL	0	8	Standard	NULL	884
TipperStation8b	43	42	39	NULL	0	8	Standard	NULL	884
TipperStation1a	4	3	1	NULL	0	1	Standard	NULL	884
TipperStation1b	6	5	2	NULL	0	1	Standard	NULL	884
TipperStation6a	30	29	23	NULL	0	6	Standard	NULL	884
TipperStation6b	32	31	28	NULL	0	6	Standard	NULL	884
TipperStation2a	9	8	2	NULL	0	2	Standard	NULL	884
TipperStation2b	11	10	7	NULL	0	2	Standard	NULL	884
TipperStation3a	14	13	7	NULL	0	3	Standard	NULL	884
TipperStation3b	16	15	12	NULL	0	3	Standard	NULL	884
TipperStation4a	20	19	17	NULL	0	4	Standard	NULL	884
TipperStation4b	22	21	18	NULL	0	4	Standard	NULL	884
TipperStation5a	25	24	18	NULL	0	5	Standard	NULL	884
TipperStation5b	27	26	23	NULL	0	5	Standard	NULL	884
GLDStation2b	878	875	879	NULL	0	10	GLD	874	884
GLDStation2a	872	876	881	NULL	0	10	GLD	874	884
GLDStation1a	871	873	879	NULL	0	9	GLD	874	884
GLDStation1b	877	883	881	NULL	0	9	GLD	874	884

- The Sorter Side Tipper Station table stores the map for each sorter side and the sequence in which it should consider tippers.

SorterId	TipperDatabaseModelId	Sequence
2	11	10
3	8	5
2	12	12
2	7	3
2	8	4
2	4	5
2	3	7
2	6	1
2	5	2
2	9	6
2	10	8
2	2	9
2	1	11
2	13	13
2	14	14
2	15	15
2	16	16
3	9	1
3	10	2
3	11	3
3	12	4
3	13	6
3	7	7
3	14	8
3	6	9
3	15	10
3	5	11
3	16	12
3	4	13

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Robotics Database

Robot Database Model

The RobotDatabaseModel table contains the data for every robot in the system including its state, tote name, assigned sorterside, currentlocationid, destination location, and state.

Name	SorterSideId	CommandId	CurrentLocationId	DestinationLocationId	State	ToteId	ChargesSinceLastCharged	ChargeKick	Status	ErrorCode	Coordinate	CellCode	Angle	Battery	Current	Voltage	Temperature	Ip	IsToteEmpty	LastCommandGuic
1555058	3	NULL	491	NULL	SWaitingInzone	NULL	323	0	NORMAL		RoboticsModule.Messages.Location	10851070	90.08	47	330	52000	28	10.27.80.210	1	c6318105-fe57-4e
1555025	NULL	NULL	NULL	NULL	Removed	NULL	9	0	REMOVED	21085	RoboticsModule.Messages.Location	16151145	88.91	64	360	54200	22	10.27.80.203	1	eb0115aa-8c72-4
1555017	NULL	NULL	NULL	NULL	Removed	NULL	0	0	REMOVED	21085	RoboticsModule.Messages.Location	16001145	89.86	25	100	49900	20	10.27.80.198	1	5e86bf6d-a9db-47
1542196	NULL	NULL	NULL	NULL	Removed	NULL	809	0	REMOVED	12003	RoboticsModule.Messages.Location	NULL	0	48	320	52100	23	10.27.80.119	1	e28203ed-405b-4
1555022	3	NULL	629	NULL	SWaitingForExchange	99	752	0	NORMAL		RoboticsModule.Messages.Location	14301070	90.11	43	710	51400	28	10.27.80.89	1	b24cefc7-f0b2-49
1555064	2	13063907	864	NULL	GoCharge	347	730	0	NORMAL		RoboticsModule.Messages.Location	16950980	90.01	72	0	57600	36	10.27.80.82	1	a501c31e-c72d-4
1555065	3	13065501	18	859	MovingToChargerEntranceQueue	116	253	0	NORMAL		RoboticsModule.Messages.Location	17501045	179.6	40	0	51100	28	10.27.80.215	1	a4e5069b-ec41-4
1542149	NULL	NULL	NULL	NULL	Removed	NULL	288	0	REMOVED		RoboticsModule.Messages.Location	17601085	-1.17	64	420	54200	18	10.27.80.126	1	ccad2561-5ba9-4
1555048	3	13065503	444	26	MovingToTipperQueue	445	905	0	NORMAL		RoboticsModule.Messages.Location	17301085	0.16	47	410	51800	28	10.27.80.88	1	62261f5b-ffc4-4d

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Robotics Database

Robot States

- Undefined
- Removed
- ToBeInitialized
- Tipper
  - MovingToTipperQueue
  - AtTipperQueue
  - MovingToTipper
  - AtTipper
  - AtTipperClearToLeave
  - MovingToTipperExitQueue
  - AtTipperExitQueue
- Hospital
  - MovingToHospitalQueue
  - AtHospitalQueue
  - DroppingOffHospitalDropOff
  - DroppedOffHospitalDropOff
  - MovingToHospitalPickup
  - AtHospitalPickup
  - PickingUpToteHospital
  - PickedUpToteHospital
  - PickedUpClearToLeave
  - MovingToHospitalExitQueue
  - AtHospitalExitQueue
- Charger
  - MovingToChargerEntranceQueue
  - AtChargerEntranceQueue
  - DroppingOffChargerToteStand
  - DroppedOffChargerToteStand
  - CancelRequestedBeforeCharging
  - GoCharge
  - Charging
  - ChargeCancelling
  - ChargeComplete
  - LeavingCharger
  - LeftCharger
  - PickingUpToteChargerToteStand
  - PickedUpToteChargerToteStand
  - MovingToChargerExitQueue
  - AtChargerExitQueue

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Robotics Database

Robot States Sorter

- In and Out
  - MovingToSorterEntranceQueue
  - AtSorterEntranceQueue
  - MovingToSorterExitQueue
  - AtSorterExitQueue
- Sorter Queue
  - SQMovingIntoZone
  - SQWaitingInZone
  - SQStagingExchange
  - SQStagedExchange
  - SQExchanging
  - SQExchanged
  - SDroppingOffTote
- Sorter Locations
  - SWaitingInzone
  - SZoneStaging
  - SPickingUpTote
  - SPickedUpTote
  - SWaitingForExchange
- GLD
  - AtSorterExitQueue
  - AtGLDPreQueue
  - MovingToGLDPreQueue
  - AtGLDPreQueueExit
- ShutDown
  - MovingToShutDownOffTheEnd
  - AtShutDownOffTheEnd
  - MovingToShutDownQueue
  - AtShutDownQueue

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Robotics Database

Robot Tote Database Model

The RobotToteDatabseModel table contains the tote information including toteid, the state of the tote, the locationid, and the robot id for all the totes in the system.

Tote States

- PickupToteRequested
- TotePickedUp
- MoveOutSorterRequested
- AtTipper
- DropOffSorterRequested
- DroppedOffSorter
- DropOffHospitalRequested
- DroppedOffHospital
- InductToteRequested
- ToteInducted
- RemoveToteRequested
- ToteRemoved

ToteId	ToteStatus	ToteState	LocationId	CommandId	RobotId	TaskId	ExchangePriority
313	NULL	DroppedOffSorter	97	NULL	NULL	NULL	0
427	NULL	DroppedOffSorter	171	NULL	NULL	NULL	0
343	NULL	DroppedOffSorter	417	NULL	NULL	NULL	0
34	NULL	DroppedOffSorter	617	NULL	NULL	NULL	0
22	NULL	DroppedOffSorter	545	NULL	NULL	NULL	0
27	NULL	DroppedOffSorter	237	NULL	NULL	NULL	0
270	NULL	DroppedOffSorter	107	NULL	NULL	NULL	0
145	NULL	DroppedOffSorter	631	NULL	NULL	NULL	0
151	NULL	DroppedOffSorter	519	NULL	NULL	NULL	0
339	NULL	DroppedOffSorter	305	NULL	NULL	NULL	0
123	NULL	DroppedOffSorter	99	NULL	NULL	NULL	0
32	NULL	DroppedOffSorter	85	NULL	NULL	NULL	0
402	NULL	DroppedOffSorter	185	NULL	NULL	NULL	0
75	NULL	DroppedOffSorter	443	NULL	NULL	NULL	0
401	NULL	DroppedOffSorter	77	NULL	NULL	NULL	0
400	NULL	DroppedOffSorter	571	NULL	NULL	NULL	0
462	NULL	DroppedOffSorter	113	NULL	NULL	NULL	0
83	NULL	DroppedOffSorter	501	NULL	NULL	NULL	0
344	NULL	DroppedOffSorter	529	NULL	NULL	NULL	0
338	NULL	DroppedOffSorter	125	NULL	NULL	NULL	0
16	NULL	DroppedOffSorter	271	NULL	NULL	NULL	0

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Robotics Database

Robot Command Database Model

The RobotCommandDatabaseModel table tracks the status of the robot commands requested by the robotics module for AGVs.

CommandGuid	Name	Status	RobotCode	Parameters	SentTime	InProgressTime	ExecutedTime
a501c31e-c72d-410e-8f98-7d92f1945b26	GoCharge	Sent	1555064	{"ChargerId":"4"}	2026-03-09 18:31:23.0530875	NULL	NULL
5549dcfe-f38d-4dba-ad6e-81b361fff9d3	MoveTote	InProgress	1555018	{"ContainerId":"348","StationId":"670"}	2026-03-09 18:41:08.1026068	2026-03-09 18:41:11.4190116	NULL
b066c9d6-c6d0-4cf5-b097-8219ac30d89b	MoveTote	InProgress	1555067	{"ContainerId":"397","StationId":"845"}	2026-03-09 18:40:32.9806596	2026-03-09 18:40:37.0968901	NULL
7f8cc4d2-c1f9-40c1-8d29-9aa1de34eb96	MoveTote	Sent	1555061	{"ContainerId":"71","StationId":"858"}	2026-03-09 18:41:20.0576120	NULL	NULL
9c4093c4-5b2b-4d47-b014-1d34db6ac63c	MoveTote	InProgress	1542159	{"ContainerId":"309","StationId":"1"}	2026-03-09 18:41:13.1653808	2026-03-09 18:41:16.8860476	NULL
42f452da-9ef0-46b5-a853-99e589897248	MoveTote	InProgress	1555041	{"ContainerId":"69","StationId":"866"}	2026-03-09 18:41:04.1318974	2026-03-09 18:41:09.2829645	NULL
07e82771-47db-4aa8-bd17-4d388ef15b5e	MoveTote	InProgress	1555078	{"ContainerId":"426","StationId":"853"}	2026-03-09 18:40:57.9110347	2026-03-09 18:41:02.5363510	NULL
55801381-1c2e-4408-a3ac-f9c437832b36	GoCharge	Sent	1555051	{"ChargerId":"15"}	2026-03-09 18:35:30.9120310	NULL	NULL
e8ad0434-d769-42e8-ad9b-a527907a235a	GoCharge	Sent	1542200	{"ChargerId":"1"}	2026-03-09 18:26:02.3144802	NULL	NULL
cc2f2ad8-0c69-4ccb-816a-ccb4d4d88a1f	MoveTote	InProgress	1555019	{"ContainerId":"470","StationId":"866"}	2026-03-09 18:41:04.1318257	2026-03-09 18:41:09.2830091	NULL
e0b8a9c2-9cba-4378-b738-610c4b78fae3	MoveTote	InProgress	1555057	{"ContainerId":"25","StationId":"603"}	2026-03-09 18:41:08.1026858	2026-03-09 18:41:11.4190171	NULL
bf4ba815-2234-4b94-b17b-dedfa2742f4b	MoveTote	Sent	1555055	{"ContainerId":"411","StationId":"495"}	2026-03-09 18:41:20.0576292	NULL	NULL
74025aec-8c42-43f0-b997-54a1a2c8ac53	GoCharge	Sent	1555021	{"ChargerId":"2"}	2026-03-09 18:27:08.2083977	NULL	NULL
cf949561-05be-4fb0-ad27-47ec718af62b	GoCharge	Sent	1542186	{"ChargerId":"5"}	2026-03-09 18:35:40.8647146	NULL	NULL
c9b523b0-26ca-4610-8bcb-bcc31c9b0db4	MoveTote	Sent	1555029	{"ContainerId":"131","StationId":"603"}	2026-03-09 18:41:20.0577113	NULL	NULL
23b21c3c-ef51-4bed-a4be-286c04f43f6f	GoCharge	Sent	1555046	{"ChargerId":"8"}	2026-03-09 18:34:30.3546634	NULL	NULL

Command Names

- EmptyGoTo
- PickupTote
- MoveTote
- DropOffTote
- InductTote
- RemoveTote
- ReleaseAgv
- InitializeAgv
- RecoverAgv
- SystemRecover
- SystemStop
- RemoveAgv
- UpdateBox
- GetInventoryToteLocation
- GetChargersStatus
- GoCharge
- CancelRequest

When Commands are complete. They are archived to RobotCommandActivity

CommandGuid	Name	Status	RobotCode	Parameters	SentTime	InProgressTime	ExecutedTime	Station	SortGuid	TotalTime
e4ba1595-e0ca-4d2e-b89a-f1fd980e3fa6	MoveTote	Executed	1542161	{"ContainerId":"43","StationId":"678"}	2026-03-09 18:39:53.7837680	2026-03-09 18:39:57.0340723	2026-03-09 18:39:58.9770087	678	d1419b69-e68f-4d30-b9c8-0b8e1da65419	00:00:05.1932407
1d070a1d-93b2-4fde-a19d-81dc1277cdae	MoveTote	Executed	1555020	{"ContainerId":"66","StationId":"853"}	2026-03-09 18:39:25.3363317	2026-03-09 18:39:28.9903012	2026-03-09 18:39:58.9770630	853	d1419b69-e68f-4d30-b9c8-0b8e1da65419	00:00:33.6407313
7b3d0205-f3fe-4d1e-9d64-d94fb0a94352	MoveTote	Executed	1542191	{"ContainerId":"184","StationId":"867"}	2026-03-09 18:39:53.7841476	2026-03-09 18:39:57.0340826	2026-03-09 18:39:58.9770659	867	d1419b69-e68f-4d30-b9c8-0b8e1da65419	00:00:05.1929183
bc8adb2a-38ed-4a47-85d8-9681ada1cf47	MoveTote	Executed	1542153	{"ContainerId":"395","StationId":"437"}	2026-03-09 18:39:51.9902533	2026-03-09 18:39:58.9770613	2026-03-09 18:39:58.9770928	437	d1419b69-e68f-4d30-b9c8-0b8e1da65419	00:00:06.9868395
f0d4dc45-e6dd-4a44-818c-3d50d56d5d2c	MoveTote	Executed	1555030	{"ContainerId":"110","StationId":"555"}	2026-03-09 18:39:42.6049668	2026-03-09 18:39:46.7147268	2026-03-09 18:39:58.9771036	555	d1419b69-e68f-4d30-b9c8-0b8e1da65419	00:00:16.3721368
379bf4bf-9e83-4b6d-bece-7a29fefc91cf	MoveTote	Executed	1555023	{"ContainerId":"277","StationId":"850"}	2026-03-09 18:39:51.9904515	2026-03-09 18:39:57.0341365	2026-03-09 18:39:58.9771217	850	d1419b69-e68f-4d30-b9c8-0b8e1da65419	00:00:06.9866702
c9dc1b1c-f5e7-4ad7-ac68-b03c61401a0d	MoveTote	Executed	1555068	{"ContainerId":"421","StationId":"847"}	2026-03-09 18:39:46.4022145	2026-03-09 18:39:50.4558095	2026-03-09 18:39:57.0341266	847	d1419b69-e68f-4d30-b9c8-0b8e1da65419	00:00:10.6319121
0a69f93d-1266-4eb9-8c5b-e87c9947eafb	MoveTote	Executed	1555077	{"ContainerId":"286","StationId":"858"}	2026-03-09 18:39:27.1668695	2026-03-09 18:39:30.7618436	2026-03-09 18:39:57.0341357	858	d1419b69-e68f-4d30-b9c8-0b8e1da65419	00:00:29.8672662
e545f993-70e6-4497-ab9b-a775dcaae9d8	MoveTote	Executed	1555065	{"ContainerId":"301","StationId":"11"}	2026-03-09 18:39:51.9903881	2026-03-09 18:39:57.0341851	2026-03-09 18:39:57.0342128	11	d1419b69-e68f-4d30-b9c8-0b8e1da65419	00:00:05.0438247
ae3615ba-bb9d-4c33-b026-582f42942c63	MoveTote	Executed	1542166	{"ContainerId":"461","StationId":"590"}	2026-03-09 18:39:46.4029624	2026-03-09 18:39:50.4557817	2026-03-09 18:39:54.7871319	590	d1419b69-e68f-4d30-b9c8-0b8e1da65419	00:00:08.3841695
e2c6a20e-340a-4efb-937e-5ccddcc73611	MoveTote	Executed	1555010	{"ContainerId":"21","StationId":"853"}	2026-03-09 18:39:27.1658749	2026-03-09 18:39:30.7618385	2026-03-09 18:39:54.7872464	853	d1419b69-e68f-4d30-b9c8-0b8e1da65419	00:00:27.6213715
c96a9b97-8998-4db9-98d3-7d5a7715538e	MoveTote	Executed	1542191	{"ContainerId":"184","StationId":"866"}	2026-03-09 18:39:37.6679465	2026-03-09 18:39:41.1496062	2026-03-09 18:39:52.2830217	866	d1419b69-e68f-4d30-b9c8-0b8e1da65419	00:00:14.6150752

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Robotics Database

Robot Task Database Model

- The RobotTaskDatabaseModel table tracks the status of the tote task requested by the sorter sweeper module.

- When tasks are complete. They are archived to RobotTaskActivity

Task Name	Task Type	Tote	Station	Task Status
DropOff Sorter	DropOff Sorter	51	0	Executed
DropOff Sorter	DropOff Sorter	129	0	Executed
MoveOut Sorter	MoveOut Sorter	429	0	Executed
Pickup Tote	Pickup Tote	457	503	Executed
Pickup Tote	Pickup Tote	48	790	Executed
Pickup Tote	Pickup Tote	470	777	Executed
DropOff Sorter	DropOff Sorter	460	0	Executed
DropOff Sorter	DropOff Sorter	411	0	Executed
DropOff Sorter	DropOff Sorter	98	0	Executed
Pickup Tote	Pickup Tote	39	765	Executed
Pickup Tote	Pickup Tote	158	504	Executed
MoveOut Sorter	MoveOut Sorter	332	0	Executed
Pickup Tote	Pickup Tote	57	477	Executed
MoveOut Sorter	MoveOut Sorter	48	0	Executed
MoveOut Sorter	MoveOut Sorter	457	0	Executed

Task Name	Task Type	Tote	Station	Task Status
Remove Tote	Remove Tote	294	0	Not Executed
DropOff Sorter	DropOff Sorter	282	0	Not Executed
DropOff Sorter	DropOff Sorter	159	0	Not Executed
DropOff Sorter	DropOff Sorter	336	0	Not Executed
DropOff Sorter	DropOff Sorter	347	0	Not Executed
Remove Tote	Remove Tote	78	0	Not Executed
DropOff Sorter	DropOff Sorter	46	0	Not Executed
Remove Tote	Remove Tote	48	0	Not Executed
Pickup Tote	Pickup Tote	139	443	Not Executed
DropOff Sorter	DropOff Sorter	331	0	Not Executed
DropOff Sorter	DropOff Sorter	93	0	Not Executed
Pickup Tote	Pickup Tote	390	426	Not Executed
Remove Tote	Remove Tote	96	0	Not Executed
MoveOut Sorter	MoveOut Sorter	452	0	Not Executed
DropOff Sorter	DropOff Sorter	411	0	Not Executed
DropOff Sorter	DropOff Sorter	86	0	Not Executed

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Robotics Database

AGV Error Model

The AgvErrorModel tables stores the list of errors and their codes as well as provides a description of the error.

For metrics purposes errors are classified as critical and noncritical.

Code	Description	Cause	Critical
22000	RMS communication interruption	RMS system check	1
22001	CAN communication interrupt (charging module)	Charger Check	1
22002	Screen communication is interrupted	RMS system check	1
22003	RMS data abnormal	RMS system check	1
22004	RMS command exception	RMS system check	1
22007	The charging module is overtemperature	Charger Check	1
22008	The automatic mode charging current is 0	Charger Check	1
22009	Charging module warning status	Charger Check	1
22010	Charging module error status	Charger Check	1
22011	The charging station reports the unavailability status...	Charger Check	1
13001	The charging station is offline (lost contact for a long...	Charger Check	1
21000	The lateral deviation of the 2D code positioning at t...	QR code check	1
21057	Front bumper trigger	Robot Reboot	0
21059	The rear bumper is triggered	Robot Reboot	1
21060	Robot obstacle avoidance	Robot Reboot	0
21041	The drive is out of contact	Robot Reboot	1
21042	The encoder is out of contact	Robot Reboot	1
21044	The gyro temperature changes too much	Robot Reboot	1
21061	The lifting motor cannot be lowered. Such as: can n...	Robot Reboot	1
21067	The drive wheel motor is overcurrent	Robot Reboot	1
21068	Lift motor overcurrent	Robot Reboot	1
21071	Shelf QR code decoding failed. For example, even t...	QR code check	1
21072	Shelf QR code decoding timeout	QR code check	1
21073	The drive is overtemperatured	Robot Reboot	1
21074	Lidar data loss, or device failure	Robot Reboot	1
21081	The communication between the main control and t...	Robot Reboot	1
12003	The robot is not on the map	Robot Reboot	1
12007	No paths can be planned	RMS system check	1
21085	Drive instruction error	Robot Reboot	1
21086	Drive motor is out of phase	Robot Reboot	1

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Robotics Database

AGV Error

- The AgvActiveErrorDatabaseModel table logs any AGV active errors that are currently in the system. The table includes the name (robot code), error code, and a description of the error. Lastly it also provides the sort guid which is used to identify during what sort the error occurred.
- When error is resolved or the AGV is removed they are archived to the AGVErrorActivityDatabaseModel

Name	ErrorCode	Error	SortGuid
1555024	21057	Front bumper trigger	d1419b69-e68f-4d30-b9c8-0b8e1da65419

Name	ErrorCode	Error	SortGuid	StartTime	FinishTime	TotalTime
1555037	12114	The robot task is not completed for a long time	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-07 10:20:07.1898196	2026-03-07 10:20:37.3094656	00:00:30.1196460
1555030	12011	The waiting point has not been reached for a lo...	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-07 10:41:54.1235916	2026-03-07 10:42:40.5312680	00:00:46.4076764
1555073	12011	The waiting point has not been reached for a lo...	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-07 10:46:10.2797905	2026-03-07 10:46:24.9582723	00:00:14.6784818
1555050	21060	Robot obstacle avoidance	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-07 10:52:56.8578526	2026-03-07 10:53:10.5610810	00:00:13.7032284
1555021	12011	The waiting point has not been reached for a lo...	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-07 10:53:26.2744513	2026-03-07 10:53:56.6386126	00:00:30.3641613
1555021	21060	Robot obstacle avoidance	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-07 10:52:25.6972579	2026-03-07 10:53:56.6386126	00:01:30.9413547
1555046	12011	The waiting point has not been reached for a lo...	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-07 10:55:55.8082631	2026-03-07 10:56:11.9407932	00:00:16.1325301
1555011	12011	The waiting point has not been reached for a lo...	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-07 10:56:56.9985704	2026-03-07 10:57:12.0925836	00:00:15.0940132
1555043	21091	Driver overcurrent	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-07 11:14:43.7090518	2026-03-07 11:15:29.7225338	00:00:46.0134820
1555077	21060	Robot obstacle avoidance	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-07 11:15:29.7232470	2026-03-07 11:15:43.9170107	00:00:14.1937637
1555073	12011	The waiting point has not been reached for a lo...	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-07 11:16:31.1993194	2026-03-07 11:17:00.2980439	00:00:29.0987245
1555043	12011	The waiting point has not been reached for a lo...	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-07 11:17:59.5755320	2026-03-07 11:19:14.1038177	00:01:14.5282857
1555043	12003	The robot is not on the map	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-07 11:19:29.5637230	2026-03-07 11:19:44.1220227	00:00:14.5582997
1555043	12011	The waiting point has not been reached for a lo...	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-07 11:19:29.5637031	2026-03-07 11:19:44.1220227	00:00:14.5583196
1555052	12011	The waiting point has not been reached for a lo...	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-07 11:31:45.8643110	2026-03-07 11:32:01.1775991	00:00:15.3132881

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Robotics Database

Location Active Dwell Model

- The LocationActiveDwellModel table displays the active dwells at locations marked for DwellTracking
- One the AGV leaves it is done dwelling and the activity is archived to the LocationDwellActivity.

AgvName	Station	SortGuid
1555070	858	d1419b69-e68f-4d30-b9c8-0b8e1da65419
1555048	853	d1419b69-e68f-4d30-b9c8-0b8e1da65419

AgvName	Station	SortGuid	StartTime	FinishTime	TotalTime
1555029	858	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:04:11.3008705	2026-03-09 11:04:16.2811987	00:00:04.9803282
1542163	866	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:04:23.6873759	2026-03-09 11:04:28.4294208	00:00:04.7420449
1542161	866	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:04:30.1707546	2026-03-09 11:04:35.9143833	00:00:05.7436287
1555026	866	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:04:47.5454565	2026-03-09 11:04:51.8305933	00:00:04.2851368
1555037	866	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:05:09.9362412	2026-03-09 11:05:15.1491171	00:00:05.2128759
1542166	858	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:05:15.1502661	2026-03-09 11:05:21.4890778	00:00:06.3388117
1542159	866	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:05:21.4902969	2026-03-09 11:05:26.6115929	00:00:05.1212960
1555064	858	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:05:31.6381310	2026-03-09 11:05:36.1230452	00:00:04.4849142
1555074	858	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:05:39.3919818	2026-03-09 11:05:44.2175061	00:00:04.8255243
1555079	866	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:05:42.6282045	2026-03-09 11:05:47.4303631	00:00:04.8021586
1555045	853	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:06:17.4548943	2026-03-09 11:06:21.3286158	00:00:03.8737215
1555024	866	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:06:48.7779144	2026-03-09 11:06:54.0722165	00:00:05.2943021
1555028	858	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:06:52.3507789	2026-03-09 11:06:57.2510505	00:00:04.9002716
1555039	866	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:07:12.8014411	2026-03-09 11:07:18.9248475	00:00:06.1234064
1555059	858	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:07:16.3010798	2026-03-09 11:07:20.7782573	00:00:04.4771775

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Robotics Database

System Error

- SystemErrorDefinition stores the definition of the system errors.
- The SystemActiveErrorModel shows any active errors that are currently in the system. The information includes the name, area, and error code as well as a description of the error.
- Once an active error is resolved it is archived to the SystemErrorActivityModel table.

Error	Area	SystemErrorDescription	SystemErrorResolution
AgvHasNotMoved30Seconds	AGV	AGV has not moved in 30 seconds	Reboot AGV
AgvOutOfSequence	AGV	AGV is out of sequence	Remove AGV
AgvCanNotMoveToHospital	AGV	Cannot move to Hospital	Clear Hospital
AgvCommandErrored	AGV	AGV Command has Errored	Remove AGV
AgvCanNotMoveToCharger	AGV	Cannot move to Charger	Remove AGV
ZoneMissingAgv	AGV	Missing AGV	Add AGV
Communication	System	Communication Error	Reconnect

Name	Area	Error	LastSeen	SystemErrorDescription	SystemErrorResolution
Items on Sorter	System	7	2026-03-09 21:06:35.6336606	Communication Error System Shutdown with Active Totes	Reconnect
1542203	AGV	1	2026-03-09 21:06:31.7542079	AGV has not moved in 30 seconds	Reboot AGV

Name	Area	Error	systemErrorDescription	systemErrorResolution	StartTime	FinishTime	TotalTime
1555045	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:41:59.3436330	2026-03-07 10:42:26.2690115	00:00:26.9253785
1555045	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:41:59.3436330	2026-03-07 10:42:26.2690115	00:00:26.9253785
1542158	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:29.3963747	2026-03-07 10:43:44.6185118	00:00:15.2221371
1542158	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:29.3963747	2026-03-07 10:43:44.6185118	00:00:15.2221371
1555055	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:15.6329122	2026-03-07 10:43:46.2890186	00:00:30.6561064
1555019	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:20.2066937	2026-03-07 10:43:46.2890186	00:00:26.0823249
1555024	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:29.3963631	2026-03-07 10:43:46.2890186	00:00:16.8926555
1542154	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:20.2066730	2026-03-07 10:43:46.2890186	00:00:26.0823456
1555055	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:15.6329122	2026-03-07 10:43:46.2890186	00:00:30.6561064
1542154	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:20.2066730	2026-03-07 10:43:46.2890186	00:00:26.0823456
1555019	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:20.2066937	2026-03-07 10:43:46.2890186	00:00:26.0823249
1555024	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:29.3963631	2026-03-07 10:43:46.2890186	00:00:16.8926555
1542200	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:27.9168621	2026-03-07 10:43:47.8845750	00:00:19.9677129
1542200	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:27.9168621	2026-03-07 10:43:47.8845750	00:00:19.9677129
1542191	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:20.2066871	2026-03-07 10:43:49.4134980	00:00:29.2068109
1542161	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:30.9330512	2026-03-07 10:43:49.4134980	00:00:18.4804468
1542191	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:20.2066871	2026-03-07 10:43:49.4134980	00:00:29.2068109
1542161	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:43:30.9330512	2026-03-07 10:43:49.4134980	00:00:18.4804468
1555021	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:52:37.5273449	2026-03-07 10:54:03.5140854	00:01:25.9867405
1555021	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:52:37.5273449	2026-03-07 10:54:03.5140854	00:01:25.9867405
1555020	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:59:29.6708794	2026-03-07 10:59:53.0826901	00:00:23.4118107
1555020	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 10:59:29.6708794	2026-03-07 10:59:53.0826901	00:00:23.4118107
AGV Disconnected - Turn on AGV	System	Communication	Communication Error AGV(s) disconnected	Reconnect Turn on the AGV	2026-03-07 11:08:14.9305794	2026-03-07 11:08:43.0553291	00:00:28.1247497
AGV Disconnected - Turn on AGV	System	Communication	Communication Error AGV(s) disconnected	Reconnect Turn on the AGV	2026-03-07 11:08:44.8339075	2026-03-07 11:09:15.9149833	00:00:31.0810758
1542209	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 11:09:51.2429228	2026-03-07 11:10:39.5057585	00:00:48.2628357
1542209	AGV	AgvHasNotMoved30Seconds	AGV has not moved in 30 seconds	Reboot AGV	2026-03-07 11:09:51.2429228	2026-03-07 11:10:39.5057585	00:00:48.2628357
AGV Disconnected - Turn on AGV	System	Communication	Communication Error AGV(s) disconnected	Reconnect Turn on the AGV	2026-03-07 11:09:46.3955311	2026-03-07 11:11:15.0791884	00:01:28.6836573

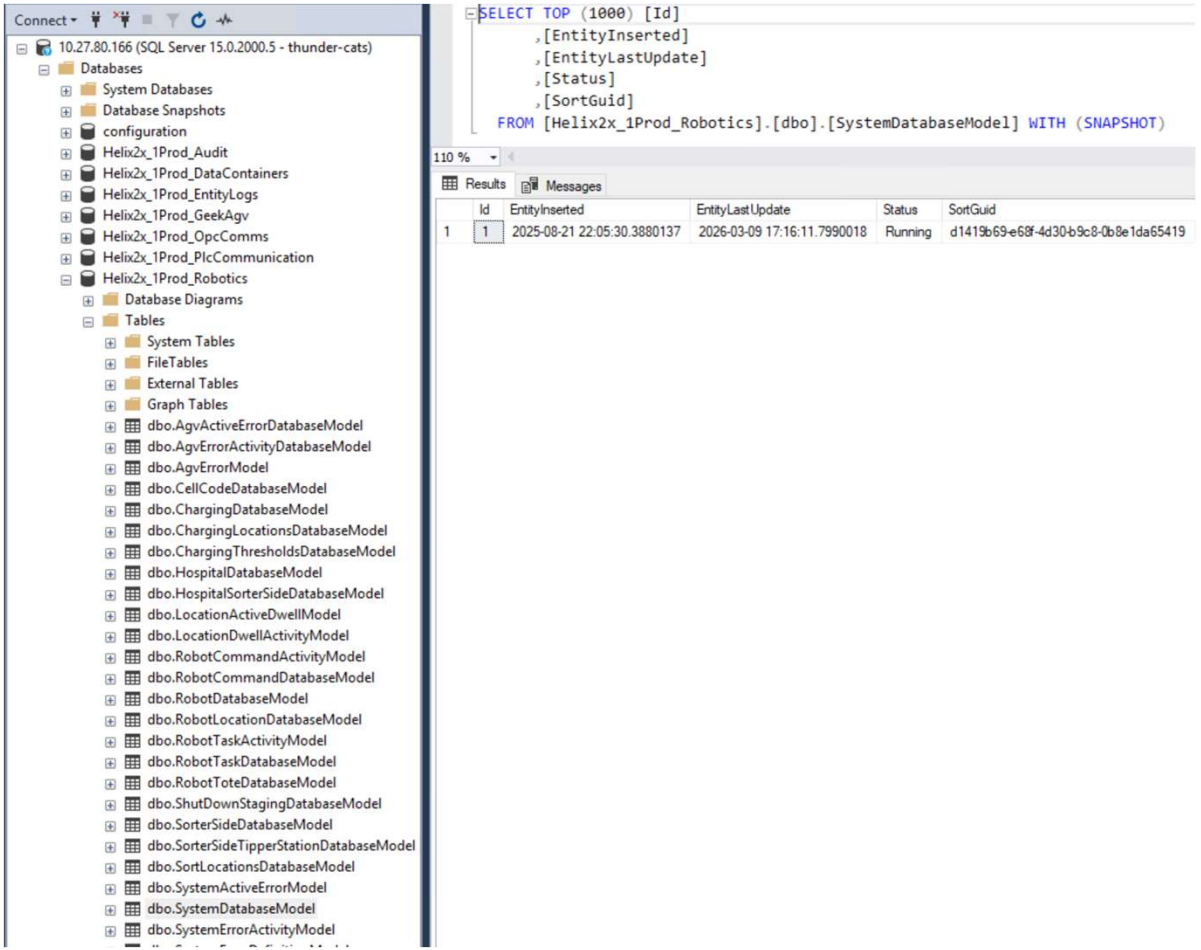
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Robotics Database

System Database Model

The SystemDatabaseModel shows the state of the system and the current sort guid.

- StartingUp
- StartUpRequested
- Running
- ShutDownRequested
- Shutdown
- ZoneConfigRequested
- ZoneConfigStaging
- CloseOutRequested
- CloseOutStaging
- CloseOutStaged



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Sorter Sweeper Database

Overview

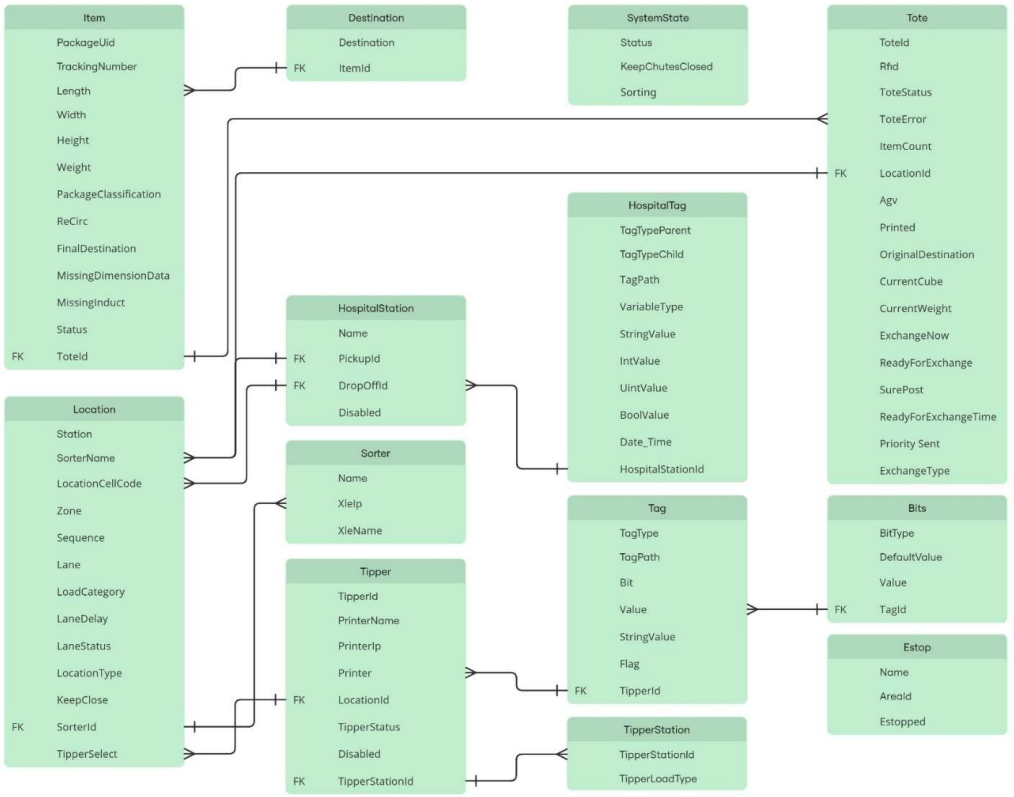


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Sorter Sweeper Database

Foundation



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Sorter Sweeper Database

Sorter Model

The Sorter Model define our sorter legs.

At Haslet we have 4 legs and only 2 at McKinney.

- Name: of the side. From RMS map we go from top to bottom.
- XleIp: Receiver Ip address when we send messages to the sorter.
- XleName: Receiver name when we send messages to the sorter.

```
SELECT TOP (1000) [Id]
      ,[EntityInserted]
      ,[EntityLastUpdate]
      ,[Name]
      ,[XleIp]
      ,[XleName]
  FROM [Helix2x_1Prod_SorterSweeper].[dbo].[SorterModel] WITH (SNAPSHOT)
```

110 %

Results | Messages

	Id	EntityInserted	EntityLastUpdate	Name	XleIp	XleName
1	1	2025-08-21 22:05:55.1815717	2025-08-21 22:05:55.1815720	SorterSide1a	10.27.8.105	PFC
2	2	2025-08-21 22:05:55.1864977	2025-08-21 22:05:55.1864979	SorterSide1b	10.27.8.105	PFC
3	3	2025-08-21 22:05:55.6364346	2025-08-21 22:05:55.6364347	SorterSide2a	10.27.8.105	PFC
4	4	2025-08-21 22:05:55.6401986	2025-08-21 22:05:55.6401988	SorterSide2b	10.27.8.105	PFC

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Sorter Sweeper Database

Location Model

Locations represent stations in the map. Locations have 4 types. Sorter, Tipper, Hospital, Induction.

- Induction: is the specified tote Induction location at the hospital.
- Hospital: is the specified tote drop off location at the hospital.
- When an AGV reports the same cell code as induction or hospital we notify the hospital controller of the presence of an AGV.
- Tippers: when a task is complete from robotics engine we use the station it reported back to determine which tipper its at and notify that specific tipper a tote has arrived.



	Id	EntityInserted	EntityLastUpdate	Station	SorterName	LocationCellCode	Zone	Sequence	Lane	LoadCategory	LaneDelay	LaneStatus	LocationType	Agv	X	Y	Z	KeepClose	SorterDatabaseModelId	TipperSelect
1	139	2025-08-21 22:05:55.1841059	2026-03-09 20:38:18.1392457	462	SorterSide1b	NULL	4	48	03446	NULL	1	Bypassed	Sorter	NULL	NULL	NULL	NULL	0	2	None
2	107	2025-08-21 22:05:55.1831792	2026-03-09 20:35:55.2599144	430	SorterSide1b	NULL	6	80	03475	NULL	1	Bypassed	Sorter	NULL	NULL	NULL	NULL	0	2	None
3	64	2025-08-21 22:05:55.1637525	2026-03-09 20:45:12.0826982	767	SorterSide1a	NULL	3	30	03028	NULL	1	Bypassed	Sorter	NULL	NULL	NULL	NULL	0	1	GLDStation1
4	267	2025-08-21 22:05:55.6351660	2026-03-09 20:45:13.9447118	392	SorterSide2a	NULL	2	14	03675	NULL	1	Bypassed	Sorter	NULL	NULL	NULL	NULL	0	3	GLDStation2
5	80	2025-08-21 22:05:55.1641925	2026-03-09 20:45:12.2155561	783	SorterSide1a	NULL	2	14	03015	NULL	1	Bypassed	Sorter	NULL	NULL	NULL	NULL	0	1	GLDStation2
6	52	2025-08-21 22:05:55.1633832	2026-03-09 20:45:11.9577407	755	SorterSide1a	NULL	4	42	03040	NULL	1	Bypassed	Sorter	NULL	NULL	NULL	NULL	0	1	GLDStation2
7	93	2025-08-21 22:05:55.1648188	2026-03-09 20:45:12.3556077	796	SorterSide1a	NULL	1	1	03002	NULL	1	Bypassed	Sorter	NULL	NULL	NULL	NULL	0	1	GLDStation1
8	186	2025-08-21 22:05:55.1854665	2026-03-09 20:45:13.1846267	509	SorterSide1b	NULL	1	1	03402	NULL	1	Bypassed	Sorter	NULL	NULL	NULL	NULL	0	2	GLDStation2
9	173	2025-08-21 22:05:55.1851098	2026-03-09 20:45:13.0486598	496	SorterSide1b	NULL	2	14	03415	NULL	1	Bypassed	Sorter	NULL	NULL	NULL	NULL	0	2	GLDStation1
10	240	2025-08-21 22:05:55.6344820	2026-03-09 20:45:13.6995735	365	SorterSide2a	NULL	4	41	03651	NULL	1	Bypassed	Sorter	NULL	NULL	NULL	NULL	0	3	GLDStation2
11	280	2025-08-21 22:05:55.6354980	2026-03-09 20:45:14.0796109	405	SorterSide2a	NULL	1	1	03688	NULL	1	Bypassed	Sorter	NULL	NULL	NULL	NULL	0	3	GLDStation1

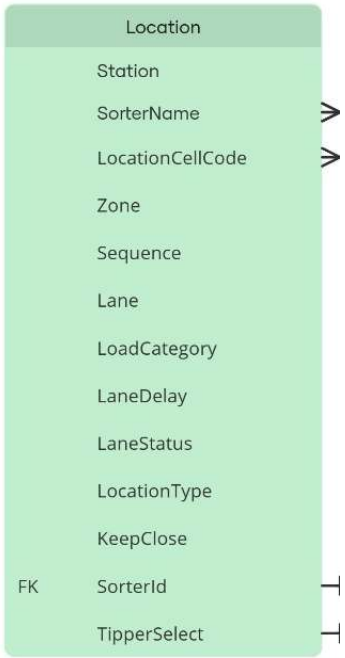
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Sorter Sweeper Database

Location Model

Sorter Location have attributes for reporting

- SorterName: what sorter side they belong to
- Zone: what zone are they in
- Sequence: starts from sorter entrance
- Lane: chute number as UPS defines
- LaneDelay: used for straight line sorters to predict when packages will divert
- LaneStatus:
  - Active: chute is open
  - Bypassed: chute is closed
  - NotTracking: destinations that are not being used yet
- KeepClose: if the chute should remain closed after the exchange
- TiperSelect: If a location has a specified tipper it needs to tip at. Used for GLD.

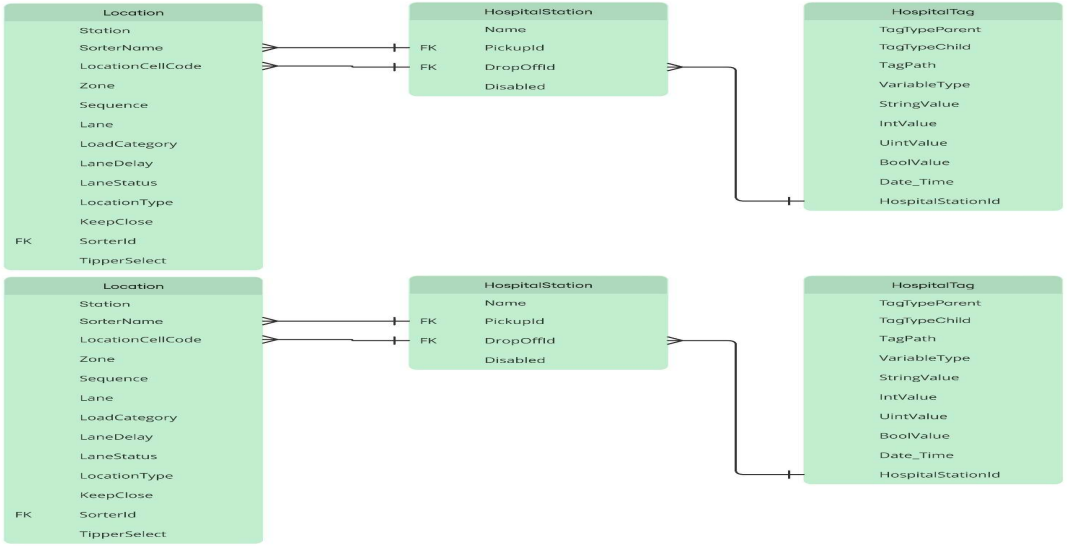


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Sorter Sweeper Database

Hospital Tags Model

To process the data from the hospital controller we have a list of tags that we subscribe to and action on based on the TagTypeParent.



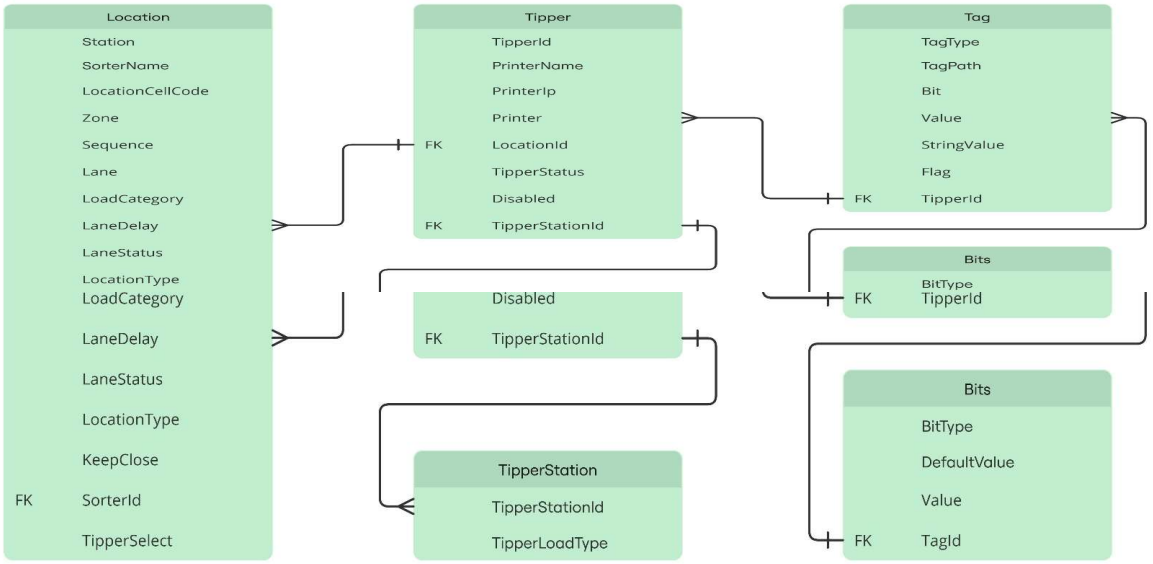
	Id	EntityInserted	EntityLastUpdate	TagTypeParent	TagTypeChild	TagPath	VariableType	StringValue	IntValue	UintValue	BoolValue	Date_Time	HospitalStationDatabaseModelId
1	27	2025-08-21 22:05:56.3157515	2026-03-09 20:50:45.5343398	stConfirmBtn1_1	xBtnState	ns=2;s=[default]/Hospital1/ConfirmBtn1_1/xBtnState	BoolVal	NULL	NULL	NULL	0	NULL	1
2	4	2025-08-21 22:05:56.3148456	2026-03-09 20:56:28.2945687	uiCurrentFunction	NULL	ns=2;s=[default]/Hospital1/uiCurrentFunction	UintVal	NULL	NULL	2	NULL	NULL	1
3	28	2025-08-21 22:05:56.3157842	2026-03-09 20:50:45.5343375	stConfirmBtn1_1	xBtnEna	ns=2;s=[default]/Hospital1/ConfirmBtn1_1/xBtnEna	BoolVal	NULL	NULL	NULL	0	NULL	1
4	6	2025-08-21 22:05:56.3149089	2026-03-09 20:56:31.3117365	sRFID_READ	NULL	ns=2;s=[default]/Hospital1/abRFID_Read	StringVal	00000336	NULL	NULL	NULL	NULL	1
5	56	2025-08-21 22:05:56.3166272	2026-03-09 02:01:41.0189972	stReInductToteBtn1_5	iBtnStatus	ns=2;s=[default]/Hospital1/ReInductToteBtn1_5/iBtn...	IntVal	NULL	0	NULL	NULL	NULL	1
6	24	2025-08-21 22:05:56.3156530	2026-03-09 20:56:12.5843051	stExtract	sStatus2	ns=2;s=[default]/Hospital1/Extract/sStatus2	StringVal	CloseOut	NULL	NULL	NULL	NULL	1
7	8	2025-08-21 22:05:56.3149693	2026-03-09 02:01:49.5580642	sAGV_ID	NULL	ns=2;s=[default]/Hospital1/sAGV_ID	StringVal	Enter AGV ID here	NULL	NULL	NULL	NULL	1
8	20	2025-08-21 22:05:56.3155338	2026-03-09 20:56:29.4754143	stExtract	sAGV_Num	ns=2;s=[default]/Hospital1/Extract/sAGV_Num	StringVal		NULL	NULL	NULL	NULL	1
9	5	2025-08-21 22:05:56.3148789	2026-03-09 20:56:31.3117383	sTOTE_ID	NULL	ns=2;s=[default]/Hospital1/sTOTE_ID	StringVal	336	NULL	NULL	NULL	NULL	1
10	57	2025-08-21 22:05:56.3166595	2026-03-09 02:01:41.0189196	stPrintLabelBtn1_6	xBtnState	ns=2;s=[default]/Hospital1/PrintLabelBtn1_6/xBtnState	BoolVal	NULL	NULL	NULL	0	NULL	1
11	65	2025-08-21 22:05:56.3168954	2026-03-09 02:01:41.0190099	stReInductToteBtn1_6	iBtnStatus	ns=2;s=[default]/Hospital1/ReInductToteBtn1_6/iBtn...	IntVal	NULL	0	NULL	NULL	NULL	1
12	60	2025-08-21 22:05:56.3167503	2026-03-09 02:01:41.0191037	stRemoveToteBtn1_6	xBtnState	ns=2;s=[default]/Hospital1/RemoveToteBtn1_6/xBtn...	BoolVal	NULL	NULL	NULL	0	NULL	1
13	9	2025-08-21 22:05:56.3151779	2026-03-09 02:01:41.0190458	stInduct	xStatusType	ns=2;s=[default]/Hospital1/Induct/xStatusType	BoolVal	NULL	NULL	NULL	0	NULL	1
14	68	2025-08-21 22:05:56.3169855	2026-03-09 02:01:41.0190129	CallMaintBtn1_7	iBtnStatus	ns=2;s=[default]/Hospital1/MaintBtn/iBtnStatus	IntVal	NULL	0	NULL	NULL	NULL	1

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Sorter Sweeper Database

Tipper Station Model

Tipper Stations represent operator stations. As of now we have our standard Stations and our GLD Stations. Each station has 2 tippers.



	Id	EntityInserted	EntityLastUpdate	TipperStationId	TipperLoadType
1	1	2025-08-21 22:05:55.8198704	2025-08-21 22:05:55.8198709	TipperStation1	Standard
2	2	2025-08-21 22:05:56.0414364	2025-08-21 22:05:56.0414368	TipperStation2	Standard
3	3	2025-08-21 22:05:56.0729935	2025-08-21 22:05:56.0729941	TipperStation3	Standard
4	4	2025-08-21 22:05:56.1071506	2025-08-21 22:05:56.1071511	TipperStation4	Standard
5	5	2025-08-21 22:05:56.1372920	2025-08-21 22:05:56.1372922	TipperStation5	Standard
6	6	2025-08-21 22:05:56.1615215	2025-08-21 22:05:56.1615217	TipperStation6	Standard
7	7	2025-08-21 22:05:56.1882565	2025-08-21 22:05:56.1882568	TipperStation7	Standard
8	8	2025-08-21 22:05:56.2328731	2025-08-21 22:05:56.2328734	TipperStation8	Standard
9	9	2025-11-12 08:39:57.1833333	2025-11-12 08:39:57.1833333	GLDStation1	GLD
10	10	2025-11-12 08:39:57.1833333	2025-11-12 08:39:57.1833333	GLDStation2	GLD

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Sorter Sweeper Database

Tipper Model

Tipper model defines all tippers in all stations.

- TipperId: Name of the tipper. We use a b to represent Right Left tipper in an operator station.
- PrinterName, PrinterIp, Printer, are settings we use when sending a PL (Print Label) directive to UPS.
- LocationId: Station in Geek to the tipper
- TipperStatus: if the tipper is Ready, NotReady, or Disabled.

	Id	EntityInserted	EntityLastUpdate	TipperId	PrinterName	PrinterIp	Printer	LocationId	TipperStatus	Disabled	TipperStationDatabaseModelId
1	6	2025-08-21 22:05:56.0766494	2026-03-09 21:03:28.3380132	TipperStation3b	TP01	127.0.0.1	00001	380	Disabled	0	3
2	20	2025-11-12 08:39:57.1933333	2026-03-09 21:01:37.5033171	GLDStation2b	TP01	127.0.0.1	00012	397	NotReady	0	10
3	5	2025-08-21 22:05:56.0732457	2026-03-09 21:03:28.0041951	TipperStation3a	TP01	127.0.0.1	00001	379	Disabled	0	3
4	4	2025-08-21 22:05:56.0433425	2026-03-09 20:58:24.8985763	TipperStation2b	TP01	127.0.0.1	00003	378	Disabled	0	2
5	3	2025-08-21 22:05:56.0419242	2026-03-09 20:58:24.8985747	TipperStation2a	TP01	127.0.0.1	00003	377	Disabled	0	2
6	17	2025-11-12 08:39:57.1933333	2026-03-09 20:59:47.9886164	GLDStation1a	TP01	127.0.0.1	00011	395	NotReady	0	9
7	18	2025-11-12 08:39:57.1933333	2026-03-09 20:59:47.9886138	GLDStation1b	TP01	127.0.0.1	00011	396	NotReady	0	9
8	15	2025-08-21 22:05:56.2330931	2026-02-06 22:44:22.8219706	TipperStation8a	TP01	127.0.0.1	00010	389	Disabled	0	8
9	12	2025-08-21 22:05:56.1628416	2026-03-09 20:56:04.4736150	TipperStation6b	TP01	127.0.0.1	00006	386	Disabled	0	6
10	16	2025-08-21 22:05:56.2347162	2026-02-06 22:44:22.8219731	TipperStation8b	TP01	127.0.0.1	00010	390	Disabled	0	8
11	19	2025-11-12 08:39:57.1933333	2026-03-08 15:19:21.8775719	GLDStation2a	TP01	127.0.0.1	00012	394	NotReady	0	10
12	2	2025-08-21 22:05:55.8737405	2026-03-09 17:12:13.7678535	TipperStation1b	TP01	127.0.0.1	00005	376	Disabled	0	1
13	8	2025-08-21 22:05:56.1095352	2026-03-09 21:04:40.8435605	TipperStation4b	TP01	127.0.0.1	00002	382	Disabled	0	4
14	11	2025-08-21 22:05:56.1616495	2026-03-09 20:56:04.4736137	TipperStation6a	TP01	127.0.0.1	00006	385	Disabled	0	6
15	10	2025-08-21 22:05:56.1384413	2026-03-09 20:56:04.2331572	TipperStation5b	TP01	127.0.0.1	00004	384	Disabled	0	5
16	1	2025-08-21 22:05:55.8438051	2026-03-09 17:12:13.7678512	TipperStation1a	TP01	127.0.0.1	00005	375	Disabled	0	1
17	7	2025-08-21 22:05:56.1073470	2026-03-09 21:04:40.8435591	TipperStation4a	TP01	127.0.0.1	00002	381	Disabled	0	4
18	9	2025-08-21 22:05:56.1374061	2026-03-09 20:56:04.2331520	TipperStation5a	TP01	127.0.0.1	00004	383	Disabled	0	5
19	13	2025-08-21 22:05:56.1883674	2026-02-06 22:43:57.8076844	TipperStation7a	TP01	127.0.0.1	00009	387	Disabled	0	7
20	14	2025-08-21 22:05:56.1900659	2026-02-06 22:43:57.8076864	TipperStation7b	TP01	127.0.0.1	00009	388	Disabled	0	7

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Sorter Sweeper Database

Tag Model

Every tipper has 12 tags that we subscribe to. 7 in and 5 out. OPTISweep Software receives messages from the tipper on the out Tags and write to the In Tags.

- TagType: Defines the Tag and is used to process the action.
- Bit: indicates if that tag should be broken up into bits when received
- Value: Shows the Current Value
- Flag: Is used to track when the tipper went into cycle.

TagType	TagPath	Bit	DefaultValue	Value	StringValue	Flag	TipperDatabaseModelId
InRFid	ns=2;s=[default]/TipperStation1a/In/abRFID	0	NULL	NULL		0	1
InDiRFid	ns=2;s=[default]/TipperStation1a/In/diRFID	0	NULL	0	NULL	0	1
InSortStatus	ns=2;s=[default]/TipperStation1a/In/wSortStatus	0	NULL	0	NULL	0	1
InHeartBeat	ns=2;s=[default]/TipperStation1a/In/iHeartBeat	0	NULL	1368	NULL	0	1
InCommand	ns=2;s=[default]/TipperStation1a/In/wCommand	1	NULL	NULL	NULL	0	1
InCommand2	ns=2;s=[default]/TipperStation1a/In/wCommand2	1	NULL	NULL	NULL	0	1
RmsStatus	ns=2;s=[default]/TipperStation1a/In/iRMS_Status	0	NULL	0	NULL	0	1
OutFault	ns=2;s=[default]/TipperStation1a/Out/wFaultCode	0	NULL	0	NULL	0	1
OutRfid	ns=2;s=[default]/TipperStation1a/Out/abRFID	0	NULL	NULL		0	1
OutState	ns=2;s=[default]/TipperStation1a/Out/wState	0	NULL	30	NULL	0	1
OutHeartBeat	ns=2;s=[default]/TipperStation1a/Out/iHeartBeat	0	NULL	1368	NULL	0	1
OutCommand	ns=2;s=[default]/TipperStation1a/Out/wCommand	1	NULL	NULL	NULL	0	1

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Sorter Sweeper Database

Bits Model

For Tags that have bits that need to be converted.

- BitType: defines the function.
- Bit: defines the bit to look for
- Value: shows if its currently a 0 (NULL) or a 1.
- DefaultValue: is used for tags we write to the tipper to reset the state.

BitType	Bit	DefaultValue	Value	TagDatabaseModelId
TipperNotInEstop	0	NULL	NULL	12
TipperInAutoCycle	1	NULL	NULL	12
TipperReadyForAgv	2	NULL	NULL	12
RfidTagReadSuccess	3	NULL	NULL	12
RfidTagMatch	4	NULL	NULL	12
TipperStopped	5	NULL	NULL	12
TipperFaulted	8	NULL	NULL	12
TipperPaused	10	NULL	1	12
NextTipSide	11	NULL	NULL	12
SendToHospital	15	NULL	NULL	12

BitType	Bit	DefaultValue	Value	TagDatabaseModelId
SystemNotInEstop	0	1	1	5
SystemInCycle	1	1	1	5
AgvInLocation	2	0	NULL	5
SystemFreehold	5	0	NULL	5
SystemContinue	6	0	NULL	5
SystemReset	7	0	NULL	5

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Sorter Sweeper Database

Tipper/Tag Logging

Tag Read and Tag Write logs all tags the systems reads and writes.

TagPath	Value
ns=2;s=[default]/GLDStation1b/Out/iHeartBeat	-28030
ns=2;s=[default]/GLDStation1b/In/wSortStatus	1
ns=2;s=[default]/GLDStation1b/In/abRFID	{0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0...
ns=2;s=[default]/GLDStation1b/In/diRFID	0
ns=2;s=[default]/GLDStation1b/In/iRMS_Status	0
ns=2;s=[default]/GLDStation1b/In/wCommand2	0
ns=2;s=[default]/TipperStation1a/Out/wCommand	3072
ns=2;s=[default]/GLDStation1b/In/wCommand	3
ns=2;s=[default]/TipperStation1a/Out/wState	30
ns=2;s=[default]/TipperStation1a/Out/wFaultCode	0
ns=2;s=[default]/GLDStation1b/Out/wFaultCode	0
ns=2;s=[default]/TipperStation1a/Out/abRFID	{0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0...
ns=2;s=[default]/TipperStation4b/Out/abRFID	{0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0...
ns=2;s=[default]/TipperStation6a/Out/wFaultCode	0
ns=2;s=[default]/TipperStation1a/Out/iHeartBeat	27748
ns=2;s=[default]/TipperStation1a/In/wSortStatus	1
ns=2;s=[default]/TipperStation1a/In/diRFID	0
ns=2;s=[default]/TipperStation1a/In/abRFID	{0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0...
ns=2;s=[default]/TipperStation1a/In/iRMS_Status	0
ns=2;s=[default]/GLDStation1b/Out/abRFID	{0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0...
ns=2;s=[default]/TipperStation1a/In/wCommand2	0
ns=2;s=[default]/TipperStation1a/In/wCommand	3
ns=2;s=[default]/TipperStation1b/Out/wCommand	1024
ns=2;s=[default]/TipperStation1b/Out/wState	30
ns=2;s=[default]/TipperStation1b/Out/abRFID	{0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0...
ns=2;s=[default]/TipperStation1b/Out/wFaultCode	0
ns=2;s=[default]/TipperStation1b/Out/iHeartBeat	27748
ns=2;s=[default]/GLDStation1b/Out/wState	30
ns=2;s=[default]/TipperStation1b/In/wSortStatus	1
ns=2;s=[default]/TipperStation1b/In/diRFID	0
ns=2;s=[default]/TipperStation1b/In/abRFID	{0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0...
ns=2;s=[default]/TipperStation1b/In/iRMS_Status	0
ns=2;s=[default]/TipperStation1b/In/wCommand2	0

TagPath	Value
ns=2;s=[default]/TipperStation1a/In/wSortStatus	1
ns=2;s=[default]/TipperStation1b/In/wSortStatus	1
ns=2;s=[default]/TipperStation2a/In/wSortStatus	1
ns=2;s=[default]/TipperStation2b/In/wSortStatus	1
ns=2;s=[default]/TipperStation3a/In/wSortStatus	1
ns=2;s=[default]/TipperStation3b/In/wSortStatus	1
ns=2;s=[default]/TipperStation4a/In/wSortStatus	1
ns=2;s=[default]/TipperStation4b/In/wSortStatus	1
ns=2;s=[default]/TipperStation5a/In/wSortStatus	1
ns=2;s=[default]/TipperStation5b/In/wSortStatus	1
ns=2;s=[default]/TipperStation6a/In/wSortStatus	1
ns=2;s=[default]/TipperStation6b/In/wSortStatus	1
ns=2;s=[default]/TipperStation7a/In/wSortStatus	1
ns=2;s=[default]/TipperStation7b/In/wSortStatus	1
ns=2;s=[default]/TipperStation8a/In/wSortStatus	1
ns=2;s=[default]/TipperStation8b/In/wSortStatus	1
ns=2;s=[default]/GLDStation1a/In/wSortStatus	1
ns=2;s=[default]/GLDStation1b/In/wSortStatus	1
ns=2;s=[default]/GLDStation2a/In/wSortStatus	1
ns=2;s=[default]/GLDStation2b/In/wSortStatus	1
ns=2;s=[default]/TipperStation1a/In/iHeartBeat	27748
ns=2;s=[default]/GLDStation1b/In/iHeartBeat	-28030
ns=2;s=[default]/TipperStation1b/In/iHeartBeat	27748
ns=2;s=[default]/TipperStation2a/In/iHeartBeat	5349
ns=2;s=[default]/TipperStation2b/In/iHeartBeat	5349
ns=2;s=[default]/TipperStation3a/In/iHeartBeat	-21563
ns=2;s=[default]/TipperStation3b/In/iHeartBeat	-21563
ns=2;s=[default]/GLDStation2a/In/iHeartBeat	-32456
ns=2;s=[default]/GLDStation1a/In/iHeartBeat	-28030
ns=2;s=[default]/TipperStation8a/In/iHeartBeat	-24479
ns=2;s=[default]/TipperStation5b/In/iHeartBeat	-22042
ns=2;s=[default]/TipperStation6a/In/iHeartBeat	-18498
ns=2;s=[default]/TipperStation6b/In/iHeartBeat	-18498
ns=2;s=[default]/TipperStation7a/In/iHeartBeat	27791

Tipper Departure Activity Model records every time an AGV is sent away from the tipper. DropOffSorter or DropOffHospital, the Tote, the Tag path and any fault code are recorded. This can be helpful in narrowing down tipper issues.

TaskName	ToteId	FaultTagPath	FaultValue
DropOffSorter	70	ns=2;s=[default]/TipperStation4b/Out/wFaultCode	0
DropOffSorter	299	ns=2;s=[default]/TipperStation4a/Out/wFaultCode	0
DropOffSorter	115	ns=2;s=[default]/TipperStation3b/Out/wFaultCode	0
DropOffSorter	128	ns=2;s=[default]/TipperStation6b/Out/wFaultCode	0
DropOffSorter	89	ns=2;s=[default]/TipperStation1b/Out/wFaultCode	0
DropOffSorter	259	ns=2;s=[default]/TipperStation6a/Out/wFaultCode	0
DropOffSorter	71	ns=2;s=[default]/TipperStation3a/Out/wFaultCode	0
DropOffSorter	315	ns=2;s=[default]/TipperStation2a/Out/wFaultCode	0
DropOffSorter	141	ns=2;s=[default]/TipperStation1a/Out/wFaultCode	0
DropOffSorter	41	ns=2;s=[default]/TipperStation5a/Out/wFaultCode	0
DropOffSorter	132	ns=2;s=[default]/TipperStation5b/Out/wFaultCode	0
DropOffSorter	466	ns=2;s=[default]/TipperStation3b/Out/wFaultCode	0
DropOffSorter	418	ns=2;s=[default]/TipperStation2b/Out/wFaultCode	0
DropOffSorter	459	ns=2;s=[default]/TipperStation4b/Out/wFaultCode	0
DropOffSorter	438	ns=2;s=[default]/TipperStation4a/Out/wFaultCode	0
DropOffSorter	64	ns=2;s=[default]/TipperStation3a/Out/wFaultCode	0
DropOffSorter	468	ns=2;s=[default]/TipperStation5a/Out/wFaultCode	0
DropOffSorter	49	ns=2;s=[default]/TipperStation5b/Out/wFaultCode	0
DropOffSorter	91	ns=2;s=[default]/GLDStation1b/Out/wFaultCode	0
DropOffSorter	43	ns=2;s=[default]/GLDStation1a/Out/wFaultCode	0
DropOffSorter	133	ns=2;s=[default]/TipperStation4b/Out/wFaultCode	0
DropOffSorter	349	ns=2;s=[default]/TipperStation4a/Out/wFaultCode	0
DropOffSorter	279	ns=2;s=[default]/TipperStation3b/Out/wFaultCode	0
DropOffSorter	345	ns=2;s=[default]/GLDStation1b/Out/wFaultCode	0
DropOffSorter	281	ns=2;s=[default]/TipperStation2a/Out/wFaultCode	0
DropOffSorter	16	ns=2;s=[default]/TipperStation3a/Out/wFaultCode	0
DropOffSorter	406	ns=2;s=[default]/TipperStation4b/Out/wFaultCode	0
DropOffHospital	152	ns=2;s=[default]/GLDStation2a/Out/wFaultCode	45
DropOffSorter	286	ns=2;s=[default]/TipperStation3b/Out/wFaultCode	0
DropOffSorter	185	ns=2;s=[default]/GLDStation1a/Out/wFaultCode	0
DropOffHospital	40	ns=2;s=[default]/TipperStation5a/Out/wFaultCode	40
DropOffSorter	90	ns=2;s=[default]/GLDStation2b/Out/wFaultCode	0
DropOffSorter	76	ns=2;s=[default]/TipperStation4a/Out/wFaultCode	0

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Sorter Sweeper Database

Item Models

Items are parcels on the sorter. We listen to Scan_Data2, Sort, and SortConf from UPS. We update the items as we receive messages and remove them when they divert to their final destination.

Scan_Data2 provides us with PackageUID, Tracking number, Length, Width, Height, Weight, and Package Classification.

Item Cube Adjustment Model stores the % of the cube we should take based on packages classification.

Item Default Dim Model stores the default dimensions when one is missing.

Recirc Destinations Model stores the destinations that are for packages that go off the end.

PackageUid	TrackingNumber	Length	Width	Height	Weight	Cube	PackageClassification	ReCirc	FinalDestination	MissingDimensionData	MissingInduct
000017187_AI01Chat	1Z51R5R19091691062	9	4.4	3.2	0.6	152.064	1	0	NULL	0	0
000016148_AI01Chat	1Z0F57320342480506	10	5.2	3.8	6.6	237.12	1	0	NULL	0	0
000016410_AI01Chat	1Z443412YW12231960	15.2	10.4	5.8	4	1100.2368	1	0	NULL	0	0
000013439_AI02Chat	1ZAC91870306444263	11.4	13.2	3	1.9	203.148	2	0	NULL	0	0
000013434_AI02Chat	1Z5369780399640489	12.2	6.2	5	0.8	453.84	1	0	NULL	0	0
000016392_AI01Chat	1Z0JJ4170304837956	14.6	11.2	6.2	5.3	1216.5888	1	0	NULL	0	0
000013770_AI02Chat	1Z247V0W0358614171	14.4	11.2	6.4	4.2	1238.6304	1	0	NULL	0	0
000016481_AI01Chat	1ZY0G8680335782728	5.2	11.6	5.4	2	390.8736	1	0	NULL	0	0
000017148_AI01Chat	1ZK118J99070156696	5.2	8.2	4.4	0.8	225.1392	1	0	NULL	0	0
000015351_AI01Chat	NULL	12.4	15	4.8	2.2	401.76	2	0	NULL	0	0
000015577_AI01Chat	1Z7749880354608215	10.6	13.6	3.2	1.8	553.5744	1	0	NULL	0	0
000015376_AI01Chat	1Z5369780395762908	12	6.2	4.8	0.9	428.544	1	0	NULL	0	0
000017613_AI01Chat	1Z7F08560377030676	10.6	13.6	3.2	1	553.5744	1	0	NULL	0	0

We track how often the item recirc, If it is missing some dimension data and if we never received an Induction message.

PackageClassification	PercentageOfCube
1	1.2
2	0.45
0	1
4	1

Length	Width	Height	Weight
12	12	1	1

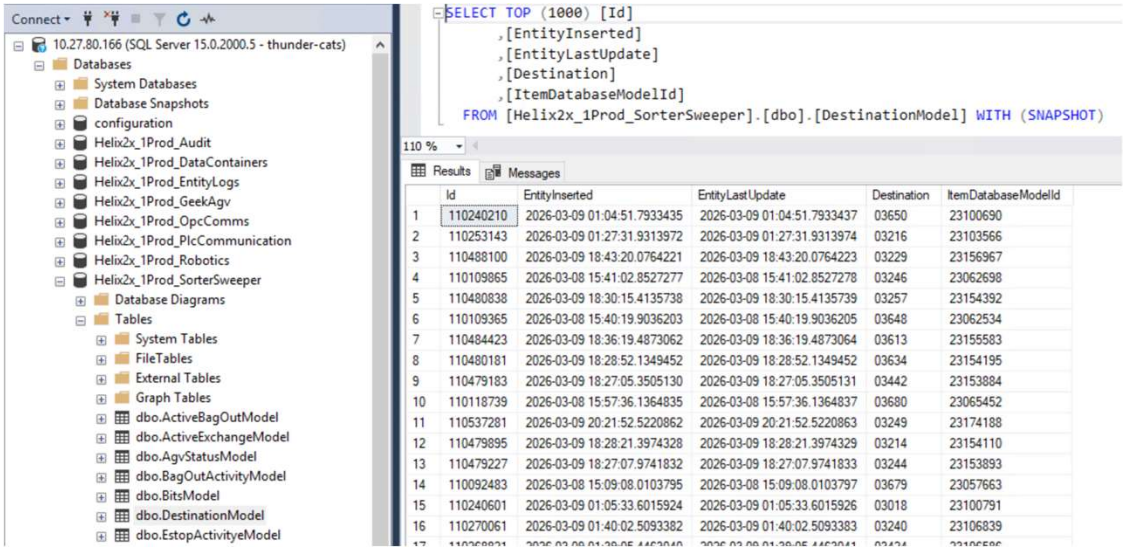
Name
01999
02999

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Sorter Sweeper Database

Destination Model

When we receive a sort message we receive a list of possible destination for the parcel. We attach that to the item.



	Id	EntityInserted	EntityLastUpdate	Destination	ItemDatabaseModelId
1	110240210	2026-03-09 01:04:51.7933435	2026-03-09 01:04:51.7933437	03650	23100690
2	110253143	2026-03-09 01:27:31.9313972	2026-03-09 01:27:31.9313974	03216	23103566
3	110488100	2026-03-09 18:43:20.0764221	2026-03-09 18:43:20.0764223	03229	23156967
4	110109865	2026-03-08 15:41:02.8527277	2026-03-08 15:41:02.8527278	03246	23062698
5	110480838	2026-03-09 18:30:15.4135738	2026-03-09 18:30:15.4135739	03257	23154392
6	110109365	2026-03-08 15:40:19.9036203	2026-03-08 15:40:19.9036205	03648	23062534
7	110484423	2026-03-09 18:36:19.4873062	2026-03-09 18:36:19.4873064	03613	23155583
8	110480181	2026-03-09 18:28:52.1349452	2026-03-09 18:28:52.1349452	03634	23154195
9	110479183	2026-03-09 18:27:05.3505130	2026-03-09 18:27:05.3505131	03442	23153884
10	110118739	2026-03-08 15:57:36.1364835	2026-03-08 15:57:36.1364837	03680	23065452
11	110537281	2026-03-09 20:21:52.5220862	2026-03-09 20:21:52.5220863	03249	23174188
12	110479895	2026-03-09 18:28:21.3974328	2026-03-09 18:28:21.3974329	03214	23154110
13	110479227	2026-03-09 18:27:07.9741832	2026-03-09 18:27:07.9741833	03244	23153893
14	110092483	2026-03-08 15:09:08.0103795	2026-03-08 15:09:08.0103797	03679	23057663
15	110240601	2026-03-09 01:05:33.6015924	2026-03-09 01:05:33.6015926	03018	23100791
16	110270061	2026-03-09 01:40:02.5093382	2026-03-09 01:40:02.5093383	03240	23106839

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Sorter Sweeper Database

Ignition Lane Activity Model

Igntion lane activity tracks all Scandata, Induct(sort), and SortConf message that fall within our working area.

SorterName	SortName	SortGuid	Zone	LaneName	PackageUid	TrackingNumber	Length	Width	Height	Weight	Cube	ToteId	Message	PackageClassification	MissingDimensionData	MissingInduct	ToteNotFound	Diverted	OffTheEnd	Recirc
NULL	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	0	NULL	000018120_AI01Chat	1ZY0G8680321998058	6.4	13	6.6	3.4	658.944	NULL	SortConf	1	0	0	1	1	0	10
SorterSide2a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	4	03645	000019020_AI01Chat	1ZB37G650366554812	10.2	12.2	3.4	0.6	507.7152	301	SortConf	1	0	0	0	1	0	1
SorterSide1b	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	2	03424	000019134_AI01Chat	1ZF20J029063934129	3	6.4	1.4	1	12.096	NULL	Induct	2	1	0	0	0	0	0
SorterSide2a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	6	03617	000019056_AI01Chat	1Z4528EV0327337339	7.8	7	6.4	0.7	157.248	459	SortConf	2	0	0	0	1	0	1
SorterSide1b	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	1	03406	000019124_AI01Chat	1ZEF89240394560200	14.6	13.6	5.6	4.4	1334.3232	260	SortConf	1	0	0	0	1	0	0
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	1	03006	000018863_AI01Chat	1ZE9A4070391439551	15.2	12	1	0.4	182.4	342	SortConf	0	1	0	0	1	0	0
NULL	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	0	NULL	000018992_AI01Chat	1Z05859BYN00497799	14.4	8.8	4	0.8	228.096	NULL	SortConf	2	0	0	1	1	0	1
SorterSide1b	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	1	03409	000019126_AI01Chat	1Z0JJ4170304855132	13	11.4	3.6	1.2	640.224	186	SortConf	1	0	0	0	1	0	0
NULL	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	0	NULL	000018131_AI01Chat	1Z0JJ4170304939800	11.4	13.2	3.8	3.1	686.1888	NULL	SortConf	1	0	0	1	1	0	10
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	5	03055	000018422_AI01Chat	1Z00VR930354880350	10	11	6.4	3.5	844.8	314	SortConf	1	0	0	0	1	0	6
SorterSide2a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	6	03617	000019067_AI01Chat	1ZV54B999010144721	17.8	12	3.2	1.1	307.584	459	SortConf	2	0	0	0	1	0	1
SorterSide1b	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	1	03409	000011333_AI02Chat	1Z0JJ4170304875209	10	7.4	6.4	1.8	568.32	186	SortConf	1	0	0	0	1	0	0
NULL	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	0	NULL	000018801_AI01Chat	1Z00VR930354880341	10	10.8	6.4	3.4	829.44	NULL	SortConf	1	0	0	1	1	0	4
NULL	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	NULL	NULL	000019136_AI01Chat	NULL	15.8	12.2	7.2	4.4	1665.4464	NULL	ScanData	1	0	0	0	0	0	0
SorterSide2a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	5	03636	000019136_AI01Chat	1Z247V0W0358597646	15.8	12.2	7.2	4.4	1665.4464	NULL	Induct	1	0	0	0	0	0	0
SorterSide2a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	6	03617	000019069_AI01Chat	1ZY228K7YN43395157	5.8	8.6	1.6	0.4	79.808	459	SortConf	4	0	0	0	1	0	1
NULL	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	NULL	NULL	000019137_AI01Chat	NULL	14.6	11.2	6.2	6.7	1216.5888	NULL	ScanData	1	0	0	0	0	0	0
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	4	03044	000019137_AI01Chat	1Z0JJ417YW04918647	14.6	11.2	6.2	6.7	1216.5888	NULL	Induct	1	0	0	0	0	0	0
NULL	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	0	NULL	000011292_AI02Chat	1Z247V0W0358609883	14.6	12	6.2	3.8	488.808	NULL	SortConf	2	0	0	1	1	0	0
NULL	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	0	NULL	000018802_AI01Chat	1Z00VR930354880038	10.6	9.2	6.4	3.2	748.9536	NULL	SortConf	1	0	0	1	1	0	4
NULL	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	NULL	NULL	000019140_AI01Chat	NULL	4.2	9.2	3.2	0.4	55.6416	NULL	ScanData	2	0	0	0	0	0	0
SorterSide2a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	5	03639	000019140_AI01Chat	1ZGH2280YW17507830	4.2	9.2	3.2	0.4	55.6416	NULL	Induct	2	0	0	0	0	0	0
NULL	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	0	NULL	000019059_AI01Chat	1Z02C4E70330088823	8	5	1.8	0.3	72	NULL	SortConf	4	0	0	1	1	0	1
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	4	03050	000019120_AI01Chat	1Z036E5H0321933783	12.2	14.4	3	3.2	632.448	41	SortConf	1	0	0	0	1	0	0
NULL	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	NULL	NULL	000019141_AI01Chat	NULL	9.4	14	2.6	0.3	153.972	NULL	ScanData	2	0	0	0	0	0	0
NULL	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	NULL	NULL	000019142_AI01Chat	NULL	15.6	12.4	1.2	1.4	104.4576	NULL	ScanData	2	0	0	0	0	0	0
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	1	03002	000019142_AI01Chat	1Z0699430326474401	15.6	12.4	1.2	1.4	104.4576	NULL	Induct	2	0	0	0	0	0	0
NULL	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	NULL	NULL	000019143_AI01Chat	NULL	18	8.4	0.8	0.2	54.432	NULL	ScanData	2	0	0	0	0	0	0
SorterSide1b	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	4	03448	000019143_AI01Chat	1ZY0G8680318263348	18	8.4	0.8	0.2	54.432	NULL	Induct	2	0	0	0	0	0	0

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Sorter Sweeper Database

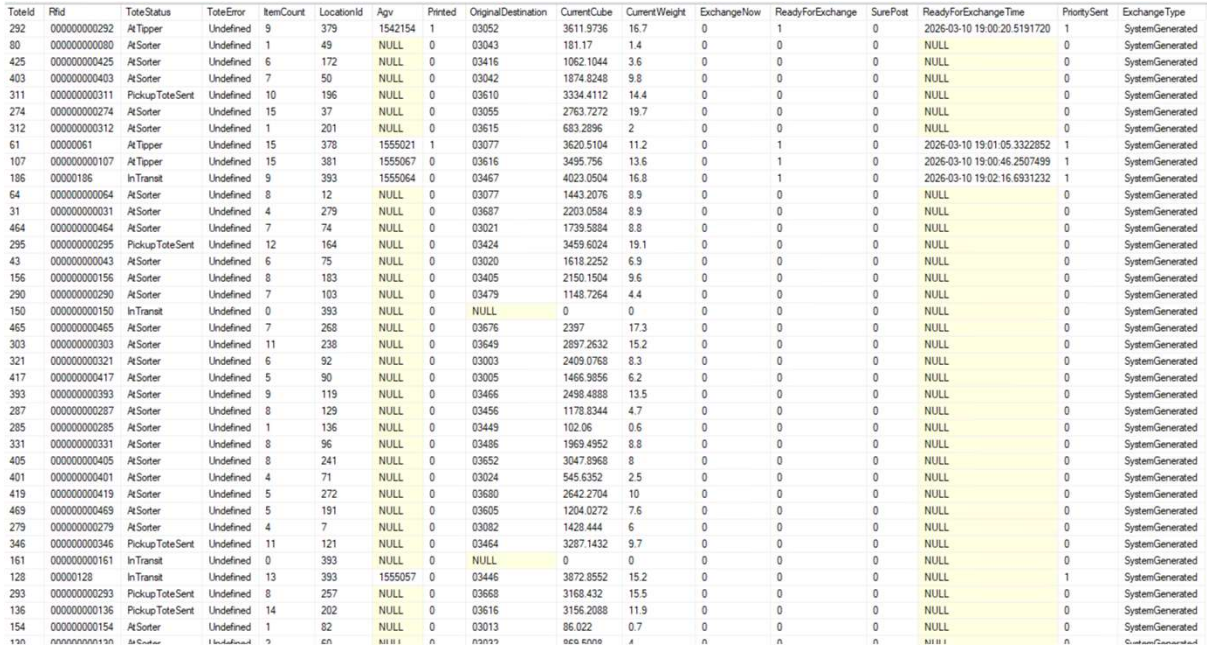
Tote Model

Tote Status

- InTransit
- AtTipper
- AtSorter
- PickupToteSent
- Removed
- AtSorterLifted
- AtInduction
- AtHospital

ExchangeType

- SystemGenerated
- MissSort
- HotBin
- HotBinCloseOut
- HotBinDisabled
- CloseOut
- Purge
- UserSent
- CloseOutLowest ExchangedLanes
- LaneCloseout
- SorterSideCloseout



ToteId	Rfid	ToteStatus	ToteError	ItemCount	LocationId	Agv	Printed	OriginalDestination	CurrentCube	CurrentWeight	ExchangeNow	ReadyForExchange	SurePost	ReadyForExchangeTime	PriorityySent	ExchangeType
292	000000000292	AtTipper	Undefined	9	379	1542154	1	03052	3611.9736	16.7	0	1	0	2026-03-10 19:00:20.5191720	1	SystemGenerated
80	000000000080	AtSorter	Undefined	1	49	NULL	0	03043	181.17	1.4	0	0	0	NULL	0	SystemGenerated
425	000000000425	AtSorter	Undefined	6	172	NULL	0	03416	1062.1044	3.6	0	0	0	NULL	0	SystemGenerated
403	000000000403	AtSorter	Undefined	7	50	NULL	0	03042	1874.8248	9.8	0	0	0	NULL	0	SystemGenerated
311	000000000311	PickupToteSent	Undefined	10	196	NULL	0	03610	3334.4112	14.4	0	0	0	NULL	0	SystemGenerated
274	000000000274	AtSorter	Undefined	15	37	NULL	0	03055	2763.7272	19.7	0	0	0	NULL	0	SystemGenerated
312	000000000312	AtSorter	Undefined	1	201	NULL	0	03615	683.2896	2	0	0	0	NULL	0	SystemGenerated
61	00000061	AtTipper	Undefined	15	378	1555021	1	03077	3620.5104	11.2	0	1	0	2026-03-10 19:01:05.3322852	1	SystemGenerated
107	000000000107	AtTipper	Undefined	15	381	1555067	0	03616	3495.756	13.6	0	1	0	2026-03-10 19:00:46.2507499	1	SystemGenerated
186	00000186	InTransit	Undefined	9	393	1555064	0	03467	4023.0504	16.8	0	1	0	2026-03-10 19:02:16.6931232	1	SystemGenerated
64	000000000064	AtSorter	Undefined	8	12	NULL	0	03077	1443.2076	8.9	0	0	0	NULL	0	SystemGenerated
31	000000000031	AtSorter	Undefined	4	279	NULL	0	03687	2203.0584	8.9	0	0	0	NULL	0	SystemGenerated
464	000000000464	AtSorter	Undefined	7	74	NULL	0	03021	1739.5884	8.8	0	0	0	NULL	0	SystemGenerated
295	000000000295	PickupToteSent	Undefined	12	164	NULL	0	03424	3459.6024	19.1	0	0	0	NULL	0	SystemGenerated
43	000000000043	AtSorter	Undefined	6	75	NULL	0	03020	1618.2252	6.9	0	0	0	NULL	0	SystemGenerated
156	000000000156	AtSorter	Undefined	8	183	NULL	0	03405	2150.1504	9.6	0	0	0	NULL	0	SystemGenerated
290	000000000290	AtSorter	Undefined	7	103	NULL	0	03479	1148.7264	4.4	0	0	0	NULL	0	SystemGenerated
150	000000000150	InTransit	Undefined	0	393	NULL	0	NULL	0	0	0	0	0	NULL	0	SystemGenerated
465	000000000465	AtSorter	Undefined	7	268	NULL	0	03676	2397	17.3	0	0	0	NULL	0	SystemGenerated
303	000000000303	AtSorter	Undefined	11	238	NULL	0	03649	2897.2632	15.2	0	0	0	NULL	0	SystemGenerated
321	000000000321	AtSorter	Undefined	6	92	NULL	0	03003	2409.0768	8.3	0	0	0	NULL	0	SystemGenerated
417	000000000417	AtSorter	Undefined	5	90	NULL	0	03005	1466.9856	6.2	0	0	0	NULL	0	SystemGenerated
393	000000000393	AtSorter	Undefined	9	119	NULL	0	03466	2498.4888	13.5	0	0	0	NULL	0	SystemGenerated
287	000000000287	AtSorter	Undefined	8	129	NULL	0	03456	1178.8344	4.7	0	0	0	NULL	0	SystemGenerated
285	000000000285	AtSorter	Undefined	1	136	NULL	0	03449	102.06	0.6	0	0	0	NULL	0	SystemGenerated
331	000000000331	AtSorter	Undefined	8	96	NULL	0	03486	1969.4952	8.8	0	0	0	NULL	0	SystemGenerated
405	000000000405	AtSorter	Undefined	8	241	NULL	0	03652	3047.8968	8	0	0	0	NULL	0	SystemGenerated
401	000000000401	AtSorter	Undefined	4	71	NULL	0	03024	545.6352	2.5	0	0	0	NULL	0	SystemGenerated
419	000000000419	AtSorter	Undefined	5	272	NULL	0	03680	2642.2704	10	0	0	0	NULL	0	SystemGenerated
469	000000000469	AtSorter	Undefined	5	191	NULL	0	03605	1204.0272	7.6	0	0	0	NULL	0	SystemGenerated
279	000000000279	AtSorter	Undefined	4	7	NULL	0	03082	1428.444	6	0	0	0	NULL	0	SystemGenerated
346	000000000346	PickupToteSent	Undefined	11	121	NULL	0	03464	3287.1432	9.7	0	0	0	NULL	0	SystemGenerated
161	000000000161	InTransit	Undefined	0	393	NULL	0	NULL	0	0	0	0	0	NULL	0	SystemGenerated
128	00000128	InTransit	Undefined	13	393	1555057	0	03446	3872.8552	15.2	0	0	0	NULL	1	SystemGenerated
293	000000000293	PickupToteSent	Undefined	8	257	NULL	0	03668	3168.432	15.5	0	0	0	NULL	0	SystemGenerated
136	000000000136	PickupToteSent	Undefined	14	202	NULL	0	03616	3156.2088	11.9	0	0	0	NULL	0	SystemGenerated
154	000000000154	AtSorter	Undefined	1	82	NULL	0	03013	86.022	0.7	0	0	0	NULL	0	SystemGenerated

Tote Error

- NoRead
- MissMatch
- AgvRemoved
- AgvCalled
- FlaggedByBagger
- TipperDisabled

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FORTNA

Sorter Sweeper Database

Tote Config Model

ToteTypeName	Length	Width	Height	MaxWeight	WeightThreshold	Cube	CubeThrosholdPickUp	CubeThrosholdPriority	CubeThrosholdClose	CountPickup	CountPriority	CountClose	Method
21x29x14	18	13	37	70	0.9	6926	0.45	0.5	0.55	11	14	15	Cube

Tote config determines what settings are used to trigger Pickups, Priority, or Close chute.

Method

- Cube
- Count
- CubeAndCount

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Sorter Sweeper Database

Exhange

When a chute is closed an active exchange is created in the ActiveExchangeModel.

SorterSideName	SortName	SortGuid	Zone	LaneName	ToteName	ExchangeType	TotalSorted	VolumetricFullPercentage	WeightFullPercentage	StartTime
SorterSide1b	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	5	03459	156	SystemGenerated	19	56.5569448455097	33.7142857142857	2026-03-09 19:05:46.4504603
SorterSide2a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	5	03636	412	SystemGenerated	9	56.3201963615362	22.7142857142857	2026-03-09 19:08:09.5559106
SorterSide2a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	5	03629	408	SystemGenerated	7	58.2730811435172	17.4285714285714	2026-03-09 19:07:16.8727963
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	4	03042	91	SystemGenerated	14	62.2397343343921	31.4285714285714	2026-03-09 19:08:06.2467262

When a empty tote replaces it, the exchange is closed and archived in Exchange activity

SorterSideName	SortName	SortGuid	Zone	LaneName	ToteName	ExchangeType	TotalSorted	VolumetricFullPercentage	WeightFullPercentage	StartTime	FinishTime	ExchangeTime
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	6	03068	418	SystemGenerated	6	57.6151949177014	33.2857142857143	2026-03-09 19:07:50.6787223	2026-03-09 19:08:15.0106790	00:00:24.3319568
SorterSide2a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	5	03636	412	SystemGenerated	9	56.3201963615362	22.7142857142857	2026-03-09 19:08:09.5559106	2026-03-09 19:09:19.5589881	00:01:10.0030776
SorterSide1b	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	5	03457	420	SystemGenerated	18	64.1419462893445	30.7142857142857	2026-03-09 19:08:07.3970069	2026-03-09 19:09:21.6948564	00:01:14.2978496
SorterSide1b	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	1	03403	446	SystemGenerated	12	52.7369852728848	21.7142857142857	2026-03-09 19:09:15.7127663	2026-03-09 19:09:25.6307059	00:00:09.9179397
SorterSide2a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	7	03609	125	SystemGenerated	7	48.646347097892	25.8571428571429	2026-03-09 19:09:30.3499812	2026-03-09 19:09:40.3266593	00:00:09.9766782
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	2	03020	430	SystemGenerated	15	51.3962056020791	22.4285714285714	2026-03-09 19:09:40.5337184	2026-03-09 19:09:51.0341288	00:00:10.5004104
SorterSide2a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	6	03620	80	SystemGenerated	8	45.2489055732024	20.7142857142857	2026-03-09 19:09:40.5337437	2026-03-09 19:09:51.0340645	00:00:10.5003209
SorterSide1b	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	7	03479	58	SystemGenerated	13	59.0764885937049	23.5714285714286	2026-03-09 19:08:10.4785751	2026-03-09 19:10:01.9207113	00:01:51.4421362
SorterSide1b	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	6	03464	57	SystemGenerated	21	57.2451978053711	22.4285714285714	2026-03-09 19:09:15.7127652	2026-03-09 19:10:03.7192757	00:00:48.0065106

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Sorter Sweeper Database

Active Bag Out Model

When a BagOut/CloseOut is requested all open chutes are closed. Any tote with an item in it is marked for exchange and similar to the exchange tables, an entry is made in Active Bag out. The difference is it closes the bagout once the tote has been tipped. The record is archived to bagout activity

SorterSideName	SortName	SortGuid	Zone	LaneName	ToteName	ExchangeType
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	7	03080	32	CloseOut
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	7	03077	129	CloseOut
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	7	03076	296	CloseOut
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	6	03068	467	CloseOut
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	6	03066	39	CloseOut
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	6	03065	282	CloseOut
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	5	03054	315	CloseOut
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	5	03053	430	CloseOut
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	4	03040	149	CloseOut
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	3	03029	74	CloseOut
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	2	03017	187	CloseOut
SorterSide1a	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	2	03016	464	CloseOut
SorterSide1b	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	7	03481	25	CloseOut
SorterSide1b	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	7	03480	71	CloseOut
SorterSide1b	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	7	03477	277	CloseOut
SorterSide1b	NewSort3/9/2026 5:04:39 PM	d1419b69-e68f-4d30-b9c8-0b8e1da65419	6	03467	306	CloseOut

SorterSideName	SortName	SortGuid	Zone	LaneName	ToteName	StartTime	FinishTime	TotalTime	Exchang
SorterSide2a	NewSort3/7/2026 9:07:14 AM	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	1	03676	412	2026-03-07 13:52:07.1612446	2026-03-07 13:53:19.5857576	00:01:12.4245130	CloseOu
SorterSide2a	NewSort3/7/2026 9:07:14 AM	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2	03664	131	2026-03-07 13:52:07.0356400	2026-03-07 13:53:33.9232454	00:01:26.8876054	CloseOu
SorterSide2a	NewSort3/7/2026 9:07:14 AM	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	4	03642	320	2026-03-07 13:52:06.8122332	2026-03-07 13:53:36.2750573	00:01:29.4628241	CloseOu
SorterSide1b	NewSort3/7/2026 9:07:14 AM	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	7	03487	457	2026-03-07 13:52:05.4901665	2026-03-07 13:53:45.1661143	00:01:39.6759478	CloseOu
SorterSide1a	NewSort3/7/2026 9:07:14 AM	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	1	03014	27	2026-03-07 13:52:05.3451459	2026-03-07 13:53:46.8950060	00:01:41.5498601	CloseOu
SorterSide1a	NewSort3/7/2026 9:07:14 AM	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	3	03039	89	2026-03-07 13:52:05.0782364	2026-03-07 13:53:58.4474379	00:01:53.3692015	CloseOu
SorterSide2a	NewSort3/7/2026 9:07:14 AM	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	7	03603	465	2026-03-07 13:52:06.4261954	2026-03-07 13:54:00.6520309	00:01:54.2258355	CloseOu

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Sorter Sweeper Database

AGV Status Model

Data from Geek on the AGVs are sent to us. We use the information to notify tippers if there is an RMS Fault, notify the hospital if an AGV is waiting for a tote, and send a tote to the hospital if the AGV is called to maintenance.

ToteId	Status	Task	ErrorCode	Error	ErrorCause	Coordinate	Location	LocationCellCode	Angle	Battery	Current	Voltage	Temperature	Ip	Name	CallToMaint
	DISCONNECTED			NULL	NULL	RoboticsModule.Messages.Location	NULL	17001020	-89.98	63	0	57200	32	10.27.80.93	1555047	0
51	NORMAL		12003,12011	NULL	NULL	RoboticsModule.Messages.Location	NULL		100.09	67	310	54600	28	10.27.80.92	1555071	0
	NORMAL		12011	NULL	NULL	RoboticsModule.Messages.Location	NULL	17551085	-84.32	51	130	52500	26	10.27.80.232	1555040	0
	DISCONNECTED			NULL	NULL	RoboticsModule.Messages.Location	NULL	15251145	0.13	55	170	53300	24	10.27.80.118	1542199	0
468	CONNECTED_AGAIN		12003,12011	NULL	NULL	RoboticsModule.Messages.Location	NULL		0	70	280	54800	24	10.27.80.124	1542187	0
161	CONNECTED_AGAIN		12003,12011,21085	NULL	NULL	RoboticsModule.Messages.Location	NULL		0	65	280	54400	27	10.27.80.234	1542157	0
	NORMAL			NULL	NULL	RoboticsModule.Messages.Location	NULL	17601085	-1.17	64	420	54200	18	10.27.80.126	1542149	0
	NORMAL			NULL	NULL	RoboticsModule.Messages.Location	NULL	16951150	-179.52	15	390	48900	20	10.27.80.193	1555076	0

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Sorter Sweeper Database

Estop

We receive callbacks from Geek for estops. Estop model shows the Name of the estop, Geeks AreaId, and if its actively estopped.

When an Estop is released a record is added to estop activity model.

Name	AreaId	Estopped
System	NULL	0
zone1	10805	0
zone2	10806	0
zone3	10807	0
zone4	10808	0
zone5	10813	0
zone6	10814	0
zone7	10809	0
zone8	10810	0
zone9	10811	0
zone10	10812	0
Zone11	10815	0

EstopAreaName	SortName	SortGuid	StartTime	FinishTime	StoppedTime
System	NewSort3/9/2026 7:44:34 AM	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 11:51:04.6902705	2026-03-09 12:02:40.1083794	00:11:35.4181089
System	NewSort3/8/2026 10:28:40 PM	33c8c0eb-ad2c-4236-baaf-5411e597dcd7	2026-03-09 04:38:47.0096891	2026-03-09 04:42:28.6448945	00:03:41.6352054
zone1	NewSort3/8/2026 2:08:26 PM	2e6ae530-524b-49ce-bf37-1584d5de32ca	2026-03-08 16:26:17.8414234	2026-03-08 16:26:31.5520466	00:00:13.7106232
zone8	NewSort3/8/2026 2:08:26 PM	2e6ae530-524b-49ce-bf37-1584d5de32ca	2026-03-08 16:31:30.8426722	2026-03-08 16:37:49.9758176	00:06:19.1331454
zone4	NewSort3/8/2026 2:08:26 PM	2e6ae530-524b-49ce-bf37-1584d5de32ca	2026-03-08 18:13:12.9514600	2026-03-08 18:14:28.2952828	00:01:15.3438228
System	NewSort3/8/2026 2:08:26 PM	2e6ae530-524b-49ce-bf37-1584d5de32ca	2026-03-08 15:18:49.3178610	2026-03-08 15:19:09.8110700	00:00:20.4932090
zone8	NewSort3/8/2026 2:08:26 PM	2e6ae530-524b-49ce-bf37-1584d5de32ca	2026-03-08 16:38:39.0790157	2026-03-08 16:38:59.0023883	00:00:19.9233726
System	NewSort3/7/2026 9:07:14 AM	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e	2026-03-08 14:05:08.4978255	2026-03-08 14:21:50.0819471	00:16:41.5841216
System	NewSort3/9/2026 7:44:34 AM	0e32c5d3-59d3-44f5-8021-6b2dce895142	2026-03-09 12:03:42.9520968	2026-03-09 12:16:20.4249902	00:12:37.4728934

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Sorter Sweeper Database

Sort Model

When UPS cycles a sort they call and API called Reset Metrics. This opens any closed chutes and creates a new SortName and SortGuid. All activity is logged with these fields for historical metrics sort to sort.

SortName	CurrentSort	SortGuid
NewSort3/7/2026 9:07:14 AM	0	bd4ac506-e790-4ee6-988b-fbf7fce7ad1e
NewSort3/8/2026 2:07:57 PM	0	95c7d9b1-3b3e-4952-bdb6-e45e7ac80c9d
NewSort3/8/2026 2:08:06 PM	0	e44029d3-4b81-47a2-b9c1-b67cab889363
NewSort3/8/2026 2:08:26 PM	0	2e6ae530-524b-49ce-bf37-1584d5de32ca
NewSort3/8/2026 10:28:06 PM	0	3350fb7a-58ff-46e0-897c-f4d7585af4ba
NewSort3/8/2026 10:28:18 PM	0	b22c92ef-79aa-488c-be19-448eb02702be
NewSort3/8/2026 10:28:23 PM	0	b4302f65-a186-4a1e-b7ea-695e1cc2ce6b
NewSort3/8/2026 10:28:40 PM	0	33c8c0eb-ad2c-4236-baaf-5411e597dcd7
NewSort3/9/2026 7:44:09 AM	0	908548f2-0832-4cc4-ab4e-ddddd239413f
NewSort3/9/2026 7:44:15 AM	0	949f6ac1-34bf-424d-87b0-75cedc55ca47
NewSort3/9/2026 7:44:34 AM	0	0e32c5d3-59d3-44f5-8021-6b2dce895142
NewSort3/9/2026 5:04:24 PM	0	3c08d6aa-5cc1-452c-b012-12bf42541598
NewSort3/9/2026 5:04:29 PM	0	a9f4aee0-cc28-4611-91bd-3cc3a4271892
NewSort3/9/2026 5:04:39 PM	1	d1419b69-e68f-4d30-b9c8-0b8e1da65419

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Sorter Sweeper Database

System State Model

Keeps track of the System Status, if chutes are to be kept closed after a bag out, and if we are getting sort data.

System Status

- StartingUp
- StartUpRequested
- Running
- ShutDownRequested
- Shutdown
- ZoneConfigRequested
- ZoneConfigStaging
- CloseOutRequested
- CloseOutStaging
- CloseOutStaged

Status	KeepChutesClosed	Sorting
Shutdown	1	0

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Sorter Sweeper Database

Task Model

All Task sent to robotics engine are tracked in the task model.

When a task is completed its archived to task activity

TaskId	TaskName	ToteId	Station	TaskStatus	RobotCode	ErrorCode	Parameters
NULL	ToteReadyToExchange	89	NULL	Executed	NULL	NULL	NULL
b2180d7c-e6b0-40ef-be25-454d56c6c07b	PickupTote	89	732	Executed	1542160	NULL	NULL
ac053089-c8d0-47c6-90a2-0181f5d32e64	DropOffSorter	141	732	Executed	1542159	NULL	NULL
b4b1e7f4-fdee-49c3-9e90-ad13a3018a90	MoveOutSorter	89	872	Executed	1542160	NULL	NULL
d601bc89-c20d-4ae9-b3d2-e6d08ca082b0	Not Ready Tipper Station:872	NULL	NULL	Sent	NULL	NULL	NULL
NULL	ToteReadyToExchange	128	NULL	Executed	NULL	NULL	NULL
90f18410-5465-4844-811f-d500f91a51a6	PickupTote	128	336	Executed	1555022	NULL	NULL
76433483-dafa-401d-8a5a-1d0c1b2d6ed0	DropOffSorter	49	336	Executed	1555021	NULL	NULL
NULL	ToteReadyToExchange	299	NULL	Executed	NULL	NULL	NULL
NULL	ToteReadyToExchange	70	NULL	Executed	NULL	NULL	NULL
60d6e6be-2f15-4f13-94db-068d311bbeba	PickupTote	70	481	Executed	1555012	NULL	NULL
247bfe31-2956-4ed2-af7a-7413db964ff3	PickupTote	299	454	Executed	1555057	NULL	NULL
NULL	ToteReadyToExchange	259	NULL	Executed	NULL	NULL	NULL
34a9b0fc-86aa-418d-ae38-310cdeb512bb	DropOffSorter	279	481	Executed	1555013	NULL	NULL
7511107f-6f85-4650-a3f3-5c699549dd0a	PickupTote	259	351	Executed	1555009	NULL	NULL
2444f78c-0ebd-46bc-91a4-dae3a467de9	DropOffSorter	16	454	Executed	1555055	NULL	NULL
NULL	ToteReadyToExchange	115	NULL	Executed	NULL	NULL	NULL
6ccbc498-245f-4989-9002-fa31a566e2e7	PickupTote	115	443	Executed	1542158	NULL	NULL
72c17b50-2e51-44ac-9877-3c0e1e6ad3b9	DropOffSorter	134	351	Executed	1555027	NULL	NULL
NULL	ToteReadyToExchange	71	NULL	Executed	NULL	NULL	NULL
3d9ac723-6f7e-4bbe-b185-9462cee7c0d0	PickupTote	71	445	Executed	1555055	NULL	NULL
cde43391-4220-4439-8d4d-5950f7824d2c	DropOffSorter	155	443	Executed	1555028	NULL	NULL
c3637a71-c39c-4026-9f40-154c2419c807	MoveOutSorter	70	849	Executed	1555012	NULL	NULL
f74a30ee-61f2-4433-a6c9-0d8fa2878456	Not Ready Tipper Station:849	NULL	NULL	Sent	NULL	NULL	NULL
NULL	ToteReadyToExchange	315	NULL	Executed	NULL	NULL	NULL
92e9195e-faf7-43f6-9bb7-759f9ee21c0c	MoveOutSorter	128	832	Executed	1555022	NULL	NULL
4bf7fc30-eea7-4614-8436-3f8eadd55856	Not Ready Tipper Station:832	NULL	NULL	Sent	NULL	NULL	NULL
NULL	ToteReadyToExchange	141	NULL	Executed	NULL	NULL	NULL
ae585901-0e82-465d-b266-2c44dad16fda	DropOffSorter	133	445	Executed	1555064	NULL	NULL
669c4c02-5420-4d13-ae84835f867c914	PickupTote	315	715	Executed	1555037	NULL	NULL

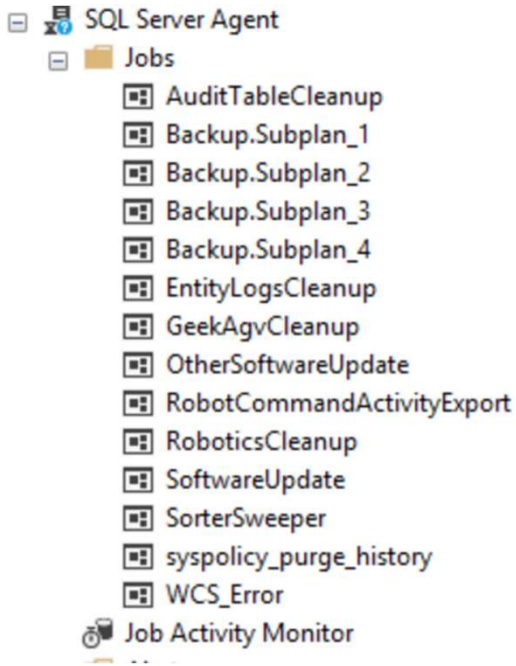
TaskId	TaskName	ToteId	Station	TaskStatus	RobotCode	ErrorCode	Parameters
ba1d702e-22db-49cf-8800-297892003b15	Resume AGV	NULL	NULL	Sent	NULL	NULL	NULL
ec210c5b-ba94-4bda-8a26-8fc383ae932c	PickupTote	450	315	Sent	NULL	NULL	NULL
c8b48500-0cba-4c59-9565-474b479a9be7	Resume AGV	NULL	NULL	Sent	NULL	NULL	NULL
6ba645fb-1081-46e3-8962-5f3d2ac69427	Resume AGV	NULL	NULL	Sent	NULL	NULL	NULL
beef2cb6-5d9e-4984-9b54-eae5a480626f	Resume AGV	NULL	NULL	Sent	NULL	NULL	NULL
22198523-e90b-4843-9528-94ca1c2f934e	DropOffSorter	299	NULL	Sent	NULL	NULL	NULL
de809acb-f08b-4f2d-bb02-2207a00c330b	DropOffSorter	314	NULL	Sent	NULL	NULL	NULL
40a74dd7-adfc-4191-a047-a3cc5bfa226b	Resume AGV	NULL	NULL	Sent	NULL	NULL	NULL
2aa13a0a-75d5-4217-b65f-44f20bb110b6	DropOffSorter	34	NULL	Sent	NULL	NULL	NULL
5d5932b3-29c2-4d85-a9c0-69b599507fe8	Pause AGV	NULL	NULL	Sent	NULL	NULL	NULL
fca0efce-4435-43ba-af34-1773d5bb6efa	DropOffSorter	423	NULL	Sent	NULL	NULL	NULL
dd6327f1-458a-4266-ac65-3e38167ce152	Resume AGV	NULL	NULL	Sent	NULL	NULL	NULL
faee9b22-bcc3-46a5-b896-2d829478b18a	Resume AGV	NULL	NULL	Sent	NULL	NULL	NULL
62f2e167-b3e8-4e79-bb9f-1b6c5bed91bb	DropOffSorter	315	NULL	Sent	NULL	NULL	NULL
4ac665cd-83d0-49a4-9c61-5987ab351746	Resume AGV	NULL	NULL	Sent	NULL	NULL	NULL
b6f46856-cdda-4b67-aef2-5c345867e7c2	Resume AGV	NULL	NULL	Sent	NULL	NULL	NULL
f0a8a6c9-bc1b-468a-a725-4d10eef367cd	DropOffSorter	430	NULL	Sent	NULL	NULL	NULL
f50ac556-1b8d-4398-af36-d59b5fdb576d	Resume AGV	NULL	NULL	Sent	NULL	NULL	NULL
3c8f9575-9bcb-48f4-bf6a-d861f077c46c	DropOffSorter	311	NULL	Sent	NULL	NULL	NULL
78a63220-45d4-475d-8ca1-79cfcd529cd3	DropOffSorter	433	NULL	Sent	NULL	NULL	NULL
0bf38abf-bc5f-4997-a641-4aacb9c904be	DropOffSorter	82	NULL	Sent	NULL	NULL	NULL
45fc0c21-186f-4632-9d92-901eba5bf33d	DropOffSorter	65	NULL	Sent	NULL	NULL	NULL
23d8c53b-51f0-4448-9e7d-36147c30d924	Resume AGV	NULL	NULL	Sent	NULL	NULL	NULL
df87d160-298b-47ea-ba70-2c29dae34209	Resume AGV	NULL	NULL	Sent	NULL	NULL	NULL
96cf1720-4b9c-46b7-9535-e0ea6711538e	DropOffSorter	80	NULL	Sent	NULL	NULL	NULL
6e905833-3aa1-4398-8813-ca9ed7ed2d36	Resume AGV	NULL	NULL	Sent	NULL	NULL	NULL
b969e05b-af40-40fb-9214-6f854de76154	DropOffSorter	111	NULL	Sent	NULL	NULL	NULL
46826e21-2741-4206-a40b-075a675cbc5a	DropOffSorter	398	NULL	Sent	NULL	NULL	NULL
c63302f5-6c64-4b8b-ba4c-dc8453d6af40	Resume AGV	NULL	NULL	Sent	NULL	NULL	NULL
01e5c331-b411-49cd-95f2-8b88dfc01484	Pause AGV	NULL	NULL	Sent	NULL	NULL	NULL
48d76981-70e6-4cf1-9309-a6deec7d5c2a	Pause AGV	NULL	NULL	Sent	NULL	NULL	NULL

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Database jobs

A DBA has setup jobs to run house keeping on the database every 12 hours



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API Walkthrough

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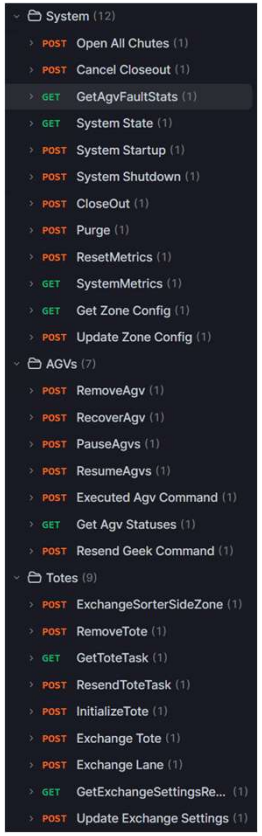
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APIs

Overview

OptiSweep provides APIs to get data and to make request.

We have Apidog installed on the primary server. The same APIs feed Ignition.



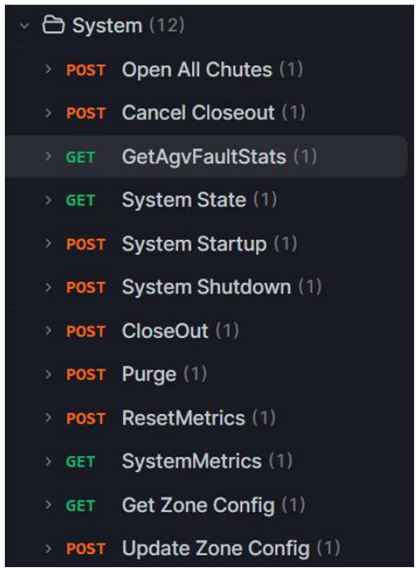
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APIs

System

- OpenAllChutes: Will open all closed chutes. This is helpful in the event they did a close out but did not cycle the sort and they want to startup again.
- CancelCloseout: Will cancel all close out exchanges that are not in process. Used when the customer does a manual bagout.
- GetAGVFaultStats: Will get the current AGV fault stats.
- System State: Will get the current state of the system.
- System Startup: Will startup the system if its in a shutdown state.

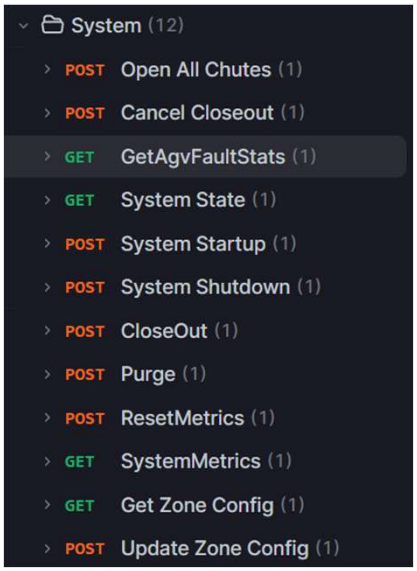


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APIs

System

- System Shutdown: Will shutdown the system if its in a running state.
- CloseOut: When the customer wants to bag out this will close all chutes and exchange any totes with parcels.
- Purge: Will exchange every tote in the system.
- ResetMetrics: Is called when UPS cycles a sort. Generates a new sort guid for metrics to be tracked.
- GetZoneConfig: shows the current zone configuration.
- UpdateZoneConfig: Allows the customer to change the zone configuration. This is never used.



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Slide 115

MF1 Missing bagouts

Mitchel Flynn, 2026-03-11T20:17:35.445

APIs

AGVs

- RemoveAgv: Requires the robotId and will send a request to Geek to remove the specified AGV from the system.
- RecoverAGV: Requires the robotId and will send a request to Geek to add the specified AGV to the system.
- PuaseAGVs: Will make a request to Geek to pause all AGV movement.
- ResumeAGVs: Will make a request to Geek to resume all AGV movement.
- ExecutedAgvCommand: Requires a robotId and will execute the active command the AGV has. This is helpful during unplanned restarts in the event that OptiSweep missed callbacks from Geek+



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APIs

AGVs

- GetAGVStatuses: Shows the current State of the AGV, its current command, vitals, any AGV errors and System errors.
- ResendGeekCommand: Requires a command guid. It will resend the request to Geek+. Helpful in the event of an unplanned restart and Geek+ missed the request.



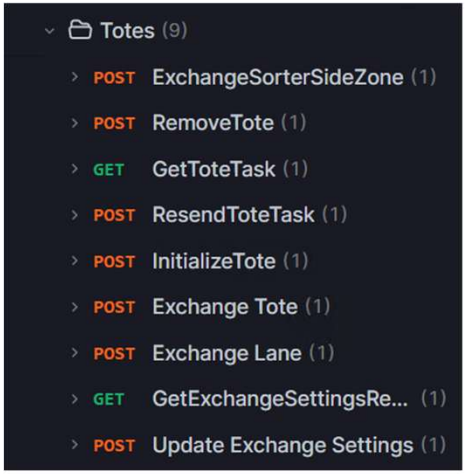


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APIs

Totes

- ExchangeSorterSideZone: Requires a sorter side and a zone. This is used during testing to send multiple exchanges to the tippers.
- RemoveTote: Requires a tote id and will remove the tote from the system. Note: the AGV must be removed before the tote can be removed.
- GetToteTask: Shows the active state of the totes.
- ResendtoteTask: Requires a task guid and will resend the task to the robotics module.

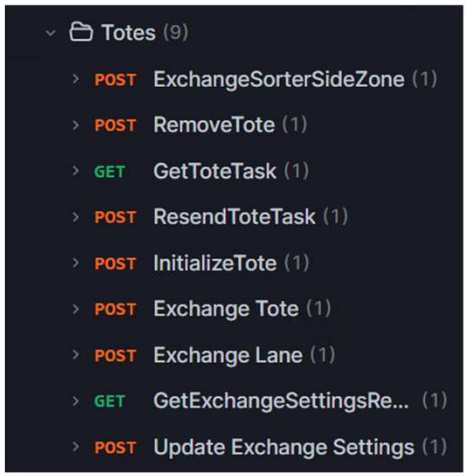


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APIs

Totes

- InitalizeTote: Requires a ToteId and a StationId to add a tote to the system.
- ExchangeTote: Requires a ToteName and will exchange that specific tote.
- ExchangeLane: requires a LaneName and will exchange that specific destination.
- GetExchangeSeettings: Will show the current exchange settings.
- UpdateExchange: Allows the customer to adjust the exchange settings.



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Ignition Walkthrough

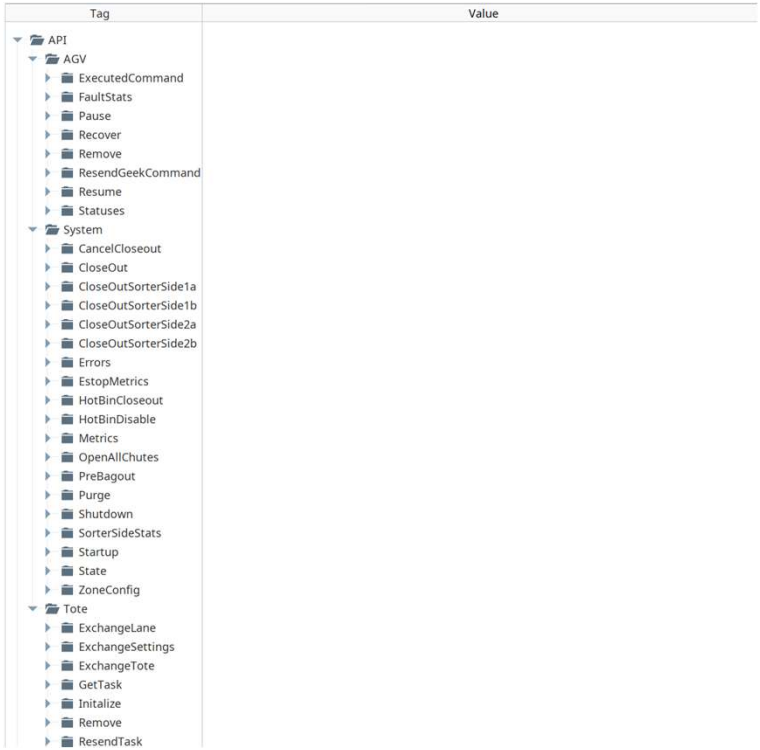
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APIs

- Ignition has the APIs outlined in the API section to give the functionality to Aveva.
- Aveva will send a request which Ignition will make an API request and recreate the data in tags.



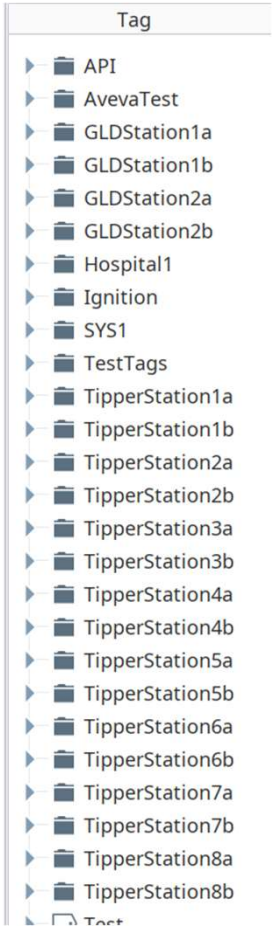


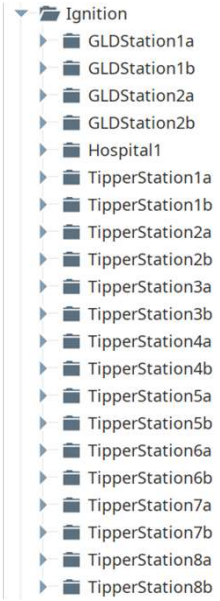
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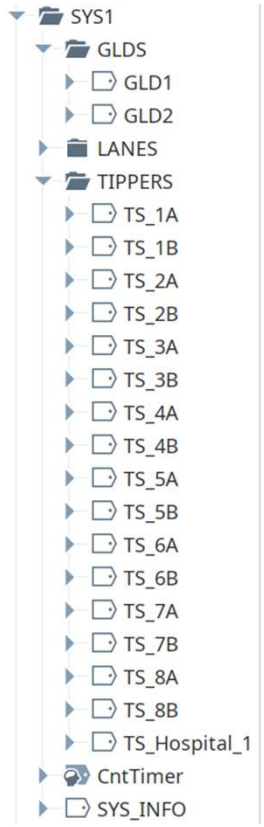
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Tags

- All tippers and Hospitals tags are mapped in Ignition.
- For OptiSweep to communicate we utilize tag provider and are mapped under Ignition.
- SYS1 is where all metrics are converted to tags for Aveva to read.





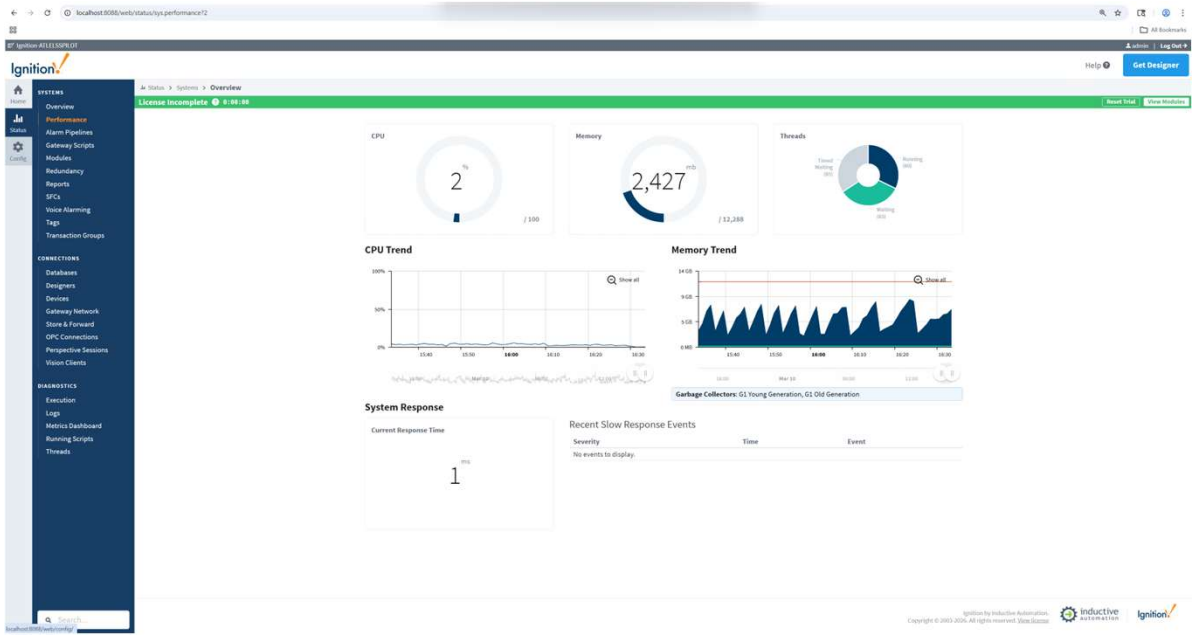


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Ignition Gateway UI

- Ignition can be accessed through the web browser. http://localhost:8088/web/status/sys.performance?2

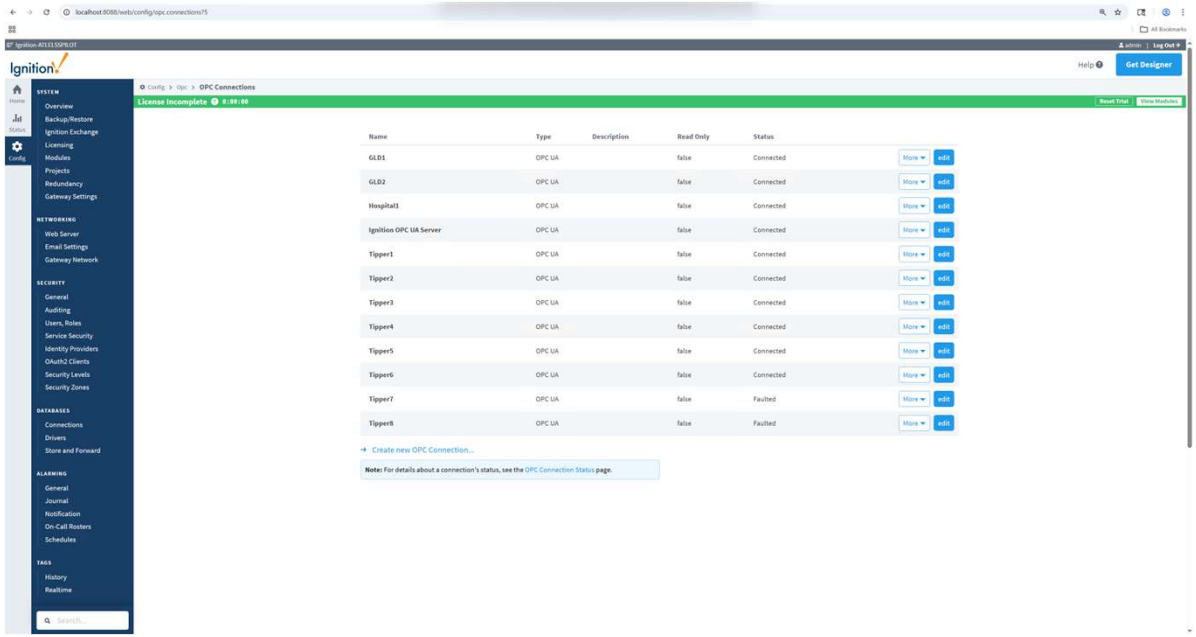


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Ignition Gateway UI

- Under Config, OPC, OPC Connections you can view the current connections.
  - In this example both tipper 7 and 8 are powered off and show Faulted.



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Service

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Services

On the primary service there are 3 programs that must be running for the system to run

- Ignition
  - Used for opc connections
- Stunnel
  - Used for secure encryption to UPS xle chat
- OptiSweep
  - Fortna's service that coordinates all operations.

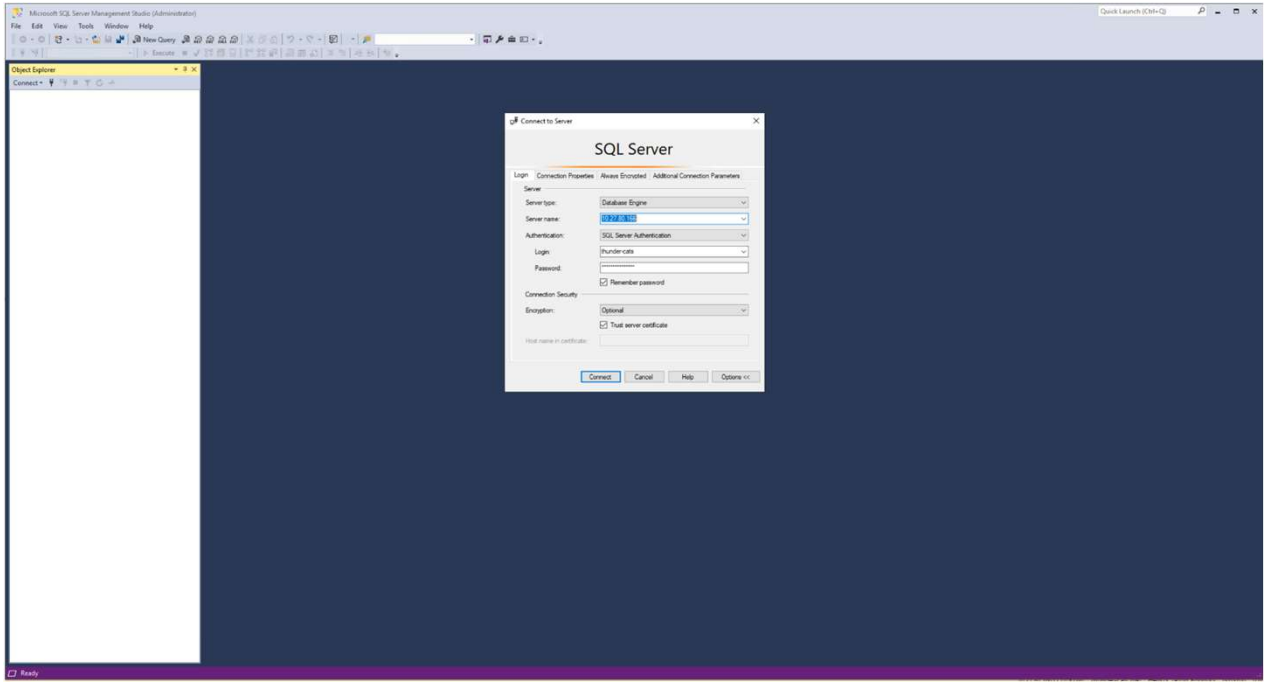
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Services

Unplanned restart

Rdp into fortna-wcs-primary.txrth.otnet.ups.com

Open Sql server management studio and check that you can connect



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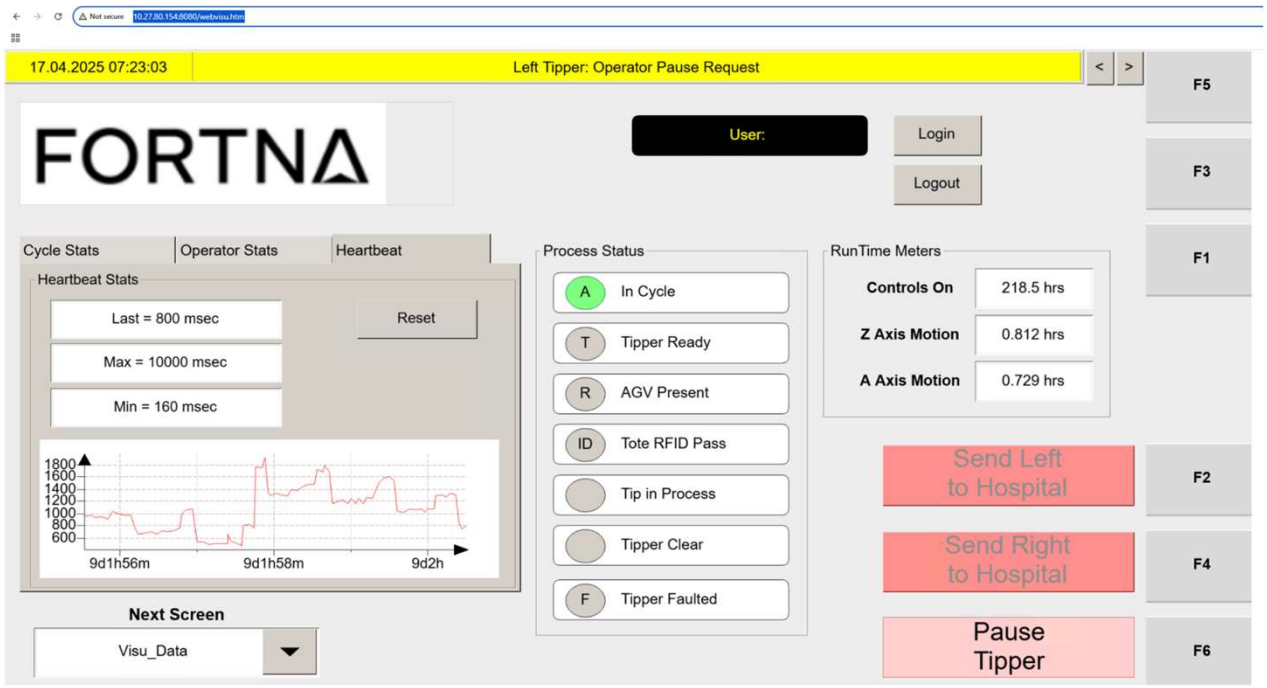
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Services

Unplanned restart

Check to make sure the tippers are getting heartbeat. You can connect via a web browser using a tipper ip.

http://10.27.80.154:8080/webvisu.htm



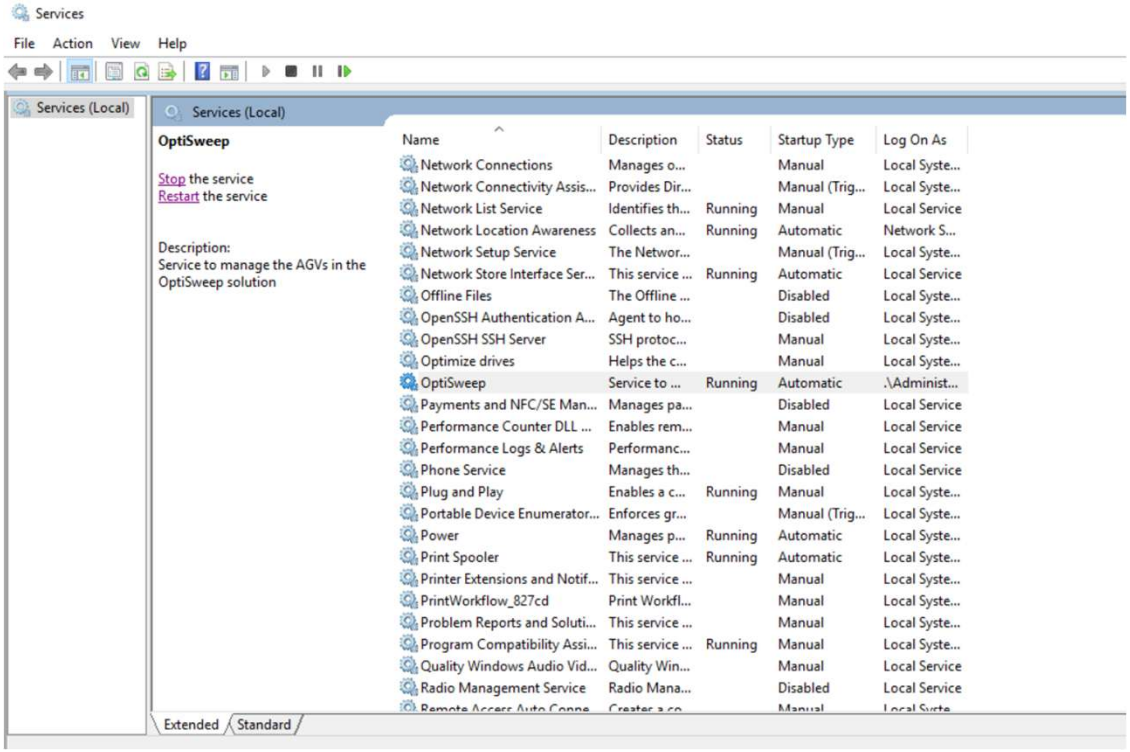
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Services

Unplanned restart

If the heartbeat is flat lined Igniton probably came up later then wcs and the service OptiSweep needs to be restarted



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Services

Unplanned restart

Open stunnel

Under file click connections

It should show 3 connections AI01, AI02, and SortationChat.



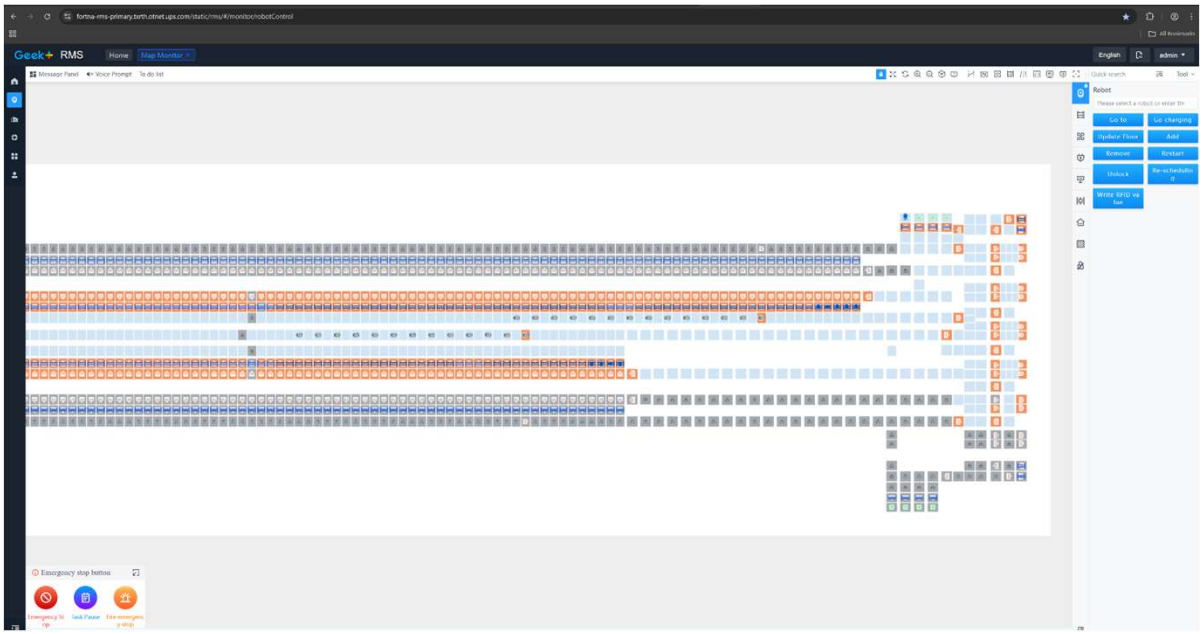
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Services

Unplanned restart

Log into RMS and confirm the screen loads before having UPS remove the master estop https://fortna-rms-primary.txrth.otnet.ups.com/static/rms/#/monitor/robotControl



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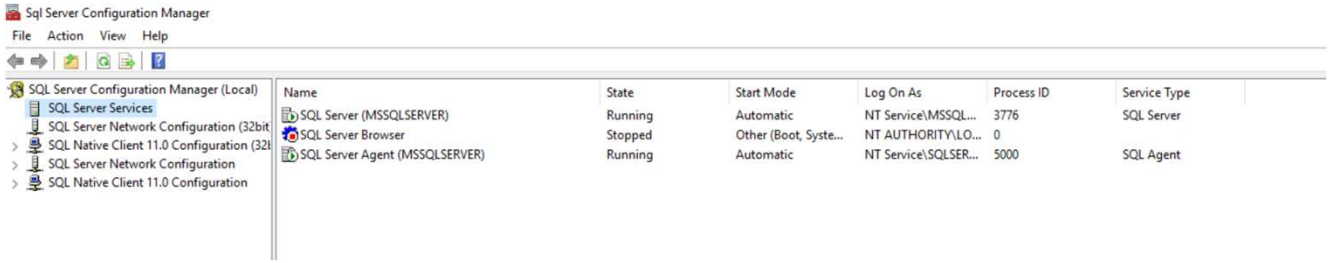
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Services

Unplanned restart

Rdp into fortna-wcs-backup.txrth.otnet.ups.com

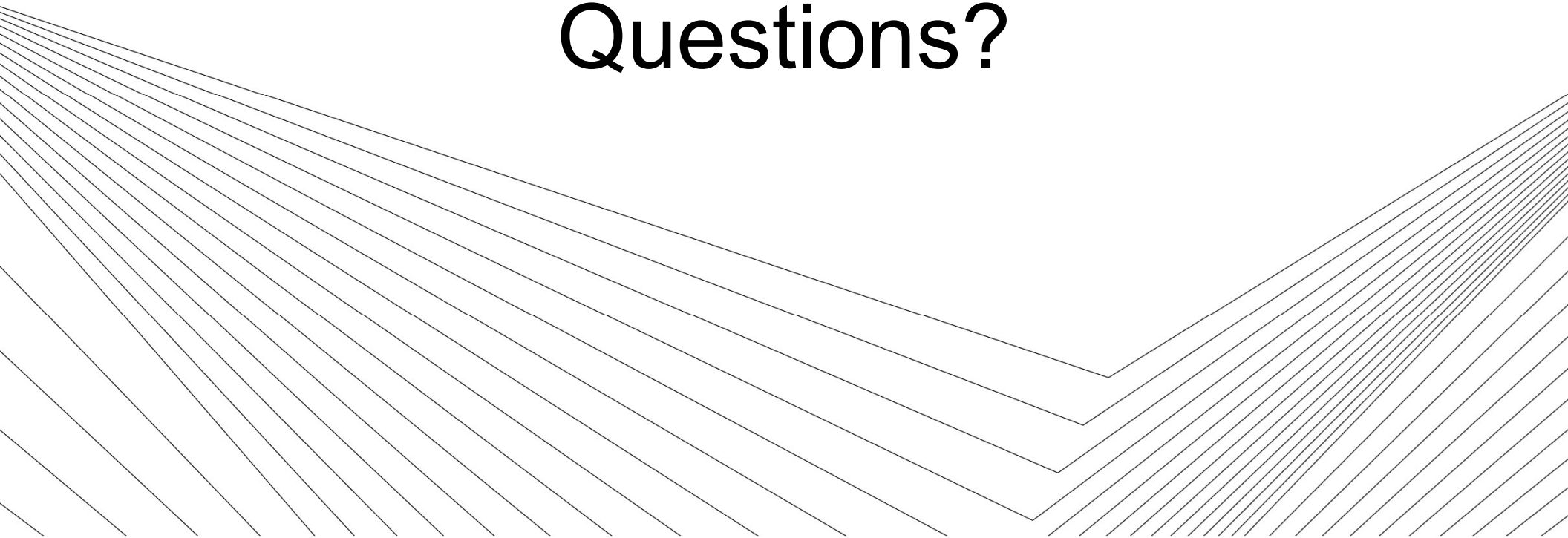
Open sql server configuration manager and make sure SLQ server agent is running



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FORTNA

Questions?





Test scenarios

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Test scenarios

1. Sorter side missing agv underneath sorter
2. Agv tracking error/lost path?
3. Object blocking agv
4. Agv losing battery?
5. Rms showing two agvs on top of each other?
   This happened recently at haslet
6. Add agv / remove agv

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